

C S C 2 4 1

Python Programming Language I

Study Manual

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```
>>> total = 0
>>> for n in scores:
    total += n
>>> print(total / len(scores))

72.5 ■
```

Contents

Every line is a link. The page numbers are read back from the rendered book, never typed by hand.

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How to use this manual

Read this page once. It is short, and it decides whether the rest of the manual works for you.

TEACH

THE BARGAIN

This manual assumes you have never written a line of code, and it carries you from there to a full mark in the examination. It is also meant to be **enough on its own**. You should never need a lecturer, a website or the official documentation to follow it. Anything it uses, it teaches first.

In exchange it asks one thing: **type the code**. Reading code creates a feeling of understanding that disappears in the examination hall. Typing it, getting it wrong, and fixing it is the thing that actually teaches. Every code block here was run on a real interpreter before it was printed, so if yours does something different, the difference is information.

The boxes

Every page is built from a small set of boxes. Each has one job, and the colour tells you which before you read a word.

TEACH

New ideas, explained from nothing. Read these properly.

● MUST MEMORISE

Python yellow means **learn this by heart**. It is the only thing this colour is ever used for in this manual. Nothing decorative is yellow. If a page has yellow on it, there is something on that page you are expected to know cold.

WORKED

A problem solved in numbered steps, then a **now you try** at the end. Do the "now you try" before moving on.

DRY RUN

A program traced by hand, one line at a time, with a table of what every variable holds. This examination asks you to state what code displays, and this is how you work it out without a computer. **Dry run** is the term used in this course.

⚠ TRAP

A mistake that is easy to make and costly. Most of these are taken from the errors the examination itself plants in its code.

RECALL

Questions with the answers printed underneath, below a dashed line. **Cover the answers.** Say your answer out loud, or write it, and only then look. Checking whether you knew something is not the same as knowing it, and the difference is the whole point of this box.

Reading the code

● MUST MEMORISE

THREE KINDS OF BLOCK

Code appears in three kinds of block, and confusing them is the most common early mistake. Each has a different coloured left edge.

1. The interactive prompt. Lines starting `>>>` are what you type. Lines without it are what Python printed back. You never type the `>>>` itself.

```
>>> 2 + 2
4
```

2. A script. The contents of a file, to be saved and run as a whole. No `>>>` anywhere.

```
print("saved in a file")
```

3. Output. What appeared on screen. You never type this.

```
saved in a file
```

Where a keystroke is meant rather than something to type, it is shown as a key: press

Enter .

TEACH

ONE PROMISE ABOUT OUTPUT

Every output shown in this manual was captured from a real Python 3.13 interpreter at the moment this page was built. None was typed by hand from memory. This matters more than it sounds: a printed output that was never actually run is just a confident guess, and where Python's real answer is surprising, a guess is exactly where it will be wrong. You will meet one such surprise in Module Two, Unit 2, and it is the reason for this promise.

MODULE ONE

Introduction to Python Programming

What a program actually is, what the machine does with it, and how to get Python running on your own computer so that every page after this one is something you can try rather than something you read.

```
>>> print( "hello" )
```

```
hello
```

Overview of Python

Start here even if you have never written a line of code. This unit assumes nothing.

By the end of this unit you can: say what a program is, explain what an interpreter does and how it differs from a compiler, state why Python is used, and name the places Python is used in practice.

What a program is

TEACH

FROM ZERO

A computer does nothing on its own. It waits to be told, in exact terms, what to do. A **program** is that telling: an ordered list of instructions, written down, that the computer carries out one after another.

Think of a recipe. A recipe is not the cooking; it is the written instructions that produce the cooking when somebody follows them. A program is not the work; it is the written instructions that produce the work when the computer follows them.

Two words that follow from this, and that you will use constantly:

- ▶ **Code** is the text you write. The instructions themselves.
- ▶ **Running** (or **executing**) a program means telling the computer to carry out those instructions now, from the top, in order.

ANCHOR

A recipe written down is a program. It lists ingredients, which are the data, and steps in an order that matters, which are the instructions. Anyone who follows it exactly gets the same cake, and that is the whole idea: the cook does no thinking, and every decision was made by whoever wrote the recipe. A computer is a cook that follows a recipe perfectly and understands nothing.

Why we need a language

TEACH

The computer's own vocabulary is numbers. Nothing else. Instructions inside the machine are patterns of the digits 0 and 1, called **machine code**. Writing anything useful directly in 0s and 1s is possible and unbearable.

So we write in a **programming language** instead: a fixed, small, precisely defined set of words and symbols that a human can read and that can be turned into machine code automatically. Python is one such language.

A programming language is deliberately far stricter than English. In English, "print hello" and "hello, print it" and "could you print hello" all land. A programming language accepts exactly one form and rejects the rest. That strictness is not the language being difficult. It is the reason the translation to machine code can be done without a human guessing at your intention.

Interpreter and compiler

Something must sit between your text and the machine's numbers. There are two ways to build that something, and the difference is examinable.

TEACH

EXAMINABLE

A **compiler** reads your *whole* program, translates all of it into machine code, and writes that machine code out as a new file. Nothing runs during translation. You run the produced file afterwards, and it runs on its own.

An **interpreter** reads your program *one instruction at a time*, and carries out each instruction as soon as it has read it. Nothing is written out. There is no separate file to run afterwards. Translation and running are the same activity.

Python is interpreted. The program that reads your Python and does what it says is called **the Python interpreter**, and installing Python (Unit 2) means installing it.

● MUST MEMORISE

COMPILER VS INTERPRETER

	COMPILER	INTERPRETER
How much it reads before acting	The whole program	One instruction
Produces a machine code file?	Yes	No
When errors surface	All at translation, before any run	When that line is reached
Speed of the finished program	Faster	Slower
Example language	C, C++	Python

Marks are lost here by naming only one difference. The question asks you to differentiate, so give the pair each time: what it reads, and what it produces.

⚠ TRAP

"Interpreted means Python has no errors until it runs" is **half true and the half that is false costs marks**. Python still checks that each line is *grammatically* possible before running anything in that file. A missing colon is caught immediately, before the first line executes. What waits until the line is reached is a *meaning* error, such as dividing by zero.

You will meet this split again as `SyntaxError` versus runtime error in Module Two, and it is worth marks on its own.

◆ KEY POINT

A compiler translates the whole program before any of it runs, so it finds faults up front. An interpreter translates and runs one statement at a time, so a fault on line 40 is only found when line 40 runs, and lines 1 to 39 have already had their effect.

BLUNT VERSION: a compiler reads the whole recipe first. An interpreter starts cooking, and it runs until it hits the mistake.

Why Python

TEACH

- ▶ **It reads close to English.** Fewer symbols stand between your intention and the page, so there is less to get wrong.
- ▶ **It is short.** Work that takes many lines elsewhere often takes one here.
- ▶ **It comes with a large standard library:** a big collection of ready made code that ships with Python, so common jobs are already solved. You will use one of these, `sqlite3`, in Module Five.
- ▶ **It is portable.** The same Python file runs on Windows, macOS and Linux.
- ▶ **It is free** and its interpreter is open source.

TEACH

WHERE PYTHON IS USED

FIELD	WHAT PYTHON DOES THERE
Web development	Runs the server behind a website
Data analysis	Reads large data files, summarises and charts them
Artificial intelligence	Builds and trains models
Automation and scripting	Repetitive machine work, such as renaming thousands of files
Scientific computing	Simulation and numerical work
Software testing	Automatically exercising other programs

1. What is a program?
2. Give the two differences between a compiler and an interpreter that earn the marks.
3. Is Python compiled or interpreted?
4. Name three fields where Python is used.

ANSWERS

1. An ordered list of instructions that a computer carries out, written down in a language it can be made to understand.
2. A compiler reads the whole program and produces a machine code file. An interpreter reads one instruction at a time and carries it out at once, producing no file.
3. Interpreted.
4. Any three of: web development, data analysis, artificial intelligence, automation, scientific computing, software testing.

MODULE ONE · UNIT 2

Setting Up Python

Do this once, on your own machine, before you read Module Two. Everything after here is meant to be typed, not watched.

By the end of this unit you can: install Python, confirm it installed, use the interactive prompt, write and run a script file, and say which of the two you are looking at.

Installing

TEACH

WINDOWS

1. Open `python.org` and choose Downloads. The site offers you the installer for the system you are on.
2. Run the installer.
3. **Tick "Add Python to PATH" on the first screen.** This is the one step that people skip and then cannot proceed. It is explained directly below.
4. Choose "Install Now" and wait for it to finish.

● MUST MEMORISE

PATH

PATH is a list your operating system keeps of the folders it will search when you type a command. If Python's folder is not on that list, then typing `python` gets you "command not found", even though Python is installed perfectly well. The program is there; the system just does not know where to look.

If you have already installed without ticking it, run the installer again, choose Modify, and add it. You do not need to uninstall anything.

Both usually ship with a Python already, but often an old one. Install the current version from `python.org` in the same way. On Linux you may instead use your system's package manager, for example `sudo apt install python3` on Ubuntu.

On macOS and Linux the command is often `python3` rather than `python`. If `python` gives you an error or a version starting with 2, use `python3` everywhere in this manual.

Confirming it worked

TEACH

Open a **terminal**, the window where you type commands to your operating system rather than clicking. On Windows it is called Command Prompt or PowerShell; on macOS and Linux it is called Terminal. Type this and press **Enter**:

IN THE TERMINAL, NOT IN PYTHON

```
python --version
```

It should answer with a version number:

```
Python 3.13.5
```

Your exact numbers will differ, and that is fine. Any version starting with **3** is fine. If it starts with 2, or you get "command not found", see the PATH box above, and try `python3 --version`.

⚠ TRAP

PYTHON 2

If your version starts with **2**, stop and fix it. Python 2 is a different language in small, silent ways. In Python 2, `5 / 2` is `2`. In Python 3 it is `2.5`. Nothing warns you; you simply get wrong answers. Every page of this manual is Python 3.

◆ KEY POINT

The interactive prompt evaluates what you type and **DISPLAYS** the result, which is why `2 + 3` shows 5 there. A script does not: an expression on its own line is computed and thrown away, so a script must call `print()` to show anything.

BLUNT VERSION: the prompt talks back, a script does not. If your file prints nothing, that is usually why.

The two ways to run Python

This is the distinction that confuses beginners more than any other, so it gets its own box. From here on, every code block in this manual is labelled as one or the other.

1. The interactive prompt (also called the REPL, or interactive mode). You type one line, press **Enter**, and Python answers immediately. You recognise it by the three symbols `>>>` at the left, which Python prints to mean "your turn". It forgets everything when you close it.

2. A script. You type many lines into a file, save it with a name ending in `.py`, and run the whole file at once. It keeps everything, because it is a file.

The prompt is for asking. A script is for building. Your exam answers are scripts.

TEACH

THE PROMPT

Type `python` on its own in the terminal and press **Enter**. The prompt appears:

```
>>>
```

Now type an instruction and press **Enter**. Python answers on the next line, then offers you the prompt again:

```
>>> print("Hello, World!")
Hello, World!
>>>
```

To leave the prompt and return to the terminal, type `exit()` and press **Enter**.

In every prompt transcript in this manual:

- ▶ Lines beginning `>>>` are what **you type**. Do not type the `>>>` itself; Python prints that.
- ▶ Lines with no `>>>` are what **Python printed back**.

So in the transcript above, you typed 21 characters and Python printed `Hello, World!`. Getting this backwards is the single most common early confusion, which is why the two are given different coloured edges throughout.

Open any plain text editor. Notepad works. So does IDLE, which was installed alongside Python: find it in your programs, then choose File, then New File.

Type these two lines:

HELLO.PY

```
print("Hello, World!")
print("I am running from a file.")
```

Save it as `hello.py`. The `.py` ending is what marks a file as Python. Then, in the terminal, move to the folder you saved it in and type:

IN THE TERMINAL

```
python hello.py
```

Python reads the file from the top, carries out each line in order, prints:

```
Hello, World!
I am running from a file.
```

and then exits. Notice there is no `>>>` anywhere: you are not having a conversation, you are handing over a finished list of instructions.

⚠ TRAP

COSTS MARKS AND HOURS

Typing `python hello.py` *while already at the `>>>` prompt* gives you a `SyntaxError`. Those are terminal commands, not Python instructions. They live in different places:

```
>>> python hello.py
File "<stdin>", line 1
  python hello.py
    ^^^^^
SyntaxError: invalid syntax
```

If you see this, you are at the Python prompt when you meant to be at the terminal. Type `exit()` first.

1. What does PATH do, and what goes wrong if Python is not on it?
 2. What do the three symbols `>>>` tell you?
 3. Which ending must a Python file's name have?
 4. You want to keep your work and hand it in. Prompt or script?
-

ANSWERS

1. PATH lists the folders the operating system searches when you type a command. If Python's folder is missing from it, typing `python` reports "command not found" even though Python is installed.
2. That you are at the interactive prompt, and that Python is waiting for you to type one instruction. You never type the symbols yourself.
3. `.py`
4. A script. The prompt forgets everything when it closes.

MODULE TWO

Python Basics and Syntax

The grammar of the language, the boxes you keep values in, and the symbols that combine them. Everything in the rest of the manual is built out of what is in these three units.

```
age = 25
```

```
name = "Ada" ■
```

Python Language Syntax

Syntax means the grammar: the rules that decide whether a line is Python at all, before anyone asks whether it is correct.

By the end of this unit you can: write and run statements, use comments, explain why indentation is not decoration in Python, print with `sep` and `end`, use escape characters, format with f-strings, and read a traceback.

Statements

TEACH

A **statement** is one complete instruction. In Python, one statement occupies one line, and the end of the line ends the statement. There is no symbol to mark the end.

```
print("first")
print("second")
```

Two lines, two statements, carried out top to bottom. If you know a language that ends statements with a semicolon, note that Python does not want one. It is not an error, but it marks your code as written by somebody thinking in another language, and the marks in this course go to Python style.

Comments

TEACH

A **comment** is a note to a human. Python ignores it completely. Everything from a `#` to the end of that line is a comment.

```
# Work out the area of the room.
area = length * width      # in square metres
```

Comments do not run, do not print, and do not affect anything. They exist so that you, in three weeks, can tell what you meant.

● MUST MEMORISE

READ THIS ONCE, CAREFULLY

A comment is **never** output. If you see

```
>>> 10 % 2
0      # Even
```

then the output is `0`. The words `# Even` are a human annotation sitting beside it, not something Python said. Python never answers you in comments.

This matters because printed material sometimes puts the annotation where the output should be, and a reader new to the language cannot tell which is which. In this manual the rule is absolute: **output is shown on its own line, in an output block, exactly as a real interpreter printed it.** Every output in this manual was captured from a run, not typed by hand.

Indentation

TEACH

PYTHON'S ONE BIG IDEA

In most languages, spaces at the start of a line are decoration, and blocks of code are marked with braces. **In Python the spaces are the structure.** There are no braces. Indentation is how Python knows which lines belong to what.

You have not met `if` yet, and it arrives properly in Module Three. Read this only for the shape:

```
if raining:
    print("take a coat")      # indented: belongs to the if
print("leave the house")    # not indented: always runs
```

The indented line runs only when it is raining. The unindented line always runs. Move that one line four spaces to the left or right and you have changed what the program does. Nothing else changed.

The rule: a colon at the end of a line means "a block follows", and that block is the lines indented beneath it. Use **four spaces** per level. Be consistent: Python accepts any consistent width, and rejects a mixture.

⚠ TRAP

EXAM FAVOURITE

Indentation errors are a standing question in this course. Two distinct failures, and the wording of the error tells you which:

```
IndentationError: expected an indented block after 'if' statement on line 1
```

means you wrote the colon and then did not indent anything beneath it.

```
IndentationError: unexpected indent
```

means you indented a line that had no colon above it to belong to.

Printing

TEACH

`print()` displays things. Give it several items separated by commas and it prints them all, putting **one space** between each:

```
>>> print("Ada", 25, "Lagos")
Ada 25 Lagos
```

It then moves to a new line. Both of those behaviours can be changed, and each has a name you must know.

`sep` sets what goes **between** the items. Default: one space.

`end` sets what goes **after** the last item. Default: a newline, written `\n`.

```
>>> print("2026", "07", "17", sep="-")
2026-07-17
>>> print("Ada", "Bob", sep="")
AdaBob
```

Setting `end` to the empty string stops the newline, so the next `print` continues on the same line:

```
>>> print("Hello", end="")
>>> print("World")
HelloWorld
```

Read that transcript carefully. The first `print` displayed `Hello` and stayed put; the second added `World` and ended the line. The output `HelloWorld` is the work of both statements together.

Quotes and escape characters

TEACH

Text in Python is written inside quotes, and is called a **string**. Single and double quotes are equally valid and mean exactly the same thing:

```
'Ada'
"Ada"
```

Having both is useful, because a string in double quotes may freely contain a single quote:

```
>>> print("It's fine")
It's fine
```

The problem arrives when the string must contain the same quote that ends it. Python reads the second quote as the end of the string and the rest becomes nonsense. The fix is an **escape character**: a backslash `\` before the quote, which tells Python "this next character is data, not punctuation".

```
>>> print("She said, \"Python is cool!\")
She said, "Python is cool!"
```

ESCAPE	MEANS
<code>\n</code>	Newline: move to the next line
<code>\t</code>	Tab: jump to the next tab stop
<code>\"</code>	A double quote, as data
<code>\'</code>	A single quote, as data
<code>\\</code>	One backslash, as data

A question asking you to "list three escape characters and their effects" has appeared in this course. The three safest to give are `\n`, `\t` and `\\`.

TEACH

ONE STRING, TWO LINES

`\n` inside a string breaks it across lines when printed. It is one string, not two:

```
>>> print("Name: Ada\nCity: Lagos")
Name: Ada
City: Lagos
```

f-strings

TEACH

You will constantly want to put a value inside a sentence. The direct way is an **f-string**: put the letter `f` immediately before the opening quote, then write `{ }` wherever a value belongs.

```
>>> name = "Ada"
>>> age = 25
>>> print(f"{name} is {age}")
Ada is 25
```

The `f` is not optional and not decoration. Without it you get the braces themselves:

```
>>> print("{name} is {age}")
{name} is {age}
```

After a colon inside the braces you may say *how* to display the value. These are the ones this course uses:

WRITTEN	MEANS	EXAMPLE IN	EXAMPLE OUT
:.2f	Fixed to 2 decimal places	123.456	123.46
:,	Comma between thousands	1250000	1,250,000
:,.2f	Both together	9876543.21	9,876,543.21
:.2%	As a percentage, 2 decimals	0.875	87.50%
:<10	Left aligned, width 10	'Item'	Item.....
:>10	Right aligned, width 10	'Price'	Price

In the last two rows each dot stands for one space, so that you can count them. Python prints spaces, not dots.

WORKED

ALIGNMENT YOU CAN ACTUALLY SEE

Field widths exist to build columns, and columns only work in a font where every character takes the same width. Every code block in this manual is set in such a font. Give three rows **the same format string** and they line up:

```
>>> print(f{'Item':<10}{'Price':>10})
Item          Price
>>> print(f{'Apple':<10}{1.5:>10.2f})
Apple         1.50
>>> print(f{'Banana':<10}{12.75:>10.2f})
Banana        12.75
```

The words start in the same column because `:<10` pins them to the left. The prices **end** in the same column because `:>10` pins them to the right, and that is true even though 1.50 and 12.75 have different lengths. That is what a field width buys you.

Spaces cannot be counted by eye, so here are those same three lines with every space drawn as a dot. Each dot is one space, and nothing else has changed:

```
Item.....Price
Apple.....1.50
Banana.....12.75
```

Count the first line. Item takes 4 of its 10 columns, so 6 spaces pad it on the right. Price is 5 characters pushed to the right hand edge of its own 10 columns, so 5 spaces come first. That is $6 + 5 = 11$ dots, which is what you counted. Every line is 20 characters wide.

Now you try: work out the dots for a row holding 'Fig' and 3.5 in the same format, before you run it. (3 + 7 pad, then 6 + '3.50': Fig then 13 dots then 3.50.)

Reading a traceback

TEACH

WORTH MANY MARKS

When Python cannot do what you asked, it prints a **traceback**. Beginners see a wall of red and look away. It is the most useful text on the screen, and half of this course's code marks come from finding errors, so learn to read it.

```
Traceback (most recent call last):
  File "C:\Users\you\hello.py", line 3, in <module>
    print(10 / 0)
    ~~~^~~
ZeroDivisionError: division by zero
```

Read it **from the bottom up**:

- ▶ **Last line** is what went wrong: `ZeroDivisionError`, and in plain words, division by zero. This is the line that tells you the problem.
- ▶ **Above it** is the offending code, with the marks `~~~^~~` pointing at the exact part that failed. The `^` sits under `/`, the operation that could not be done.
- ▶ **Above that** is where: the full path of the file, and line 3.

Error type, then message, then place. Three facts, always in that order.

Python prints the file's whole location, so your first line will be longer than the one above and will show wherever you saved it. Only the file name and the line number matter to you.

● MUST MEMORISE

SYNTAX BEFORE RUNTIME

Errors come in two kinds and they happen at **different times**:

A `SyntaxError` is a grammar failure. Python finds it while reading the file, *before running a single line*. Nothing in the file executes. There is no traceback of the kind above, because nothing ran.

A runtime error is a meaning failure in a line that is grammatically fine. Python only finds it when it reaches that line, so everything before it has already run and printed.

The consequence, which is examinable: a `SyntaxError` anywhere in a file hides every other error in it. If a file has a missing bracket on line 2 and a misspelled name on line 9, you are told about line 2 only. Fix it, run again, and only then hear about line 9.

1. What does a colon at the end of a line announce?
2. What are `sep` and `end`?
3. Which end of a traceback do you read first?
4. A file has a `SyntaxError` on line 2 and a `NameError` on line 9. What does Python report?

ANSWERS

1. That a block follows, made of the lines indented beneath it.
2. `sep` is what `print` puts between items (default one space); `end` is what it puts after the last one (default a newline).
3. The bottom. The last line names the error and states it in words.
4. The `SyntaxError` on line 2, and nothing else. The file never runs, so line 9 is never reached.

MODULE TWO · UNIT 2

Variables and Data Types

Where values are kept, what kinds of value exist, and the one thing about numbers that surprises everybody exactly once.

By the end of this unit you can: create and reassign variables, name them legally, list the four basic types, check a type, read input, convert between types, and explain why `0.1 + 0.2` is not `0.3`.

ANCHOR

A variable is a label on a box in a storeroom. Writing `total = 5` puts a value in a box and hangs the label "total" on it. Hanging the same label on a different box later is allowed, and the old box is simply forgotten. What the label can never do is change the thing inside the box while still pointing at it: that is a different idea, and it is what mutability means.

Variables

TEACH

A **variable** is a name that refers to a value. You make one by writing the name, then `=`, then the value:

```
>>> age = 25
>>> age
25
```

The `=` here does **not** mean equals. It means *assign*: work out the value on the right, then make the name on the left refer to it. It is an instruction, not a statement of fact. This is why

```
count = count + 1
```

is perfectly sensible, though as mathematics it would be nonsense. It reads: take what `count` is now, add 1, and make `count` refer to that.

TEACH

REASSIGNMENT

A variable can be given a new value at any time, including a value of a different kind. The old value is simply forgotten:

```
>>> x = 99
>>> print(x)
99
>>> x = "Take me to your leader."
>>> print(x)
Take me to your leader.
```

Python does not object. The name is a label, not a box of a fixed size.

● MUST MEMORISE

NAMING RULES

Must:

- ▶ Start with a letter or an underscore. **Never** a digit.
- ▶ Contain only letters, digits and underscores. No spaces, no hyphens.
- ▶ Not be a Python keyword (`if` , `for` , `class` , and the rest).

Should: be lowercase with underscores between words (`total_score`), and say what it holds. A name like `x` tells the reader nothing.

Case matters. `age` , `Age` and `AGE` are three different variables. This is a standing source of exam errors: a program that defines `square` and calls `Square` is broken.

The four basic types

● MUST MEMORISE

TYPES

TYPE	NAME IN PYTHON	WHAT IT HOLDS	EXAMPLES
Integer	int	Whole numbers, no decimal point	25, 0, -7
Float	float	Numbers with a decimal point	3.14, -0.5, 2.0
String	str	Text, in quotes	"Ada", 'x', ""
Boolean	bool	Truth, one of exactly two values	True, False

True and False are capitalised and are not strings. "True" in quotes is a string.

⚠ TRAP

2.0 IS NOT AN INT

2 and 2.0 are equal in value but are different types. The decimal point makes it a float, even when there is nothing after it. This decides what division gives you back, as Unit 3 shows.

TEACH

TYPE()

type() tells you what something is. When in doubt, ask:

```
>>> type(25)
<class 'int'>
>>> type(3.14)
<class 'float'>
>>> type("Ada")
<class 'str'>
>>> type(True)
<class 'bool'>
```

Read <class 'int'> as "this is an int". The word class is Python's internal vocabulary for a type.

◆ KEY POINT

Python is dynamically typed: the type belongs to the VALUE, not to the name. No declaration is needed, and rebinding a name to a value of another type is legal, which is why type() reports on what a name currently holds rather than on what it was first given.

BLUNT VERSION: the value carries the type. The name is only a sticker.

Input

● MUST MEMORISE

INPUT ALWAYS GIVES A STRING

`input()` stops the program, waits for the user to type something and press **Enter**, and hands back what they typed. The text you pass to it is shown as the prompt.

What it hands back is always a string. Always. Even when the user types digits.

```
age = input("How old are you? ")
print(type(age))
```

```
How old are you? 25
<class 'str'>
```

The user typed 25 and Python is holding the two characters "25", not the number. This single fact causes more broken exam answers than anything else in the course.

⚠ TRAP

THE CLASSIC

Forget to convert, and arithmetic does something silently absurd rather than failing:

```
a = input("First number: ") # user types 5
b = input("Second number: ") # user types 3
print(a + b)
```

```
First number: 5
Second number: 3
53
```

53, not 8. Because `+` on two strings joins them end to end. The program did exactly what you asked and nothing reported an error. The fix is to convert.

Converting between types

● MUST MEMORISE

CASTING

FUNCTION	TURNS A VALUE INTO	EXAMPLE	RESULT
<code>int()</code>	an integer	<code>int("25")</code>	25
<code>float()</code>	a float	<code>float("3.5")</code>	3.5
<code>str()</code>	a string	<code>str(100)</code>	'100'

So the fix for the trap above, and the pattern you will write hundreds of times:

```
age = int(input("How old are you? "))
```

Read it from the inside out: `input` collects text, `int` turns that text into a number, `=` stores the number.

`int()` on a float throws the decimal part away. It does not round:

```
>>> int(3.9)
3
>>> int(-3.9)
-3
```

It cuts towards zero. Use `round()` if you want rounding.

⚠ TRAP

INT() IS FUSSY ABOUT STRINGS

`int()` can only convert a string that is entirely a whole number. Give it anything else and you get a runtime error, and the program stops:

```
>>> int("xyz")
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
ValueError: invalid literal for int() with base 10: 'xyz'
```

Note that `int("3.5")` fails too, for the same reason: 3.5 is not a whole number written out. Use `float("3.5")`, or `int(float("3.5"))` if you truly want 3.

◆ KEY POINT

`input()` returns a string every time, whatever the user typed. Any arithmetic on it must convert first, with `int()` for a whole number or `float()` for a decimal, or the plus operator will join text instead of adding numbers.

BLUNT VERSION: wrap it in `int()` or `float()` the moment you read it. Every program on this paper needs that line.

Floats are not exact

● MUST MEMORISE

EVERYBODY MEETS THIS ONCE

Type this at the prompt. It is not a trick:

```
>>> 0.1 + 0.2
0.30000000000000004
```

Python is not broken and you did not make a mistake. A computer stores numbers in binary, in a fixed amount of space. Some numbers that are short and exact in decimal, such as 0.1, cannot be written exactly in binary at all, in the same way that 1/3 cannot be written exactly in decimal: 0.333... never finishes. Python stores the closest value it can fit, and the tiny difference shows up when you add.

This is not a Python quirk. It is how nearly every language on nearly every machine stores decimals.

It hides until it does not. Watch an ordinary looking interest calculation:

```
>>> INTEREST_RATE = 0.069
>>> balance = 10000
>>> amount = balance * INTEREST_RATE
>>> amount
690.00000000000001
```

Not 690.0 . And yet change the rate to 0.072 and it comes out perfectly clean:

```
>>> INTEREST_RATE = 0.072
>>> amount = balance * INTEREST_RATE
>>> amount
720.0
```

Same code, same shape, one is ugly and one is clean. That is exactly what makes this dangerous: you cannot predict which by looking, so you must handle it always rather than when it looks likely.

The rule: never show a raw float to a user, and never compare two floats with `==` . Format it on the way out.

● MUST MEMORISE

THE FIX

Format the float when you display it, and the problem disappears at the only point where it mattered:

```
>>> amount = 10000 * 0.069
>>> print(f"Interest: {amount:.2f}")
Interest: 690.00
```

Any exam program that reports an average, a percentage or an amount of money should print it through `:.2f` . It is one keystroke's worth of effort and it is the difference between a professional answer and a puzzling one.

1. What type does `input()` return?
2. The user types 5 and 3 into two `input()` calls and you print `a + b`. What appears, and why?
3. What is `int(3.9)`?
4. Why is `0.1 + 0.2` not `0.3`?

ANSWERS

1. A string, always, even if the user typed digits.
2. 53. Both are strings, and `+` on strings joins them. No error is reported.
3. 3. `int()` truncates towards zero, it does not round.
4. Floats are stored in binary in fixed space. 0.1 has no exact binary form, so Python stores the nearest value it can, and the small error becomes visible when adding. Format with `:.2f` on output.

MODULE TWO · UNIT 3

Python Operators

The symbols that combine values. Two of them, `//` and `%`, carry more exam marks than the rest put together.

By the end of this unit you can: use every arithmetic operator, predict what division returns, use `%` to test divisibility, apply precedence correctly, and use comparison and logical operators.

Arithmetic

● MUST MEMORISE

ALL SEVEN

OPERATOR	NAME	EXAMPLE	RESULT
+	Add	<code>7 + 5</code>	12
-	Subtract	<code>7 - 5</code>	2
*	Multiply	<code>7 * 5</code>	35
/	Divide	<code>7 / 5</code>	1.4
//	Floor divide	<code>7 // 5</code>	1
%	Remainder (modulus)	<code>7 % 5</code>	2
**	Exponent (power)	<code>7 ** 2</code>	49

● MUST MEMORISE

/ ALWAYS GIVES A FLOAT

/ gives a **float every time**, even when the division is exact:

```
>>> 10 / 2
5.0
>>> 5 / 2
2.5
```

10 / 2 is 5.0, not 5. If you need a whole number, use // or wrap it in int().

// divides and throws away the fractional part, giving the whole number of times the divisor fits:

```
>>> 7 // 5
1
>>> 17 // 3
5
```

● MUST MEMORISE

% IS THE WORKHORSE

% gives the **remainder** after division. 17 % 3 is 2, because 3 fits into 17 five times with 2 left over.

// and % are the two halves of one division:

```
>>> 17 // 3
5
>>> 17 % 3
2
```

17 divided by 3 is **5 remainder 2**. The first gives the 5, the second the 2.

The use that appears in this exam every year: a number is divisible by n exactly when `number % n == 0`.

```
>>> 10 % 2
0
>>> 7 % 2
1
```

A remainder of 0 on division by 2 means even. A remainder of 1 means odd. Read those two transcripts again: the outputs are 0 and 1. That is what Python printed.

This is `//` and `%` working together, and it is the standard demonstration of both. Turn 11730 seconds into hours, minutes and seconds.

- 1 An hour is 3600 seconds. **How many whole hours fit?** That is floor division.
11730 // 3600 gives 3
- 2 **What is left over after those hours?** Take the remainder on 3600, then ask how many whole minutes fit in it. Both steps at once: take the remainder on 60 of the total whole minutes.
*(11730 // 3600) * 3600 + (11730 // 60) % 60 gives 15*
- 3 **Seconds left over** is the remainder on 60.
11730 % 60 gives 30

TIME_CONVERTER.PY

```
total_seconds = 11730
hours = total_seconds // 3600
minutes = (total_seconds // 60) % 60
seconds = total_seconds % 60
print("Hours:", hours)
print("Minutes:", minutes)
print("Seconds:", seconds)
```

```
Hours: 3
Minutes: 15
Seconds: 30
```

Now you try: change 11730 to 7325 and predict the three numbers before running it. (2 hours, 2 minutes, 5 seconds.)

● MUST MEMORISE

**** IS RIGHT ASSOCIATIVE**

Nearly every operator groups from the left. `**` groups from the **right**:

```
>>> 2 ** 3 ** 2
512
```

This is `2 ** (3 ** 2)`, which is `2 ** 9`, which is 512. It is **not** `(2 ** 3) ** 2`, which would be 64.

`**` also does roots, because a square root is the power one half:

```
>>> 9 ** 0.5
3.0
```

Precedence

● MUST MEMORISE

ORDER OF OPERATIONS

When several operators appear in one expression, Python applies them in this order, highest first:

ORDER	OPERATORS
1	() brackets
2	**
3	* / // %
4	+ -

Within one row, work left to right (except **, which goes right to left).

```
>>> 2 + 3 * 4
14
>>> (2 + 3) * 4
20
```

When in doubt, add brackets. They cost nothing and they make your intention unmistakable to a marker.

◆ KEY POINT

A single slash is true division and always produces a float, even when the division is exact: $6 / 2$ is 3.0, not 3. Double slash floors to an integer, and flooring goes DOWN, so $-7 // 2$ is -4 rather than -3.

BLUNT VERSION: one slash, one point. Two slashes, no point, and it rounds down even for negatives.

Comparison operators

● MUST MEMORISE

THESE GIVE TRUE OR FALSE

OPERATOR	MEANS	EXAMPLE	RESULT
==	Equal to	$5 == 5$	True
!=	Not equal to	$5 != 3$	True
>	Greater than	$7 > 5$	True
<	Less than	$4 < 10$	True
>=	Greater than or equal to	$7 >= 7$	True
<=	Less than or equal to	$2 <= 4$	True

⚠ TRAP**= AGAINST ==**

= assigns. == asks. They are unrelated, and confusing them is the most common single error in this course's error-spotting questions.

```
age = 18      # assigns 18 to age
age == 18     # asks: is age 18? Gives True.
```

Write `if age = 18:` and Python stops you with a `SyntaxError`, because a question was expected and you gave it an instruction.

TEACH**CHAINING**

Python lets you write a range check the way mathematics does, and it means what it looks like:

```
>>> bmi = 22.4
>>> 18.5 <= bmi <= 24.9
True
```

This is worth remembering: a question in this course asks for exactly this test on a BMI value, and the chained form is shorter and clearer than joining two comparisons with `and`.

Logical operators

● MUST MEMORISE**AND, OR, NOT**

OPERATOR	TRUE WHEN	EXAMPLE	RESULT
and	Both sides are true	True and False	False
or	Either side is true	True or False	True
not	The other way round	not True	False

These are English words in Python, not symbols. If you have seen `&&` and `||` elsewhere, they do not exist here.

```
>>> n = 15
>>> n % 3 == 0 and n % 5 == 0
True
```

Fix that line in your memory. It asks "is `n` divisible by both 3 and 5", and it is the first test in the FizzBuzz style question this course sets.

Assignment operators

● MUST MEMORISE

SHORTHAND

Changing a variable using its own current value is so common it has shorthand:

SHORTHAND	MEANS
<code>x += 1</code>	<code>x = x + 1</code>
<code>x -= 1</code>	<code>x = x - 1</code>
<code>x *= 2</code>	<code>x = x * 2</code>
<code>x /= 2</code>	<code>x = x / 2</code>

`x += 1` is read "increment x". You will write it in every counting loop in Module Three, and forgetting it is what makes a loop run forever.

DRY RUN

TRACE IT BY HAND

Tracing a program by hand, one line at a time, writing down what each variable holds, is the skill this course tests as "what will the following code display". Practise it now on something small, so that it is automatic later.

```
a = 7
b = 2
a += b
c = a // b
d = a % b
print(a, b, c, d)
```

Take one line at a time and never look ahead:

LINE	A	B	C	D	OUTPUT
<code>a = 7</code>	7	-	-	-	
<code>b = 2</code>	7	2	-	-	
<code>a += b</code>	9	2	-	-	
<code>c = a // b</code>	9	2	4	-	
<code>d = a % b</code>	9	2	4	1	
<code>print(...)</code>	9	2	4	1	9 2 4 1

A dash means the variable does not exist yet. Using one before it exists is a `NameError`.

CONFIRMED AGAINST A REAL INTERPRETER

```
9 2 4 1
```

1. What is `10 / 2`, and what type is it?
2. How do you test whether `n` is divisible by 5?
3. What is `2 ** 3 ** 2`?
4. What is the difference between `=` and `==`?
5. What is `17 // 3` and `17 % 3`?

ANSWERS

1. `5.0`, a float. `/` always gives a float.
2. `n % 5 == 0`
3. `512`. `**` groups from the right, so it is `2 ** 9`, not `(2 ** 3) ** 2`.
4. `=` assigns a value to a name. `==` asks whether two values are equal and gives back `True` or `False`.
5. `5` and `2`. Seventeen is three fives remainder two.

MODULE TWO · UNIT 4

Working with Strings

Text, in depth. An entire question on your paper is nothing but this unit, so it earns the space.

By the end of this unit you can: index and slice a string, explain why a string cannot be changed, use every string method this course examines, and write the text analysis programs it sets.

A string is a sequence

● MUST MEMORISE

INDEXING AND SLICING

A string is a sequence of characters, and everything you learned about indexing a list in Module Three applies here unchanged. The first character is number 0.

```
>>> name = "Python"
>>> name[0]
'P'
>>> name[-1]
'n'
>>> name[0:3]
'Pyt'
>>> len(name)
6
```

Negative indexes count from the right. Slices exclude the stop. Same rules, same traps, and `name[6]` is an `IndexError` just as a list's would be.

A list can be changed in place. **A string cannot.** This property is called being **immutable**, and it is asked directly.

```
>>> name = "Python"
>>> name[0] = "J"
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
TypeError: 'str' object does not support item assignment
```

Compare that message with the tuple's in Module Three: they are the same sentence with a different type name, because they are the same idea.

The consequence, and this is the important part: every string method that looks like it changes a string actually **returns a new string** and leaves the original untouched.

```
>>> name = "Python"
>>> name.upper()
'PYTHON'
>>> name
'Python'
```

`name` is **unchanged**. To keep the result you must catch it:

```
>>> name = name.upper()
>>> name
'PYTHON'
```

Calling `name.upper()` on its own line and expecting `name` to change is one of the most common errors in this course. It is the mirror image of the `sort()` trap from Module Three: a list method changes the list and returns `None`, a string method changes nothing and returns the answer.

◆ KEY POINT

Strings are immutable: no string method changes the string it was called on. Each one returns a NEW string, so `s.upper()` on its own line accomplishes nothing at all unless the result is assigned or used.

BLUNT VERSION: a new string comes back and the old one is untouched. If you did not catch the result, nothing happened.

The methods

● MUST MEMORISE

EVERY ONE OF THESE IS EXAMINED

METHOD	DOES	EXAMPLE	GIVES
<code>upper()</code>	All capitals	<code>"ada".upper()</code>	<code>'ADA'</code>
<code>lower()</code>	All small letters	<code>"ADA".lower()</code>	<code>'ada'</code>
<code>strip()</code>	Removes whitespace from both ends	<code>" hi ".strip()</code>	<code>'hi'</code>
<code>replace(a, b)</code>	Every a becomes b	<code>"a-b".replace("-", " ")</code>	<code>'a b'</code>
<code>split(sep)</code>	Cuts into a list	<code>"a,b".split(",")</code>	<code>['a', 'b']</code>
<code>index(sub)</code>	Where sub starts	<code>"Ada".index("d")</code>	1
<code>find(sub)</code>	Same, but -1 if absent	<code>"Ada".find("z")</code>	-1
<code>count(sub)</code>	How many times	<code>"banana".count("a")</code>	3
<code>startswith(s)</code>	True or False	<code>"https://x".startswith("https")</code>	True
<code>isalpha()</code>	True only if every character is a letter	<code>"Ada".isalpha()</code>	True
<code>join(list)</code>	Glues a list into a string	<code>", ".join(["a", "b"])</code>	<code>'a, b'</code>

A question asks you to "list and briefly explain four common string methods, with an example of each". The four safest are `upper`, `lower`, `strip` and `replace`: each is one line to explain and impossible to get wrong.

● MUST MEMORISE

INDEX VS FIND

They do the same job and fail differently, which is the whole reason both exist.

```
>>> "Ada".find("z")
-1
>>> "Ada".index("z")
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
ValueError: substring not found
```

find returns -1 when it is not there. **index** raises a **ValueError**. Use **find** when absence is normal, **index** when absence means something has gone wrong.

`split()` behaves differently depending on whether you give it a separator, and both forms are examined.

```
>>> "70 55 90".split()
['70', '55', '90']
>>> "ada@mail.com".split("@")
['ada', 'mail.com']
```

- ▶ **No argument:** cuts at every run of whitespace, and ignores extra spaces. This is the one used for "values separated by spaces".
- ▶ **With an argument:** cuts at exactly that text, which is then thrown away.

It always returns a list of strings, never numbers. That is the split-and-convert pattern from Module Three: `split` then `int`.

Note what happens when the separator is not there at all. You get a list of one:

```
>>> "nothing here".split("@")
['nothing here']
```

No error. So `parts[1]` on that would be an `IndexError`, which is worth remembering when a question asks you to split an email at the `@`.

Meant to extract the first name and put it in capitals. Find the errors:

```
full_name = "Ada Lovelace"
first_name = full_name[0:full_name.index()]
print("First name:", first_name.uppercase())
```

Two errors, and both are invented method calls:

- ▶ `index()` is called with **nothing to look for**. It needs the text you want the position of. Here that is the space.
- ▶ `uppercase()` **does not exist in Python**. The method is `upper()`. Nothing in the language is called `uppercase`.

Only the first speaks, because the program stops there:

```
TypeError: index expected at least 1 argument, got 0
```

Fix that, run again, and only then do you meet the second:

```
AttributeError: 'str' object has no attribute 'uppercase'
```

Note what you do **not** get here: a suggestion. Python offers "Did you mean" only when your typo is close enough to a real name. Write `uppe()` and it says *Did you mean: 'upper'?*; write `uppercase()` and it just says no. **The help arrives when you least need it.** Do not rely on it.

The corrected program:

```
full_name = "Ada Lovelace"
first_name = full_name[0:full_name.index(" ")]
print("First name:", first_name.upper())
```

```
First name: ADA
```

`full_name.index(" ")` is **3**, the position of the space, so the slice takes characters 0 to 2. The space itself is excluded, because a slice stops before its stop.

A neater answer, worth mentioning to a marker: `full_name.split()[0]` does the same job in fewer characters and does not break when there is no space.

A "what is the output, explain each line" question. It looks gentle and has a trap in the middle.

```
text = "  Hello, Python!  "  
print(text.strip())  
print(text.lower().replace("python", "world"))  
print(len(text))
```

```
Hello, Python!  
  hello, world!  
18
```

Line by line, which is what "explain the result of each line" wants:

- ▶ `text.strip()` removes the two spaces at each end, giving `Hello, Python!`. **text itself is unchanged**, which the next line proves.
- ▶ `text.lower()` gives `" hello, python! "`. **Now** `replace("python", "world")` matches, because the text is lowercase by the time `replace` sees it. The result keeps its spaces, because `strip` on the line before returned a new string and changed nothing.
- ▶ `len(text)` is **18**, counting the original with all four spaces: 2 at the front, 14 for `Hello, Python!`, 2 at the back.

The trap is the middle line. Answer it without noticing the `lower()` and you will say `replace` finds nothing and the text is unchanged. It does match, and the answer is `hello, world!`. Chained methods run left to right, and each one works on what the one before it returned.

Process a list of student email addresses and print the usernames, where the username is the part before the @.

- 1 Put the addresses in a list.
- 2 Loop through it.
- 3 Split each at "@" and take item 0.
split gives a list; [0] is before the @, [1] is after
- 4 Print in the format the question shows.

EMAILS.PY

```
emails = [  
    "azu.ezenwoke@university.edu.ng",  
    "odunayo.osofuye@university.edu",  
    "emmanuel.franklin@university.edu",  
    "ada.lovelace@university.edu",  
    "bola.ahmed@university.edu",  
]  
  
for email in emails:  
    username = email.split("@")[0]  
    print(f>Email: {email} | Username: {username})
```

```
Email: azu.ezenwoke@university.edu.ng | Username: azu.ezenwoke  
Email: odunayo.osofuye@university.edu | Username: odunayo.osofuye  
Email: emmanuel.franklin@university.edu | Username: emmanuel.franklin  
Email: ada.lovelace@university.edu | Username: ada.lovelace  
Email: bola.ahmed@university.edu | Username: bola.ahmed
```

Now you try: the question asks for "at least five addresses". Why does `split("@")[0]` still work on an address with a dot in the domain, such as `university.edu.ng`? (Because it splits on @, not on dots. The domain's dots are all in item 1, which we never look at.)

The extension: read a paragraph, split it into sentences at `"."`, and for each one strip it, skip it if empty, count the words, put it in capitals, and count the vowels.

- 1 Split at `"."`. This is where the empty pieces come from: a paragraph ending in a full stop leaves an empty string after the last one.
"A. B.".split(".") gives ['A', ' B', '']
- 2 `strip()` each piece, then **skip the empties**. That is what the requirement is protecting you from.
- 3 `split()` with no argument to count words.
- 4 Count vowels by walking the characters and testing membership with `in`.

ANALYSE.PY

```
paragraph = input("Enter a paragraph: ")

for sentence in paragraph.split("."):
    sentence = sentence.strip()
    if sentence == "":
        continue

    words = len(sentence.split())

    vowels = 0
    for ch in sentence.lower():
        if ch in "aeiou":
            vowels += 1

    print(f"Sentence: {sentence.upper()}")
    print(f" Words: {words}, Vowels: {vowels}")
```

Entering Python is fun. I like code.

```
Enter a paragraph: Python is fun. I like code.
Sentence: PYTHON IS FUN
Words: 3, Vowels: 3
Sentence: I LIKE CODE
Words: 3, Vowels: 5
```

Three details that decide whether this works:

- ▶ `ch in "aeiou"` asks whether that character appears in the string `"aeiou"`. It is a neat way to test membership without five comparisons joined by `or`.
- ▶ **Count vowels on `sentence.lower()`**, not on the sentence. The `I` in "I like" is a vowel, and testing it against lowercase `"aeiou"` would miss it. Count first, then display in capitals.
- ▶ **"Python is fun" has three vowels, not four.** They are the `o`, the `i` and the `u`. The `y` in Python is *not* counted, because the question defines the vowels as (a, e, i, o, u) and `y` is not in that list. Follow the question's definition, not your instinct about English.

Now you try: why is the `continue` needed at all? (The paragraph ends with a full stop, so splitting leaves an empty final piece. Without the skip you would print a blank sentence with 0 words.)

Ask for the **consonants** instead and that loop quietly gives the wrong answer. A consonant is a letter that is not a vowel, so it is tempting to flip the test to `ch not in "aeiou"`. But the space is not in "aeiou" either, and neither is the full stop, so both would be counted as consonants.

You have to ask the other half of the question: **is this character a letter at all?** That is `isalpha()`, and it is true only when every character it is given is a letter.

```
>>> "Ada".isalpha()
True
>>> "Ada Lovelace".isalpha()
False
>>> " ".isalpha()
False
>>> "!".isalpha()
False
```

The space makes the whole of "Ada Lovelace" false. That is exactly the behaviour you want one character at a time, so a consonant is both things at once:

```
consonants = 0
for ch in sentence.lower():
    if ch.isalpha() and ch not in "aeiou":
        consonants += 1
```

On "Python is fun" that gives **8**: the p, y, t, h, n, s, f and n. Eleven letters, minus the three vowels. The y is counted, because the question defines the vowels as a, e, i, o, u and y is not one of them: the same rule that makes the vowel count 3 and not 4 makes the consonant count 8.

1. What does **immutable** mean, and what follows from it for `name.upper()` ?
2. Difference between **find** and **index** ?
3. What is `"a b".split()` against `"a-b".split("-")` ?
4. What is `len(" Hello, Python! ")` ?
5. Which method does not exist: **upper** or **uppercase** ?

ANSWERS

1. A string cannot be changed in place. So `name.upper()` returns a new string and leaves `name` alone. You must write `name = name.upper()` to keep it.
2. Both give the position of the text. `find` returns -1 when it is absent; `index` raises a `ValueError`.
3. Both give `['a', 'b']`. No argument splits at whitespace; an argument splits at exactly that text.
4. **18**. Two spaces, 14 characters, two spaces.
5. `uppercase` does not exist. The method is `upper`.

MODULE THREE

Control Flow and Data Structures

Choosing, repeating, and storing many things at once. Two of the six questions on your paper come from this module alone, and the habits you build here decide how the other four go.

```
if n % 2 == 0:  
    print( "even" )  
else:  
    print( "odd" )
```

Decision Making in Python

Making the program choose. Everything so far has run top to bottom with no choices at all.

By the end of this unit you can: write `if`, `if/else` and `if/elif/else`, nest them, explain why order matters, and write the divisibility program this course sets every year.

if

TEACH

An `if` statement runs a block only when a condition is true. You met the shape in Module Two; here is the full rule.

```
if condition:
    # this block runs only if the condition is True
    print("yes")
```

Three parts, and all three are compulsory:

- ▶ The word `if`.
- ▶ A **condition**: anything that produces `True` or `False`. Usually a comparison, as in Module Two, Unit 3.
- ▶ A **colon**, then an indented block beneath.

```
>>> age = 20
>>> if age >= 18:
...     print("You may vote.")
...
You may vote.
```

At the prompt, a block makes the prompt change to `...` to show it is waiting for more. Press **Enter** on an empty line to finish the block and run it.

if / else

TEACH

`else` gives the other path. Exactly one of the two blocks runs, always.

```
n = 7
if n % 2 == 0:
    print("even")
else:
    print("odd")
```

```
odd
```

`else` takes no condition of its own. It means "in every other case", so writing a condition after it is an error.

if / elif / else

● MUST MEMORISE

THE THREE WAY CHOICE

For more than two paths, use `elif`, short for "else if". You may have as many as you like. Python tests them **in order, from the top**, and stops at the first one that is true. Every later test is skipped, even if it would also be true.

```
score = 72
if score >= 70:
    print("A")
elif score >= 60:
    print("B")
elif score >= 50:
    print("C")
else:
    print("F")
```

A

72 is greater than 70, so `A` prints and nothing else is tested. Note that 72 is *also* greater than 60 and 50. Those tests never ran, and that is the whole point of `elif`.

⚠ TRAP

ORDER DECIDES THE ANSWER

Write the same tests in the wrong order and every mark goes:

```
score = 72
if score >= 50:
    print("C")
elif score >= 70:
    print("A")
```

C

72 passes the first test, so it prints `C` and stops. The `A` branch is unreachable: no score can ever print `A`, because anything that is 70 or more is also 50 or more. **With `elif`, put the narrowest test first.**

Using separate `if` statements instead of `elif` is a different bug with the same smell: each one is tested independently, so a score of 72 would print `A`, `B` and `C`.

An `if` may live inside another. Each level indents four more spaces.

```
n = 15
if n > 0:
    if n % 2 == 0:
        print("positive and even")
    else:
        print("positive and odd")
else:
    print("not positive")
```

positive and odd

Read the indentation to see which `else` belongs to which `if`: an `else` pairs with the `if` at its own indent level. The inner `else` is at 4 spaces, so it answers the inner `if`. The outer one is at 0, so it answers the outer `if`.

Ask the user for a number of exercise sessions. If it is not positive, say so. Otherwise: divisible by both 3 and 5, print "Active - FizzBuzz"; by 3 only, "Active - Fizz"; by 5 only, "Active - Buzz"; otherwise "Delulu - Not divisible by 3 or 5".

- 1 Read the number and convert it, because `input` gives a string.
`n = int(input(...))`
- 2 Check positive first, and put everything else inside that branch. The requirement says "check if the number is positive" before any of the rest.
- 3 **Test "both 3 and 5" first.** This is the trap above. 15 is divisible by 3, so a "divisible by 3" test placed first would catch 15 and never reach FizzBuzz.
narrowest test first
- 4 Then 3 alone, then 5 alone, then the catch all `else`.

FITNESS.PY

```
n = int(input("Number of sessions: "))

if n > 0:
    if n % 3 == 0 and n % 5 == 0:
        print("Active - FizzBuzz")
    elif n % 3 == 0:
        print("Active - Fizz")
    elif n % 5 == 0:
        print("Active - Buzz")
    else:
        print("Delulu - Not divisible by 3 or 5")
else:
    print("Please enter a positive number")
```

Run it four times, with 15, 9, 10 and 7:

```
Number of sessions: 15
Active - FizzBuzz
Number of sessions: 9
Active - Fizz
Number of sessions: 10
Active - Buzz
Number of sessions: 7
Delulu - Not divisible by 3 or 5
```

Now you try: what does it print for 0, and for 30? (0 is not positive, so "Please enter a positive number". 30 is divisible by both, so "Active - FizzBuzz".)

1. In an `if/elif/elif/else`, how many blocks run?
2. Why must "`divisible by 3 and 5`" be tested before "`divisible by 3`"?
3. How do you tell which `if` an `else` belongs to?

ANSWERS

1. Exactly one. Python stops at the first true test.
2. Because anything divisible by both is also divisible by 3. The wider test would catch it first and the narrower branch could never run.
3. By indentation. An `else` pairs with the `if` at the same indent level.

MODULE THREE • UNIT 2

Loops: Repetition and Branching

Doing something many times. This unit contains the two patterns that carry a quarter of the marks on your paper.

By the end of this unit you can: write `while` and `for` loops, use `range`, use `break` and `continue`, explain `for/else`, avoid infinite loops, and write both of the "accept any number of inputs" patterns.

◆ KEY POINT

In an `if / elif / else` chain, the first test that is true runs its block and the whole chain is then finished. Replacing an `elif` with a second `if` changes that: independent `ifs` are all tested, so two blocks can run.

BLUNT VERSION: `elif` is a queue, one gets served. Separate `ifs` are a crowd, they all get in.

while

TEACH

A while loop repeats its block **as long as** a condition stays true. It checks before each pass, so a condition false at the start means the block never runs at all.

```
i = 1
while i <= 3:
    print(i)
    i += 1
```

```
1
2
3
```

Three parts, and leaving out the third is the classic disaster:

- ▶ **Set up** before the loop: `i = 1`.
- ▶ **Test** in the header: `i <= 3`.
- ▶ **Change** inside the block: `i += 1`. Without this, `i` stays 1 forever and the condition never turns false.

● MUST MEMORISE

TWO KINDS OF LOOP

This distinction is asked directly, in those words:

	CONDITION CONTROLLED	COUNT CONTROLLED
Written with	while	for
Repeats	While something stays true	Once per item in a sequence
Use when	You do not know how many times	You know how many times
Example	Read until the user types 'done'	Read exactly five scores

for and range

TEACH

A for loop walks through a sequence, giving you one item at a time. The commonest sequence is made by `range`.

```
>>> for i in range(5):
...     print(i)
...
0
1
2
3
4
```

Read it as: "for each value `i` in this sequence, do the block".

● MUST MEMORISE

RANGE

range starts at 0 and stops *before* the number you give. This off by one is deliberate and constant, and it is worth marks every year.

WRITTEN	GIVES	HOW MANY
range(5)	0, 1, 2, 3, 4	5
range(1, 5)	1, 2, 3, 4	4
range(1, 10, 2)	1, 3, 5, 7, 9	5
range(5, 0, -1)	5, 4, 3, 2, 1	5

One argument is the stop. Two are start and stop. Three are start, stop and **step**. The stop is never included, in every case.

So to count 1 to 10 inclusive you must write `range(1, 11)`.

break and continue

● MUST MEMORISE

THE DIFFERENCE

break leaves the loop entirely. Nothing further repeats.

continue skips the rest of this pass only, and goes straight back to the top for the next one.

```
>>> for i in range(5):
...     if i == 3:
...         break
...     print(i)
...
0
1
2
```

```
>>> for i in range(5):
...     if i == 3:
...         continue
...     print(i)
...
0
1
2
4
```

Same loop, one word different. **break** stops at 3 and never prints 4. **continue** skips only 3 and carries on to 4.

This is the single most instructive broken snippet in the course. It is presented as "intended to print all even numbers from 1 to 10, skipping number 6", and you are asked to find the error.

```
i = 1
while i <= 10:
    if i % 2 == 0:
        if i == 6:
            continue
        print(i)
    i += 1
```

It does not merely print the wrong numbers. **It hangs forever.** It prints 2, then 4, and then stops responding.

Trace it. When `i` is 6, `continue` jumps back to the top of the `while` immediately, which means it **skips** `i += 1` on its way past. So `i` is still 6 at the next test, and 6 again, forever.

The lesson, and it is general: in a `while` loop, `continue` skips everything below it, including the line that moves the loop forward. In a `for` loop this cannot happen, because the `for` advances the counter itself.

The safe fix removes the need for `continue` at all:

```
i = 1
while i <= 10:
    if i % 2 == 0 and i != 6:
        print(i)
    i += 1
```

```
2
4
8
10
```

If a question insists you keep `continue`, then `i += 1` must come *before* it, and you must be careful not to skip the increment.

for / else

● MUST MEMORISE

UNUSUAL AND EXAMINED

A loop may have an `else`. This surprises nearly everyone, because it does not mean what it looks like.

The `else` block runs when the loop finishes normally. It is skipped if the loop was stopped by `break`.

Read `else` here as "if no break happened".

```
>>> for i in range(5):
...     if i == 3:
...         break
...     print(i)
... else:
...     print("Loop finished")
...
0
1
2
```

"Loop finished" never prints. The `break` at 3 stopped the loop, so the `else` was skipped. Remove the `break` and it would print after 0 1 2 3 4.

This exact snippet has been set as a "what is the output" question. The wrong answer, and the one most candidates give, is to include "Loop finished".

Accepting any number of inputs

Part (e) of every question on the most recent paper asks you to extend a fixed size program to handle any number of inputs. There are exactly two patterns, and between them they are worth a quarter of the marks you can attempt. Learn both cold.

Keep asking until the user types a particular word. That word is called a **sentinel**. Here it is `done`.

```
names = []
while True:
    name = input("Name (or 'done'): ")
    if name == "done":
        break
    names.append(name)

print("You entered:", names)
```

```
Name (or 'done'): Ada
Name (or 'done'): Bola
Name (or 'done'): done
You entered: ['Ada', 'Bola']
```

The shape is always the same:

- ▶ `while True:` means "loop forever". It is not a mistake here, because the `break` is the exit.
- ▶ **Test for the sentinel first**, before doing anything with the value. Otherwise 'done' gets stored as a name.
- ▶ `break` leaves the loop.

Note that the sentinel is checked with `==` against a string, so it is case sensitive: 'Done' would not stop this loop.

Take one line holding many values separated by spaces, and turn it into a list of numbers.

```
line = input("Enter scores separated by spaces: ")
parts = line.split()
scores = []
for p in parts:
    scores.append(int(p))

print(scores)
```

```
Enter scores separated by spaces: 70 55 90
[70, 55, 90]
```

`split()` with no argument cuts a string at every run of whitespace and gives back a **list of strings**. It does not convert anything, which is why the loop and `int()` are still needed.

```
>>> "70 55 90".split()
['70', '55', '90']
```

Those are strings, in quotes. Forget the conversion and you will be sorting "90" before "70" alphabetically, or joining them end to end instead of adding.

1. What does `range(1, 5)` give?
2. Difference between `break` and `continue`?
3. When does a loop's `else` block run?
4. Why can `continue` hang a `while` loop but not a `for` loop?
5. What does `"70 55".split()` give, and what type are the items?

ANSWERS

1. 1, 2, 3, 4. The stop value is never included.
2. `break` leaves the loop completely. `continue` skips the rest of this pass and goes to the next one.
3. When the loop finishes normally. It is skipped if `break` stopped the loop. Read it as "if no break happened".
4. Because `continue` skips the rest of the block, including the line that increments the counter. A `for` loop advances its own counter, so nothing can skip it.
5. `['70', '55']`. They are strings, and must be converted with `int()` before arithmetic.

MODULE THREE • UNIT 3

Lists and Tuples

Holding many values under one name.

By the end of this unit you can: create and index lists, slice them, use the list methods this course examines, explain negative indexing, and say what a tuple is and how it differs from a list.

Lists

TEACH

A **list** holds many values in order, under one name. Write it with square brackets, values separated by commas.

```
>>> scores = [70, 55, 90, 45]
>>> scores
[70, 55, 90, 45]
>>> len(scores)
4
```

A list can hold anything, including a mixture, and including other lists. An empty list is `[]`, which is where most programs start.

● MUST MEMORISE

INDEXING STARTS AT 0

Reach one item with square brackets. **The first item is number 0**, not 1.

```
>>> scores = [70, 55, 90, 45]
>>> scores[0]
70
>>> scores[3]
45
```

So a list of 4 items has indexes 0, 1, 2, 3. **The last index is always len minus 1.** Asking for `scores[4]` is the commonest runtime error in this course:

```
>>> scores[4]
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
IndexError: list index out of range
```

Negative indexes count from the right, starting at -1 for the last item. This is the neat way to get the end without knowing the length:

```
>>> scores[-1]
45
>>> scores[-2]
90
```

TEACH

LISTS CAN BE CHANGED

Unlike a string, a list can be modified in place. This property is called being **mutable**.

```
>>> scores = [70, 55, 90, 45]
>>> scores[1] = 65
>>> scores
[70, 65, 90, 45]
```

Slicing

● MUST MEMORISE

SLICES

A **slice** takes several items at once, written `[start:stop]`. As with `range`, **the stop is not included**.

```
>>> nums = [1, 2, 3, 4, 5]
>>> nums[1:3]
[2, 3]
>>> nums[:2]
[1, 2]
>>> nums[2:]
[3, 4, 5]
>>> nums[-3:]
[3, 4, 5]
```

Leave out the start to mean "from the beginning". Leave out the stop to mean "to the end". A slice always gives back a **new list**; the original is untouched.

A **method** is a function that belongs to a value, called by writing a dot after it. These are the ones this course uses:

METHOD	DOES	NOTE
<code>append(x)</code>	Adds x to the end	The one you use in every loop
<code>insert(i, x)</code>	Puts x at index i	Shifts the rest right
<code>remove(x)</code>	Removes the first x by value	ValueError if x is absent
<code>pop(i)</code>	Removes and returns the item at i	No i means the last
<code>index(x)</code>	Gives the index of the first x	ValueError if x is absent
<code>sort()</code>	Sorts the list in place	Returns None
<code>reverse()</code>	Reverses in place	Returns None

And three built in functions that take a list: `len()` , `max()` , `min()` , plus `sum()` for numbers.

⚠ TRAP

SORT() RETURNS NONE

`sort()` changes the list and gives back **nothing**. So assigning its result destroys your list:

```
>>> scores = [70, 55, 90]
>>> scores = scores.sort()
>>> print(scores)
None
```

Correct is to call it on its own line:

```
>>> scores = [70, 55, 90]
>>> scores.sort()
>>> scores
[55, 70, 90]
```

To sort the other way, pass `reverse=True` :

```
>>> scores.sort(reverse=True)
>>> scores
[90, 70, 55]
```

If you want a sorted copy and the original kept, use `sorted(scores)` , which *does* return a new list.

⚠ TRAP

REMOVE TAKES A VALUE, NOT AN INDEX

`remove` is given the **value** to delete. `pop` is given the **index**. Confusing them is a set error in this course:

```
>>> items = ["apples", "bananas", "carrots"]
>>> items.remove("banana")
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
ValueError: list.remove(x): x not in list
```

The value is `"bananas"`, not `"banana"`. The message says exactly that: `x` is not in the list.

TEACH

+ AND * ON LISTS

`+` joins two lists into a new one. `*` repeats a list.

```
>>> [1, 2] + [3, 4]
[1, 2, 3, 4]
>>> [1, 2] * 2
[1, 2, 1, 2]
```

Both appear in "what is the output" questions. Note `[1, 2] * 2` is `[1, 2, 1, 2]`, not `[2, 4]`: it repeats the list, it does not multiply the numbers.

`+=` extends a list in place. `list2 += list1` means `list2 = list2 + list1`, so it tacks every item of `list1` onto the end of `list2`:

```
>>> list2 = [3]
>>> list2 += [1, 2, 1, 2]
>>> list2
[3, 1, 2, 1, 2]
```

A list item can itself be a list, so you can build a grid, a table, or a tree. To reach inside, put one pair of brackets after another: the first picks the row, the second picks within it.

```
>>> table = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]
>>> table[0]
[1, 2, 3]
>>> table[0][2]
3
>>> table[2][0]
7
>>> len(table[0])
3
```

Read `table[0][2]` left to right: `table[0]` is the list `[1, 2, 3]`, and `[2]` then takes item 2 **of that list**, which is 3. Note `len(table)` is 3, the number of rows, while `len(table[0])` is 3, the length of the first row: `len` counts only the top level it is given.

The nesting can go deeper, and you follow it one bracket per level:

```
>>> deep = [10, [20, [30, 40]]]
>>> deep[1]
[20, [30, 40]]
>>> deep[1][1]
[30, 40]
>>> deep[1][1][0]
30
```

Each bracket peels off one layer. `deep[1]` is a list, so it can take another `[1]`, which is again a list, so it can take a `[0]`. Work strictly left to right and write down what each step gives, and a chain like `[5][1][1][0]` holds no fear.

◆ KEY POINT

A slice and a range both EXCLUDE their stop. `xs[1:4]` gives the items at 1, 2 and 3, and `range(1, 4)` gives 1, 2 and 3. A negative index counts back from the end, and `[::-1]` walks the whole thing backwards.

BLUNT VERSION: up to, never including. Every off-by-one on this paper is that sentence not yet learned.

Tuples

● MUST MEMORISE

LIST VS TUPLE

A **tuple** is a list that cannot be changed. Write it with round brackets.

```
>>> point = (3, 4)
>>> point[0]
3
>>> point[0] = 9
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
TypeError: 'tuple' object does not support item assignment
```

	LIST	TUPLE
Written with	[]	()
Can be changed?	Yes, it is mutable	No, it is immutable
Has append, remove, sort?	Yes	No
Indexing and slicing	Yes	Yes, both work the same
Use for	Data that changes	Data that must not change

Indexing and slicing work identically on both. Only *changing* is forbidden.

```
nums = [1, 2, 3, 4, 5]
total = 0
for n in nums:
    if n % 2 == 0:
        continue
    total += n
print(total)
```

The `continue` skips the even numbers, so only the odd ones are added. Take it one pass at a time:

PASS	N	N % 2 == 0?	ACTION	TOTAL
start	-	-	-	0
1	1	False	add	1
2	2	True	skip	1
3	3	False	add	4
4	4	True	skip	4
5	5	False	add	9
print	-	-	-	9

CONFIRMED AGAINST A REAL INTERPRETER

9

Note this `continue` is safe: it is in a `for` loop, which advances its own counter. The same code as a `while` loop would need care.

1. In `nums = [1,2,3,4,5]` , what is `nums[-2]` ?
2. What is `nums[1:3]` ?
3. What does `scores = scores.sort()` leave in `scores` ?
4. Give two differences between a list and a tuple.
5. What is `[1, 2] * 2` ?

ANSWERS

1. 4 . Negative indexes count from the right, starting at -1.
2. [2, 3] . The stop index is not included.
3. None . `sort()` sorts in place and returns nothing. Call it on its own line.
4. A list uses square brackets and is mutable; a tuple uses round brackets and is immutable. A list has `append` , `remove` and `sort` ; a tuple has none of them.
5. [1, 2, 1, 2] . It repeats the list.

Sets and Dictionaries

The other two built in data structures. One removes duplicates for free; the other looks things up by name.

By the end of this unit you can: create sets, use them to remove duplicates, apply the four set operations, create and read dictionaries, change them, and explain what happens when a dictionary holds a list.

Sets

● MUST MEMORISE

A SET HAS NO DUPLICATES

A **set** is an unordered collection with **no duplicates**. Write it with curly brackets, or build one from a list with `set()`.

```
>>> students = ["Alice", "Bob", "Alice", "Eve", "Bob"]
>>> len(set(students))
3
```

Five names in, three unique out, with no loop and no test. **This is the standard way to remove duplicates and it is set as an examination question.**

Two consequences of being a set, both examinable:

- ▶ **No order.** There is no first item, so `s[0]` is an error.
- ▶ **No duplicates, ever.** Adding something already present does nothing.

To see the contents in a form you can rely on, sort them. `sorted()` gives back a **list**, in order:

```
>>> sorted(set(students))
['Alice', 'Bob', 'Eve']
```

Note the square brackets in that answer: `sorted` handed back a list, not a set. That is what makes the order meaningful.

"Unordered" is stronger than it sounds. Printing a set of strings gives an order that **varies from one run to the next**, on the same machine, with the same code. Running this same line in three fresh sessions:

```
{'Alice', 'Eve', 'Bob'}
{'Bob', 'Alice', 'Eve'}
{'Alice', 'Bob', 'Eve'}
```

Python deliberately randomises how it stores text each time it starts, so the order a set happens to print in is an accident of that run. It is not alphabetical, it is not insertion order, and it is not stable.

What this means for you: never write a set's printed order as your answer, and never rely on it in a program. If a question asks what `print(set([...]))` displays, the honest answer names the members and says the order is not defined. If you need a definite order, use `sorted()`.

Sets of small whole numbers do tend to print in ascending order, which is why `{1, 2, 3}` looks so well behaved. That is a side effect of how integers are stored, not a promise, so do not lean on it either.

This is presented as code that "should remove duplicates and print the number of unique students". Find the issue:

```
students = ["Alice", "Bob", "Alice", "Eve", "Bob"]
students = list(students)
unique_students = students
print("Total unique students:", len(unique_students))
```

```
Total unique students: 5
```

There is no error message. It runs perfectly and prints the wrong number, which is what makes it hard. Two things are wrong, and neither raises anything:

- ▶ `list(students)` makes a copy of a list that is already a list. It does nothing useful and removes no duplicates.
- ▶ `unique_students = students` just gives the same list a second name. Nothing was deduplicated.

The fix is one word:

```
students = ["Alice", "Bob", "Alice", "Eve", "Bob"]
unique_students = set(students)
print("Total unique students:", len(unique_students))
```

```
Total unique students: 3
```

```
>>> a = {1, 2, 3, 4}
>>> b = {3, 4, 5, 6}
```

NAME	SYMBOL	METHOD	MEANS	RESULT
Union	$a \mid b$	<code>a.union(b)</code>	In either	<code>{1, 2, 3, 4, 5, 6}</code>
Intersection	$a \& b$	<code>a.intersection(b)</code>	In both	<code>{3, 4}</code>
Difference	$a - b$	<code>a.difference(b)</code>	In a but not b	<code>{1, 2}</code>
Symmetric difference	$a \wedge b$	<code>a.symmetric_difference(b)</code>	In one but not both	<code>{1, 2, 5, 6}</code>

Note `a - b` is not the same as `b - a`: difference has a direction. Here `b - a` is `{5, 6}`.

◆ KEY POINT

Assignment binds a name; it never copies. After `b = a` there is ONE list with two names, so a change made through either name is visible through the other. To get a genuine copy, use `list(a)`, `a[:]` or `a.copy()`.

BLUNT VERSION: one object, two labels. If you did not ask for a copy, you did not get one.

ANCHOR

A hotel guest register is a dictionary. You look a guest up by room number, not by walking the corridor. The room number is the key, the guest is the value, one room holds one guest at a time, and asking for a room the hotel does not have is an error rather than an empty answer. That last part is exactly what a `KeyError` is.

Dictionaries

● MUST MEMORISE

KEY AND VALUE

A **dictionary** stores pairs: a **key** and its **value**. You look things up by key rather than by position.

```
>>> person = {"name": "John", "age": 25}
>>> person["name"]
'John'
```

Curly brackets like a set, but with `key: value` pairs. The key goes in the square brackets, and it is **not** a number counting positions. There is no `person[0]` here.

Assigning to a key adds it if new, and replaces it if it exists:

```
>>> person["city"] = "Lagos"
>>> person["age"] = 26
>>> person
{'name': 'John', 'age': 26, 'city': 'Lagos'}
```

Asking for a key that does not exist is an error:

```
>>> person["height"]
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
KeyError: 'height'
```

● MUST MEMORISE

DEL REMOVES A KEY

del is a keyword, not a method, and it removes a key and its value from the dictionary:

```
>>> person = {"name": "John", "age": 26, "city": "Lagos"}
>>> del person["city"]
>>> person
{'name': 'John', 'age': 26}
```

You write `del` in front, then the same `name[key]` you would use to read it. **Deleting a key that is not there is the same `KeyError`** as reading one that is not there, so the case must match exactly: `del person["Email"]` and `del person["email"]` are not the same request. That trap is set directly on your paper.

`del` also removes a list item by index, and here it shifts the rest left, exactly as `pop` does but without handing the value back:

```
>>> nums = [10, 20, 30, 40]
>>> del nums[1]
>>> nums
[10, 30, 40]
```

Three methods give you the parts, and each is examinable:

```
>>> person = {"name": "John", "age": 25}
>>> list(person.keys())
['name', 'age']
>>> list(person.values())
['John', 25]
>>> for k, v in person.items():
...     print(k, "is", v)
...
name is John
age is 25
```

METHOD	GIVES YOU	EXAMPLE	RESULT
keys()	Every key	list(person.keys())	['name', 'age']
values()	Every value	list(person.values())	['John', 25]
items()	Each key with its value	for k, v in person.items()	a pair each time round

A plain `for` over a dictionary gives you the **keys**, not the values. That catches people out, so say it to yourself now.

● MUST MEMORISE

A DICTIONARY HOLDING A LIST

A value can be any type, including a list. This exact code has been set as a "what is the output, and explain each stage" question:

```
person = {
    "name": "John",
    "age": 25,
    "skills": ["Python", "Java"]
}
person["skills"].append("C++")
print(person["skills"])
```

```
['Python', 'Java', 'C++']
```

Take the middle line apart, because that is what "explain each stage" is asking for:

- ▶ `person["skills"]` looks up the key and gives back **the list itself**, not a copy.
- ▶ `.append("C++")` is then called on that list, so the list inside the dictionary is changed.

No reassignment happened, and none was needed. The list is mutable, and you reached it through the dictionary.

1. How do you remove duplicates from a list in one step?
 2. Why is `s[0]` an error on a set?
 3. What is `{1,2,3}` & `{3,4}` ?
 4. What does a plain `for` over a dictionary give you?
 5. Name the four built in data structures and one use for each.
-

ANSWERS

1. Pass it to `set()` . Wrap that in `len()` to count them.
2. A set is unordered, so there is no item at a position. Indexing has no meaning.
3. `{3}` . & is intersection, the items in both.
4. The keys. Use `.values()` or `.items()` for the rest.
5. List (ordered, changeable: a class register). Tuple (ordered, fixed: map coordinates). Set (unique, unordered: the distinct courses a student takes). Dictionary (lookup by key: a student's name against their score).

MODULE FOUR

Functions, Modules and File Handling

Packaging work under a name, borrowing work somebody else has done, and making data outlive the program that made it. Questions Three and Five come from here.

```
def square(x):  
    return x * x
```

```
print(square( 4 ))    16 ■
```

Functions in Python

Giving a piece of work a name, so you can ask for it by name.

By the end of this unit you can: define and call a function, explain parameters against arguments, use `return`, name the two types of user defined function, use default parameters, and say why functions matter.

What a function is

TEACH

A **function** is a block of code with a name, which runs only when you ask for it. You have been using functions since page one: `print`, `len`, `int` and `input` are all functions.

Now you write your own.

```
def greet():
    print("Hello!")

greet()
```

```
Hello!
```

Two separate things happened, and keeping them apart matters:

- ▶ **Defining** the function, with `def`. This only tells Python the name and the recipe. **Nothing runs.**
- ▶ **Calling** the function, by writing its name with brackets. **Now** the block runs.

Define it once, call it as many times as you like. Delete the last line above and the program prints nothing at all, because a definition on its own does nothing.

● MUST MEMORISE

WHY FUNCTIONS MATTER

Asked two ways in those words: "why are functions important in programming", and "list five benefits of **modularizing** a program with functions". Modularizing just means building the program out of functions, so these are the same list. Five reasons:

- ▶ **Reuse.** Write it once, call it anywhere. No copying.
- ▶ **Readability.** A name says what a block does, so the program reads as steps rather than detail.
- ▶ **Maintenance.** Fix a bug in one place and every caller is fixed.
- ▶ **Decomposition.** A large problem splits into small named pieces, each testable on its own.
- ▶ **Abstraction.** A caller uses a function by knowing what it does, not how it does it, so the detail stays hidden and out of the way.

A sixth, if you need one: **collaboration**, since a team can split the functions between them and work in parallel.

● MUST MEMORISE

BUILT IN VS USER DEFINED

	BUILT IN	USER DEFINED
Who wrote it	Comes with Python	You did, with def
Needs defining?	No, always available	Yes, before you call it
Examples	print, len, int, max, range	greet, square, is_healthy

Parameters and arguments

● MUST MEMORISE

THE TWO WORDS

A function can be handed values to work on.

```
def greet(name):          # name is a PARAMETER
    print("Hello,", name)

greet("Ada")              # "Ada" is an ARGUMENT
```

Hello, Ada

A parameter is the name in the definition. An argument is the value you pass in at the call. Same slot, two words, depending on which side you are standing on.

Several parameters are separated by commas, and the arguments line up with them by position:

```
>>> def add(a, b):
...     print(a + b)
...
>>> add(2, 3)
5
```

return

● MUST MEMORISE

THE TWO TYPES OF FUNCTION

This is asked in those words: **explain the two types of user defined function.**

1. A void function does something and hands nothing back. It has no `return`.

```
def greet(name):  
    print("Hello,", name)
```

2. A value returning function works something out and hands it back with `return`.

```
def square(x):  
    return x * x
```

The difference is what you can do with the call. A value returning function's call is a value, so it can go anywhere a value can:

```
>>> def square(x):  
...     return x * x  
...  
>>> square(4)  
16  
>>> y = square(4) + 1  
>>> y  
17
```

⚠ TRAP

PRINT IS NOT RETURN

These look alike and are not alike at all.

```
>>> def shout(x):  
...     print(x * 2)  
...  
>>> result = shout(5)  
10  
>>> print(result)  
None
```

`shout` displayed 10, then handed back **nothing**. So `result` is `None`.

A function with no `return` returns `None`. Always. `print` shows a value to a human; `return` gives a value back to the program. If a question says "returns Healthy", it means `return`, and `print` will not earn the mark.

Also: `return` ends the function at once. Anything written after it in the same block never runs.

Meant to define a function that squares a number and call it. Find the errors:

```
def square(x)
    return x * x
print(Square(4))
```

Two errors, and you only ever hear about the first:

- ▶ **No colon** after `def square(x)`. This is a `SyntaxError`, caught while reading the file, so nothing runs.
- ▶ **Square with a capital S** does not exist. The function is `square`. Names are case sensitive.

Fix the colon and run again, and only then do you meet the second:

```
Traceback (most recent call last):
  File "C:\Users\you\sq.py", line 3, in <module>
    print(Square(4))
          ^^^^^
NameError: name 'Square' is not defined. Did you mean: 'square'?
```

That is the syntax before runtime rule from Module Two doing exactly what it promised. The corrected program:

```
def square(x):
    return x * x

print(square(4))
```

16

◆ KEY POINT

`print()` DISPLAYS a value and returns `None`. `return` HANDS A VALUE BACK to whoever called the function and displays nothing. A function that only prints returns `None`, so assigning its result stores `None`.

BLUNT VERSION: show it or give it back. Those are two different jobs and mixing them is the commonest function bug on this paper.

Default parameters

● MUST MEMORISE

THIS EXACT CODE HAS BEEN SET

A parameter may be given a value to fall back on, used when the caller does not supply one.

```
>>> def greet(name="Guest"):
...     print("Hello,", name)
...
>>> greet()
Hello, Guest
>>> greet("Ada")
Hello, Ada
```

Called empty, `name` takes its default. Called with an argument, the argument wins. Note the comma and space in the output: that is `print` putting one space between its two items, not part of the string.

The rule: parameters with defaults must come after those without. Writing `def f(a=1, b)` is a `SyntaxError`, because Python could not tell what a single argument was meant to fill.

Define `is_healthy(bmi)` returning "Healthy" if the BMI is between 18.5 and 24.9, and "Unhealthy" otherwise. Read three patients and display each result as `Patient: John | BMI: 22.4 | Status: Healthy`.

- 1 **Return, do not print.** The question says "returns", so the function hands the word back and the caller displays it.
`return "Healthy"`
- 2 The range test is a chained comparison, from Module Two, Unit 3.
`18.5 <= bmi <= 24.9`
- 3 Loop three times with `range(3)`, reading a name and a BMI each pass. BMI is a decimal, so convert with `float`, not `int`.
- 4 Display with an f-string. The vertical bars are literal text.

HEALTH.PY

```
def is_healthy(bmi):
    if 18.5 <= bmi <= 24.9:
        return "Healthy"
    else:
        return "Unhealthy"

for i in range(3):
    name = input("Patient name: ")
    bmi = float(input("BMI: "))
    status = is_healthy(bmi)
    print(f"Patient: {name} | BMI: {bmi} | Status: {status}")
```

With John 22.4, Ada 17.9 and Bola 24.9:

```
Patient: John | BMI: 22.4 | Status: Healthy
Patient: Ada | BMI: 17.9 | Status: Unhealthy
Patient: Bola | BMI: 24.9 | Status: Healthy
```

Now you try: what does `is_healthy(24.95)` return, and why is that worth thinking about? ("Unhealthy", because 24.95 is above 24.9. The question's own boundary is 24.9, so anything between 24.9 and 25 is Unhealthy. Say the boundary you used and you cannot be marked wrong.)

The follow up asks for any number of patients, stopping at 'done'. That is the sentinel loop from Module Three, wrapped around the same function:

```
def is_healthy(bmi):
    if 18.5 <= bmi <= 24.9:
        return "Healthy"
    else:
        return "Unhealthy"

healthy = 0
total = 0
while True:
    name = input("Patient name (or 'done'): ")
    if name == "done":
        break
    bmi = float(input("BMI: "))
    status = is_healthy(bmi)
    total += 1
    if status == "Healthy":
        healthy += 1
    print(f"Patient: {name} | BMI: {bmi} | Status: {status}")

print(f"Total: {total}, Healthy: {healthy}, Unhealthy: {total - healthy}")
```

The function did not change at all. Only the loop around it did. **That is the point of a function, and it is the point of part (e).**

RECALL

COVER THE ANSWERS

1. Difference between a parameter and an argument?
2. Name the two types of user defined function.
3. What does a function without return give back?
4. Why does `print(Square(4))` fail when `square` is defined?

ANSWERS

1. A parameter is the name in the definition. An argument is the value passed at the call.
2. Void (returns nothing) and value returning (hands a value back with `return`).
3. None .
4. Names are case sensitive. `Square` and `square` are different names, and only one exists, so it is a `NameError`.

MODULE FOUR · UNIT 2

Modules and Packages

Using code somebody else already wrote.

By the end of this unit you can: import a module, use the two import forms, and name the modules this course uses.

- ▶ A **module** is a single file of Python code you can use from your own.
- ▶ A **package** is a folder of modules grouped together.
- ▶ The **standard library** is the large set of modules that ships with Python. Nothing to install; it is already there.

You use a module by **importing** it, and there are two forms:

```
>>> import math
>>> math.sqrt(16)
4.0
```

```
>>> from math import sqrt
>>> sqrt(16)
4.0
```

`import math` brings in the whole module, and you reach into it with a dot. `from math import sqrt` brings in that one name, which you then use bare. Both give `4.0`, a float, because `sqrt` always returns a float.

TEACH

THE ONES THIS COURSE USES

MODULE	GIVES YOU	EXAMPLE
math	Mathematics	<code>math.sqrt(16)</code> , <code>math.pi</code>
random	Random numbers	<code>random.randint(1, 6)</code>
sqlite3	Databases	Module Five
tkinter	Windows and buttons	Module Five

`random.randint(a, b)` is worth a note: **both ends are included**, unlike `range`. So `random.randint(1, 6)` can give 6.

MODULE FOUR · UNIT 3

File Handling in Python

Making data survive the program. Everything you have stored so far vanished the moment the program ended.

By the end of this unit you can: name the four access modes, open, write, read and close a file, explain reading against appending, use `with`, and write the record keeping programs this course sets.

◆ KEY POINT

Assigning to a name inside a function creates a LOCAL name, which disappears when the function ends and leaves any global of the same name untouched. Changing a global from inside requires the global keyword.

BLUNT VERSION: what happens in the function stays in the function, unless you say global.

The three steps

● MUST MEMORISE

ASKED IN THESE WORDS

Using a file is always three steps, and the question "state the three steps of file use" wants exactly these:

1. **Open** the file, which connects your program to it and says what you intend to do.
2. **Process** it: read from it, or write to it.
3. **Close** it, which finishes the job and makes sure everything you wrote is actually saved.

● MUST MEMORISE

THE FOUR ACCESS MODES

`open` takes the file name and the **mode**, a short string saying what you intend.

MODE	NAME	IF THE FILE EXISTS	IF IT DOES NOT
"r"	Read	Reads it. Cannot write.	FileNotFoundError
"w"	Write	Erases it completely , then writes	Creates it
"a"	Append	Adds to the end, keeps what is there	Creates it
"x"	Exclusive create	FileExistsError	Creates it

"r" is the default, so `open("f.txt")` means read.

● MUST MEMORISE

READING AGAINST APPENDING

Asked directly. Give the pair:

- ▶ **Reading** ("r") takes data *out* of the file and changes nothing. The file must already exist.
- ▶ **Appending** ("a") puts data *into* the file, at the end, keeping everything already there. It creates the file if needed.

And the distinction that actually loses data: "w" **erases the whole file the moment you open it**, before you write a single character. If you meant to add to a file and opened it with "w" , everything in it is already gone. "a" is the one that adds.

Writing

TEACH

```
f = open("note.txt", "w")
f.write("This is a note.")
f.close()
```

Two things surprise people about `write` :

- ▶ **It adds no newline.** `print` does; `write` does not. Two writes in a row land on the same line unless you put `\n` in yourself.
- ▶ **It only writes strings.** Passing a number is a `TypeError`, so convert with `str()` first.

```
f = open("names.txt", "w")
f.write("Ada\n")
f.write("Bola\n")
f.close()
```

gives a file holding two lines. Without the `\n` it would hold `AdaBola` on one line.

⚠ TRAP

THIS EXACT CODE HAS BEEN SET

Meant to write a message to `note.txt` . Find the errors:

```
file = open("note.txt", "w")
file.write("This is a note."
file.close
```

Two errors, and the first hides the second:

- ▶ The `write` line is **missing its closing bracket**. That is a `SyntaxError`, so the file never runs.
- ▶ `file.close` has **no brackets**, so it is not a call. It names the method and throws it away, doing nothing. The file is never closed.

The second is the nastier one, because it is silent: no error, no message, and possibly no data in the file. Fixed:

```
file = open("note.txt", "w")
file.write("This is a note.")
file.close()
```

Brackets mean "do it". No brackets means "the thing itself". The same rule catches `conn.cursor` against `conn.cursor()` in Module Five.

with

● MUST MEMORISE

THE SAFE WAY

Forgetting `close()` is easy, and a crash between opening and closing skips it entirely. `with` closes the file for you, whatever happens.

```
with open("example.txt", "a") as f:
    f.write("New line added.\n")

print("Done writing.")
```

```
Done writing.
```

Read it as: open the file, call it `f`, and when the indented block ends close it automatically. There is no `close()` to write and none to forget.

What this code does, if asked: it opens `example.txt` in append mode, creating it if absent; adds one line to the end without disturbing what is already there; closes it; then prints `Done writing.` to the screen. Note that **only** `"Done writing."` appears on screen. The written line went into the file, not to the display.

◆ KEY POINT

The file mode decides what happens to what is already there. `"r"` reads, `"w"` creates or TRUNCATES, so an existing file is emptied the moment it is opened, and `"a"` appends. `with open(...)` guarantees the file is closed on the way out, even if the block raises.

BLUNT VERSION: `w` wipes. If you meant to add to the file, the mode is `"a"`.

Reading

● MUST MEMORISE

THREE WAYS TO READ

Given a `names.txt` holding Ada and Bola on two lines:

METHOD	GIVES	RESULT HERE
<code>f.read()</code>	The whole file as one string	<code>'Ada\nBola\n'</code>
<code>f.readline()</code>	The next line only, as a string	<code>'Ada\n'</code>
<code>f.readlines()</code>	A list of all lines	<code>['Ada\n', 'Bola\n']</code>

The usual way is to loop over the file directly, which gives one line at a time and does not load a large file into memory at once:

```
with open("names.txt") as f:
    for line in f:
        print(line.strip())
```

```
Ada
Bola
```

Every line you read keeps its `\n` on the end. That is why `strip()` is there: without it you would print a blank line after each name, because the line's own newline and `print`'s newline both land.

Store student records (name and score) until 'done' is entered, in `students.txt`, one per line as `Name, Score`. Then read it back, print each record, and count passes and fails. **The pass mark is 45 and above.**

- 1 Sentinel loop to collect, writing as you go. Mode `"w"` because we are creating the file fresh.
Test for 'done' before asking for the score
- 2 `write` needs a string and adds no newline, so build the line with an f-string and put `\n` on the end yourself.
- 3 Close, then reopen in `"r"`. The question asks for this explicitly.
- 4 Split each line at the comma, convert the score, count.
45 and above passes

STUDENTS.PY

```
with open("students.txt", "w") as f:
    while True:
        name = input("Name (or 'done'): ")
        if name == "done":
            break
        score = int(input("Score: "))
        f.write(f"{name}, {score}\n")

passes = 0
fails = 0
with open("students.txt", "r") as f:
    for line in f:
        name, score = line.strip().split(", ")
        score = int(score)
        print(f"{name} scored {score}")
        if score >= 45:
            passes += 1
        else:
            fails += 1

print(f"Passes: {passes}, Fails: {fails}")
```

Entering John 76, Ada 45, Bola 30, then done:

```
John scored 76
Ada scored 45
Bola scored 30
Passes: 2, Fails: 1
```

Ada passes. The mark is 45 *and above*, so the test is `>=`, not `>`. Getting that boundary wrong is the difference between 2 passes and 1.

`name, score = ...split(", ")` unpacks a two item list into two names at once. It works because `split` returned exactly two pieces.

4. Display:

- The sorted list of scores,
- The highest and lowest scores,
- The average score,
- The number of passes and fails.

⚠ TRAP

THE PASS MARK IS NOT PRINTED

Read that last line again. The paper asks for "The number of passes and fails." and nowhere in its printed text is a pass mark defined. As printed, the requirement cannot be answered.

On the real script the threshold appears only in **handwriting**, added beneath the requirement: *Pass mark is 45 and above*. The same note is repeated on page 5 beside Question Five (e). It is not quoted in the box above because it is not printed text, and this book does not present handwriting as though the examiner had set it.

If you meet a question like this with no threshold given, **state the one you are using** in a comment and in your output. A marker cannot fault an assumption you declared. They can fault a number that appeared from nowhere.

RECALL

COVER THE ANSWERS

1. Name the four access modes and what each does to an existing file.
2. What does "w" do to a file that already has data?
3. What is wrong with `file.close`?
4. Why does reading a line usually need `strip()`?
5. What does `with` save you from?

ANSWERS

1. "r" read, fails if absent. "w" write, erases it. "a" append, adds to the end. "x" create, fails if it exists.
2. Erases it completely, the moment it is opened, before anything is written.
3. No brackets, so it is not a call. Nothing happens and the file is never closed. No error is reported.
4. Each line keeps the `\n` that ended it in the file.
5. Forgetting `close()`, and skipping it when something goes wrong part way through.

MODULE FOUR • UNIT 4

Exception Handling

Catching a runtime error instead of letting it kill the program.

By the end of this unit you can: explain what an exception is, use `try` and `except`, catch the right error, and use `finally`.

TEACH

An **exception** is a runtime error, in the Module Two sense: the line was grammatically fine, but doing it turned out to be impossible. Unhandled, it prints a traceback and the program stops dead.

You met this one already:

```
>>> int("xyz")
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
ValueError: invalid literal for int() with base 10: 'xyz'
```

In a program that asks the user for a number, that is not a bug in your code. It is a user typing badly. The program should say so and carry on, not crash.

● MUST MEMORISE

TRY AND EXCEPT

```
try:
    age = int(input("Your age: "))
    print("Next year you are", age + 1)
except ValueError:
    print("That is not a whole number.")
```

Run it and type xyz :

```
Your age: xyz
That is not a whole number.
```

No traceback, no crash. How it works:

- ▶ Python runs the `try` block.
- ▶ If nothing goes wrong, the `except` block is skipped entirely.
- ▶ If a `ValueError` happens, Python **abandons the rest of the `try` block at once** and runs the `except` block instead.

Note the second line inside the `try` never ran. That is why the block should hold only the risky work and what depends on it.

● MUST MEMORISE

THE EXCEPTIONS TO KNOW BY NAME

EXCEPTION	HAPPENS WHEN	SEEN IN
ValueError	<code>int("xyz")</code>	Module Two, Unit 2
ZeroDivisionError	<code>10 / 0</code>	Module Two, Unit 1
IndexError	<code>[1,2][5]</code>	Module Three, Unit 3
KeyError	<code>d["absent"]</code>	Module Three, Unit 4
TypeError	<code>"a" + 1</code>	Module Two, Unit 2
NameError	Using a name that does not exist	Module Four, Unit 1
FileNotFoundError	<code>open("gone.txt")</code>	Module Four, Unit 3

Every one of these has appeared in this manual before this page. That is the point of the table: you have met them all as accidents, and now you can catch them on purpose.

⚠ TRAP

DO NOT CATCH EVERYTHING

A bare `except:` catches every error there is, including the ones caused by your own bugs, and hides them:

```
try:
    age = int(input("Age: "))
except:
    print("Something went wrong.")
```

Misspell a name inside that `try` and you get "Something went wrong" instead of a `NameError` telling you exactly which line and which name. You have thrown away the most useful message on the screen.

Name the exception you expect. `except ValueError:` catches bad input and leaves your own mistakes visible.

The exam's programs all begin by reading a number. Combining this unit with the loops from Module Three gives input that cannot be broken:

```
while True:
    try:
        n = int(input("Enter a whole number: "))
        break
    except ValueError:
        print("Not a whole number. Try again.")

print("You entered", n)
```

Typing `xyz`, then `3.5`, then `7`:

```
Enter a whole number: xyz
Not a whole number. Try again.
Enter a whole number: 3.5
Not a whole number. Try again.
Enter a whole number: 7
You entered 7
```

The `break` is **inside the `try`, after the conversion**. So it is only reached when the conversion worked. Put it after the `try` block and the loop would exit even on bad input.

Note `3.5` is rejected: `int()` only accepts a whole number written out, as Module Two, Unit 2 showed.

● MUST MEMORISE

ELSE AND FINALLY

Two optional extras complete the shape:

- ▶ `else` runs **only if no exception happened**. Same idea as the loop `else` from Module Three: "if nothing interrupted".
- ▶ `finally` runs **always**, exception or not. It is for cleaning up.

```
try:
    f = open("data.txt")
except FileNotFoundError:
    print("No such file.")
else:
    print("Opened it.")
    f.close()
finally:
    print("Finished trying.")
```

With no `data.txt` present:

```
No such file.
Finished trying.
```

`finally` is what `with` does for you automatically, which is why `with` is preferred for files.

1. What is an exception?
 2. What happens to the rest of the `try` block when an exception fires on its first line?
 3. Why is a bare `except:` a bad idea?
 4. When does `finally` run?
-

ANSWERS

1. A runtime error: the line was grammatically valid but could not be carried out. Unhandled, it stops the program with a traceback.
2. It is abandoned. Python jumps straight to the matching `except`.
3. It catches your own bugs too and replaces their messages with yours, hiding exactly the information you need.
4. Always, whether an exception happened or not.

MODULE FIVE

Databases and GUI Development

Storing data so it can be searched rather than merely saved, and putting a window on the front of a program. Question Six comes from here.

```
import sqlite3
conn = sqlite3.connect("school.db" )
cur.execute("SELECT * FROM students" )
conn.commit()
```

Python and Databases

A file holds data. A database organises it, so you can ask questions of it.

By the end of this unit you can: say what a database is and why it is used, explain what `sqlite3` does, name the SQL commands the course wants, filter rows with `WHERE`, change them with `UPDATE` and `DELETE`, summarise a column with `COUNT`, `MAX` and `AVG`, and write the programs it sets.

ANCHOR

A text file of records is a shoebox of receipts: everything is in there, and finding one means going through them all. A database is a filing cabinet with an index, where you ask for what you want and the cabinet finds it. That difference, asking rather than searching, is what SQL is for, and it is why a database is worth the extra setup.

What a database is

● MUST MEMORISE

ASKED IN THESE WORDS

A **database** is an organised collection of data, stored so that it can be searched, changed and retrieved efficiently. Data lives in **tables**: a table has named **columns**, and each **row** is one record.

Why it is used in programming, which is the second half of the question:

- ▶ **It lasts.** Like a file, the data outlives the program.
- ▶ **It can be searched.** Ask for one record out of a million without reading the other 999,999. A text file makes you read the lot.
- ▶ **It has structure.** Columns have types, so a score cannot quietly become text.
- ▶ **It handles many users at once** without them overwriting each other.
- ▶ **It keeps data consistent**, through rules such as a primary key that forbid nonsense.

Compare with Module Four's `students.txt`: to find one student there you read every line and split every one. In a database you ask for that student.

● MUST MEMORISE

WHAT SQLITE3 IS

SQL (say "sequel", or spell it out) is the language you ask a database questions in. It is not Python. You write it as a *string* inside your Python and hand it over to be run.

SQLite is a small database that keeps everything in a single ordinary file on your disk. There is no server to install and no password.

sqlite3 is the Python module that lets your program talk to an SQLite database. It **ships with Python**, so there is nothing to install: it is part of the standard library from Module Four, Unit 2.

That is the answer to "what role does the `sqlite3` module play in Python": it is the built in interface between your Python code and an SQLite database file, letting you create tables, insert rows, and query them from Python.

The three SQL commands

● MUST MEMORISE

NAME THESE THREE

COMMAND	FUNCTION	EXAMPLE
CREATE TABLE	Makes a new table and names its columns	CREATE TABLE students (id INTEGER PRIMARY KEY, name TEXT)
INSERT INTO	Adds a row of data	INSERT INTO students (name) VALUES ('Ada')
SELECT	Fetches rows back out	SELECT * FROM students

Two more are worth a sentence each, in case the question asks for more than three: `UPDATE` changes rows that already exist, and `DELETE` removes rows.

SQL keywords are conventionally written in capitals. It is not required, but it is what a marker expects. The `*` in `SELECT *` means "every column".

TEACH

COLUMN TYPES AND THE PRIMARY KEY

In `CREATE TABLE students (id INTEGER PRIMARY KEY, name TEXT)`, each column is a name then a type:

- ▶ `INTEGER` whole numbers, `TEXT` for text, `REAL` for decimals.
- ▶ **PRIMARY KEY** marks the column that identifies a row uniquely. Declared as `INTEGER PRIMARY KEY`, SQLite fills it in for you, counting up, so you never supply an `id` yourself.

That is why the example inserts only the name: `INSERT INTO students (name) VALUES ('Ada')` names the one column it is supplying, and `id` appears by itself.

The five steps

● MUST MEMORISE

EVERY DATABASE PROGRAM

Every program in this unit is the same five steps:

```
import sqlite3

conn = sqlite3.connect("school.db")    # 1 connect (creates the file if absent)
cursor = conn.cursor()                 # 2 get a cursor
cursor.execute("...")                  # 3 run SQL
conn.commit()                          # 4 save the changes
conn.close()                           # 5 disconnect
```

What each one is, since the question may ask:

- ▶ **connect** opens the database file, and **creates it if it does not exist**. It never fails for a missing file, unlike `open` in read mode.
- ▶ **A cursor** is the object you actually run SQL through and read results back from. Think of it as the hand that reaches into the database.
- ▶ **execute** runs one SQL statement, given as a string.
- ▶ **commit** makes your changes permanent. **Without it, your inserts are thrown away when the program ends**. No error, no warning, no data.
- ▶ **close** ends the connection.

Meant to create a table named students. Find the errors:

```
import sqlite3
conn = sqlite3.connect("school.db")
cursor = conn.cursor
cursor.execute("CREATE TABLE students (id INTEGER PRIMARY KEY, name TEXT)")
conn.commit
conn.close()
```

Two errors, and both are missing brackets. Exactly the `file.close` bug from Module Four, wearing a different hat.

- ▶ `conn.cursor` without `()` does not call the method. It hands you the method itself, so `cursor` is not a cursor at all. The next line then fails.
- ▶ `conn.commit` without `()` does nothing at all, silently. The table is never saved.

Only the first one speaks up:

```
Traceback (most recent call last):
  File "C:\Users\you\db.py", line 4, in <module>
    cursor.execute("CREATE TABLE students (id INTEGER PRIMARY KEY, name TEXT)")
AttributeError: 'builtin_function_or_method' object has no attribute 'execute'
```

Read that message closely, because it names the bug precisely: the thing you called `cursor` is a *method*, not a cursor, and methods have no `execute`. Corrected:

```
import sqlite3
conn = sqlite3.connect("school.db")
cursor = conn.cursor()
cursor.execute("CREATE TABLE students (id INTEGER PRIMARY KEY, name TEXT)")
conn.commit()
conn.close()
```

Watch out: run that corrected version **twice** and the second run fails, because the table already exists. See the next box.

● MUST MEMORISE

IF NOT EXISTS

Creating a table that is already there is an error:

```
sqlite3.OperationalError: table students already exists
```

Since `connect` keeps the file between runs, this bites the second time you run any program that creates a table. The fix is to say so in the SQL:

```
fields = "(id INTEGER PRIMARY KEY, name TEXT)"
cursor.execute("CREATE TABLE IF NOT EXISTS students " + fields)
```

Now it creates the table the first time and quietly does nothing afterwards. The examination asks for a table "if it doesn't already exist" in as many words, so **IF NOT EXISTS is the phrase that earns the mark.**

A question shows this and asks three things: explain it, say what happens when it runs, and state what must be true for it to succeed.

```
import sqlite3
conn = sqlite3.connect("school.db")
cursor = conn.cursor()
cursor.execute("INSERT INTO students (name) VALUES ('Ada')")
conn.commit()
conn.close()
```

What it does: connects to `school.db`, creating the file if absent; gets a cursor; adds one row to the `students` table with the name `Ada`, letting `id` fill itself; saves the change; closes.

What happens when it runs: nothing appears on screen. It is not a `SELECT`, so there is nothing to display. If it works you see no output at all, which is the correct answer and an easy mark to miss.

What must be true for it to succeed:

- ▶ The `students` table **must already exist**, with a `name` column. This code never creates it, so if the table is absent you get `sqlite3.OperationalError: no such table: students`.
- ▶ `id` must fill itself, i.e. be an `INTEGER PRIMARY KEY`, since no `id` is supplied.
- ▶ The file must be writable.

The single quotes around `'Ada'` are **SQL's** quotes, inside Python's double quotes. Two languages, one line.

◆ KEY POINT

Changes made through a cursor live in a transaction and are not permanent until `commit()` is called. Closing the connection without committing rolls them back, and nothing is reported: the program appears to have worked.

BLUNT VERSION: no commit, no data, and no error message to tell you.

Reading rows back

● MUST MEMORISE

SELECT AND FETCHALL

`execute` runs the `SELECT`, but it does not hand you the rows. You fetch them from the cursor:

```
cursor.execute("SELECT * FROM students")
rows = cursor.fetchall()
for row in rows:
    print(row)
```

```
(1, 'Ada')
(2, 'Bola')
```

Each row comes back as a tuple, in the column order of the table. So `row[0]` is the id and `row[1]` is the name, and the round brackets in that output are Python showing you tuples, from Module Three, Unit 3.

`fetchall()` gives every row as a list of tuples. `fetchone()` gives just the next row, or `None` when there are none left.

A `SELECT` needs no `commit`, because it changes nothing.

● MUST MEMORISE

THE ? PLACEHOLDER

Everything so far put fixed text in the SQL. Real data comes from variables, and there is a right way and a wrong way.

Right: a question mark for each value, and a tuple of the values.

```
cursor.execute("INSERT INTO contacts (name, phone) VALUES (?, ?)", (name, phone))
```

Wrong: building the SQL by joining strings.

```
sql = "INSERT INTO contacts (name, phone) VALUES ('" + name + "', '" + phone + "')"
cursor.execute(sql)
```

The joined version breaks the moment a name contains an apostrophe, which ends the SQL string early and leaves nonsense behind. A name like O'Brien is enough to do it. It is also how databases get attacked, by typing SQL into a name box.

The placeholder is a question mark, not a format specifier. Do not use an f-string here. And note the comma inside `(name,)` if you have only one value: without it, it is not a tuple.

When you have several rows ready in a list, `execute` in a loop works, but `executemany` inserts them all in one call. You give it the SQL with placeholders and a **list of tuples**, one tuple per row:

```
rows = [(101, "Ada", 78),
        (102, "Boła", 85)]
cursor.executemany("INSERT INTO student VALUES (?, ?, ?)", rows)
conn.commit()
```

It runs the same insert once for each tuple in the list, so two tuples add two rows. Use `executemany` when the data is already gathered into a list, and the `for` loop with `execute` when you are reading rows one at a time from the user. Both still need one `commit` at the end.

◆ KEY POINT

UPDATE and DELETE apply to every row that matches the WHERE clause, and a statement with no WHERE clause matches every row in the table. The clause is optional to SQL and not optional to you.

BLUNT VERSION: WHERE or everywhere. Write the WHERE before you write the SET.

Filtering, changing and summarising

● MUST MEMORISE

WHERE PICKS THE ROWS

`SELECT * FROM student` gives every row. To ask for **some** rows, add a `WHERE`, and pass the value as a placeholder exactly as an insert does. Say the table holds five students in classes SS1 to SS3:

```
cursor.execute("SELECT * FROM student WHERE Class = ?", ("SS2",))
for row in cursor.fetchall():
    print(row)
```

```
(102, 'Mary Johnson', 'SS2', 85.0)
(105, 'Esther Ade', 'SS2', 74.0)
```

Only the SS2 rows come back. `WHERE` is the same idea as an `if`: it keeps the rows the condition is true for. Note the comma in `("SS2",)`, because one value still needs a tuple.

Why the scores read 85.0 and not 85: the column is `REAL`, so SQLite hands every score back as a float. That is the column type doing its job, from the primary-key box above.

● MUST MEMORISE

UPDATE CHANGES, DELETE REMOVES

Two commands act on rows that already exist. Both take a `WHERE` that says **which** rows, and both need a `commit`, because both change the data.

UPDATE sets new values on the rows the `WHERE` picks:

```
cursor.execute("UPDATE student SET Score = ? WHERE StudentName = ?",
              (75, "Peter Okoro"))
conn.commit()
```

Peter's score is now 75.0. **The `WHERE` is not optional in practice:** leave it off and `UPDATE student SET Score = 75` sets *every* student to 75.

DELETE removes the rows the `WHERE` picks:

```
cursor.execute("DELETE FROM student WHERE StudentID = ?", (105,))
conn.commit()
```

Row 105 is gone. **DELETE FROM student with no `WHERE` empties the whole table**, silently and at once. Write the `WHERE` first, then run the command, every time.

● MUST MEMORISE

COUNT, MAX, AVG

A database can summarise a column for you, so you never loop in Python to add the scores up. These go where the column names go, inside the `SELECT`:

FUNCTION	GIVES
<code>COUNT(*)</code>	how many rows
<code>MAX(Score)</code>	the largest value in that column
<code>MIN(Score)</code>	the smallest
<code>AVG(Score)</code>	the average
<code>SUM(Score)</code>	the total

Ask for several at once, and read them back from the one row with `fetchone`:

```
cursor.execute("SELECT COUNT(*), MAX(Score), AVG(Score) FROM student")
total, highest, average = cursor.fetchone()
print(total, highest, average)
```

```
5 92.0 79.2
```

One row, three columns. A summary always comes back as a single row, so `fetchone` is the right fetch, and it unpacks straight into three names like any tuple. That is Question Four's "total number of students, highest score and average score" in one line of SQL.

A contact manager. Create `contacts.db` with a table `contacts` (`id`, `name`, `phone`), read three contacts from the user, insert each, then commit and close.

- 1 Connect and get a cursor.
connect creates `contacts.db` if it is absent
- 2 Create the table **if it does not already exist**. The question says those words, so use `IF NOT EXISTS`.
- 3 Loop three times with `range(3)`, reading name and phone.
phone stays `TEXT`: numbers can start with 0
- 4 Insert with placeholders, then commit once at the end and close.

CONTACTS.PY

```
import sqlite3

conn = sqlite3.connect("contacts.db")
cursor = conn.cursor()

cursor.execute("""CREATE TABLE IF NOT EXISTS contacts
                  (id INTEGER PRIMARY KEY, name TEXT, phone TEXT)""")

for i in range(3):
    name = input("Name: ")
    phone = input("Phone: ")
    cursor.execute("INSERT INTO contacts (name, phone) VALUES (?, ?)", (name, phone))

conn.commit()
conn.close()
print("Contacts saved.")
```

Entering Ada 08011112222, Bola 08033334444, John 08055556666:

Contacts saved.

The triple quoted string `"""..."""` lets a long piece of SQL run across two lines without fuss. It is an ordinary string, just one that may contain newlines.

Why phone is TEXT and not INTEGER: a phone number starting with 0 would lose that 0 as a number, and nobody ever adds two phone numbers together. Storing it as text is the correct choice, and saying so earns the mark.

The extension: accept contacts until the user types 'done' as the name, then fetch and display everything in the table. Sentinel loop from Module Three, plus SELECT .

CONTACTS_ANY.PY

```
import sqlite3

conn = sqlite3.connect("contacts.db")
cursor = conn.cursor()
cursor.execute("""CREATE TABLE IF NOT EXISTS contacts
                  (id INTEGER PRIMARY KEY, name TEXT, phone TEXT)""")

while True:
    name = input("Name (or 'done'): ")
    if name == "done":
        break
    phone = input("Phone: ")
    cursor.execute("INSERT INTO contacts (name, phone) VALUES (?, ?)", (name, phone))

conn.commit()

cursor.execute("SELECT * FROM contacts")
for row in cursor.fetchall():
    print(f"{row[0]} | {row[1]} | {row[2]}")

conn.close()
```

Entering Ada 08011112222, then Bola 08033334444, then done:

```
1 | Ada | 08011112222
2 | Bola | 08033334444
```

Commit before you select. It happens to work without it here, but the habit is what keeps the marks, and the question asks for the changes to be saved.

Now you try: run it a second time and add John. What ids do you see? (1, 2 and 3. The file persists, so the old rows are still there and the new one continues the count. If you expected to see only John, you have just learned what "it lasts" means.)

1. What is a database, and give two reasons to use one?
2. What role does `sqlite3` play?
3. Name three SQL commands and their functions.
4. What happens if you forget `conn.commit()` ?
5. What does a row from `fetchall()` come back as?
6. How do you fetch only the SS2 rows, change one student's score, remove one student, and count all rows?

ANSWERS

1. An organised collection of data held in tables of rows and columns, stored so it can be searched and changed efficiently. Any two of: it lasts beyond the program, it can be searched without reading everything, it enforces structure, it serves many users, it keeps data consistent.
2. It is the built in Python module that connects your program to an SQLite database file, letting you create tables, insert rows and query them from Python.
3. `CREATE TABLE` makes a table. `INSERT INTO` adds a row. `SELECT` fetches rows back.
4. Your changes are silently discarded when the program ends. No error is reported.
5. A tuple, with the columns in table order.
6. `SELECT * FROM student WHERE Class = ?` for the SS2 rows; `UPDATE student SET Score = ? WHERE ...` to change one; `DELETE FROM student WHERE ...` to remove one; `SELECT COUNT(*) FROM student` to count. `UPDATE` and `DELETE` each need a `commit`; the two `SELECT` s do not.

MODULE FIVE • UNIT 2

Python GUI Programming

Putting a window, a box and a button on the front of a program.

By the end of this unit you can: say what a GUI is, build a window with a label, an entry and a button, and wire the button to a function.

TEACH

A **GUI** is a Graphical User Interface: windows, buttons and boxes, rather than typing at a prompt. The opposite is the **command line interface**, which is everything you have written so far.

tkinter is Python's built in GUI module. Like `sqlite3`, it ships with Python.

A GUI program is inside out compared with everything so far. A normal program runs top to bottom and ends. A GUI program builds its window, then **waits**, and things happen only when the user does something. That waiting is a loop, and you start it yourself.

● MUST MEMORISE

THE FOUR WIDGETS

WIDGET	IS	MADE WITH
Window	The window itself	<code>tk.Tk()</code>
Label	Text the user reads	<code>tk.Label(window, text="...")</code>
Entry	A box the user types in	<code>tk.Entry(window)</code>
Button	Something to click	<code>tk.Button(window, text="...", command=fn)</code>

Every widget must be **placed** before it appears. `.pack()` is the simplest way: it stacks widgets down the window in the order you pack them. Create it and forget to pack it, and it simply is not there.

● MUST MEMORISE

COMMAND TAKES THE NAME, NOT THE CALL

A button is told which function to run when clicked:

```
tk.Button(window, text="Go", command=convert)
```

command=convert, not **command=convert()**. With brackets you would *call* it now, at the moment the button is built, and hand its return value to `command`. The button would then do nothing when clicked, and the work would have happened once, before the window even appeared.

Same rule as `file.close` against `file.close()` in Module Four, and `conn.cursor` against `conn.cursor()` in this module. **Brackets mean "do it now". No brackets means "the thing itself"**. Here you want the thing itself, because the button will do it later.

A window that converts Celsius to Fahrenheit. The formula is $F = C \times 9/5 + 32$.

- 1 Import `tkinter`, conventionally shortened to `tk`, and make the window.
- 2 Define what the button will do. It reads the entry with `.get()`, which hands back a **string**, exactly like `input`. Convert it.
`float(entry.get())`
- 3 Show the answer by changing the label's text with `.config(text=...)`.
- 4 Pack every widget, then start the wait with `mainloop()`.

CONVERTER.PY

```
import tkinter as tk

window = tk.Tk()
window.title("Celsius to Fahrenheit")

label = tk.Label(window, text="Celsius:")
entry = tk.Entry(window)
result = tk.Label(window, text="")

def convert():
    celsius = float(entry.get())
    fahrenheit = celsius * 9 / 5 + 32
    result.config(text=f"{fahrenheit:.1f} F")

button = tk.Button(window, text="Convert", command=convert)

label.pack()
entry.pack()
button.pack()
result.pack()

window.mainloop()
```

Type 100 and click Convert, and the bottom label reads:

212.0 F

`mainloop()` is the line people forget. Without it the program builds the window, reaches the end, and exits. The window flashes up and vanishes, or never appears at all. It must be the last line.

Now you try: what does 37 give? (98.6 F. Worth knowing: it is body temperature, so it is an easy check that your formula is the right way round.)

`entry.get()` hands back a string, always, exactly like `input()` in Module Two. Forget to convert and

```
celsius = entry.get()
fahrenheit = celsius * 9 / 5 + 32
```

fails with

```
TypeError: unsupported operand type(s) for /: 'str' and 'int'
```

Read where that error actually lands, because it is instructive. `celsius * 9` **does not fail**: `"100" * 9` is a perfectly legal string repeated nine times, giving `"100100100100100100100100100100"`. It is the `/ 5` on the next step that has nowhere to go, because you cannot divide text.

So the error names the division, several characters after the mistake. **The line Python blames is not always the line you got wrong**, and here the real fault is the missing `float()` one line earlier.

It is the 53 trap from Module Two wearing a window.

1. What is a GUI, and what is the alternative called?
2. Name four tkinter widgets.
3. Why `command=convert` and not `command=convert()`?
4. What does `mainloop()` do, and what happens without it?
5. What type does `entry.get()` return?

ANSWERS

1. A Graphical User Interface: windows, buttons and boxes. The alternative is a command line interface.
2. `Window (Tk)`, `Label`, `Entry`, `Button`.
3. Without brackets you hand over the function itself, for the button to call when clicked. With brackets you call it immediately and hand over its result, so clicking does nothing.
4. It starts the loop that waits for the user. Without it the program ends at once and the window never appears.
5. A string, like `input()`. Convert it before doing arithmetic.

REFERENCE

The Card, the List and the Cram Sheet

Everything the modules taught, compressed into what you actually carry into the hall: every function and method this course uses with what it hands back, the definitions the examiner wants word for word, a plan for the three hours, and the mistakes that cost marks. Read this part last, and read R.4 again the night before.

<code>len(x)</code>	how many items
<code>xs.sort()</code>	in place, returns None

<code>s[1:4]</code>	stops before 4
<code>7 / 2</code>	3.5, always a float

The Python Reference Card

Every keyword, built-in function and method this manual uses, in one place, with the thing people actually get wrong about each one: what it hands back.

By the end of this section you can say, for any name in this course, what it does, what it returns, and whether it changes the thing it was called on.

● MUST MEMORISE

THE QUESTION TO ASK OF EVERY CALL

Three questions, and they answer most of the paper's tracing questions before you trace anything.

1. **What does it RETURN?** A call that returns None cannot be assigned to anything useful.
2. **Does it CHANGE the thing it was called on?** A list method usually does. A string method never does.
3. **What TYPE comes back?** input() gives a string. A single slash gives a float. A slice of a list gives a list.

The 27 keywords this course uses

KEYWORD	WHAT IT DOES	TAUGHT IN
if, elif, else	Choose a branch. Only the FIRST true test in a chain runs.	M3 U1
for ... in	Repeat once per item of a sequence.	M3 U2
while	Repeat while a condition stays true. Needs something in the body to make it stop.	M3 U2
break	Leave the loop immediately.	M3 U2
continue	Abandon this pass, start the next.	M3 U2
def	Define a function.	M4 U1
return	Hand a value back and leave the function. With no value, and with no return at all, the function gives back None.	M4 U1
import, from ... import, as	Bring in a module. import math needs the math. prefix; from math import sqrt does not.	M4 U2
with	Open something and guarantee it is closed on the way out, even if the block raises.	M4 U3
try, except, else, finally	Catch an error. else runs when nothing was raised; finally runs either way.	M4 U4
raise	Signal an error yourself.	M4 U4
and, or, not	Combine conditions. Precedence: not, then and, then or.	M2 U3
in	Membership. Substring on a string, item on a list or set, KEY on a dictionary.	M3 U3
is	The very same object, not merely equal. Use it only with None.	M2 U2
del	Remove: by index on a list, by key on a dictionary.	M3 U3
class	Define a class. This course meets it mainly through the GUI widgets.	M5 U2

The 16 built-in functions

CALL	WHAT IT DOES	WHAT IT RETURNS
print(...)	Display on screen. sep and end control what goes between and after.	None. It shows, it does not hand back.
input(prompt)	Read one line typed by the user.	Always a str , however numeric it looks.
int(x)	Convert to a whole number. On a float it TRUNCATES towards zero, it does not round.	int, or raises ValueError on "abc".
float(x)	Convert to a decimal number.	float
str(x)	Convert to text so it can be joined with +.	str
len(x)	How many items, or how many characters.	int. On a nested list it counts the OUTER items.
range(a, b, step)	The whole numbers from a up to but NOT including b.	a range object; wrap in list() to see it.
list(x)	Build a list from any sequence. Also the way to COPY a list.	a new list
set(x)	Build a set, discarding duplicates.	a new set, unordered
sorted(x)	Sort ANYTHING and give back a new list. The original is untouched.	a new list, always
sum(x)	Add up the numbers in a sequence.	int or float
max(x), min(x)	Largest and smallest.	one item of the sequence
enumerate(x)	Walk a sequence giving index and item together.	pairs, used as for i, v in enumerate(xs)
type(x)	What kind of thing this is.	a type object; type(x).__name__ is the plain name
open(name, mode)	Open a file. Modes: r read, w write and TRUNCATE, a append.	a file object

⚠ TRAP

THE TWO THAT LOOK ALIKE

sorted(xs) returns a new sorted list and leaves xs alone. **xs.sort()** sorts xs itself and returns **None**. Writing `ys = xs.sort()` puts None in ys, silently, and the bug shows up several lines later.

The methods, by the type they belong to

STRING METHOD	WHAT IT DOES	CHANGES THE STRING?
.upper(), .lower(), .title()	Change case.	No. Strings are immutable: a NEW string comes back.
.strip()	Remove whitespace from both ends.	No, new string.
.split(sep)	Cut into pieces. With no argument, splits on any run of whitespace.	No. Returns a LIST.
sep.join(list)	Glue a list of strings together with sep between them. The separator is what you call it on.	No. Returns a string.
.replace(old, new)	Swap every occurrence.	No, new string.
.find(sub)	Position of sub.	No. Returns the index, or -1 if it is absent.
.startswith(x), .isalpha()	Ask a yes or no question about the text.	No. Returns True or False.
.count(x)	How many times x occurs.	No. Returns an int.

LIST METHOD	WHAT IT DOES	WHAT IT RETURNS
.append(x)	Add one item to the end.	None. It changes the list.
.remove(x)	Delete the first item EQUAL to x. Raises ValueError if there is none.	None
.sort()	Order the list in place. reverse=True for descending.	None
.index(x)	Where x is.	an int, or raises ValueError
.count(x)	How many times x appears.	an int

DICTIONARY METHOD	WHAT IT DOES	WHAT IT RETURNS
.get(key)	Look up a key WITHOUT raising if it is missing.	the value, or None. This is the safe alternative to d[key].
.keys(), .values()	All the keys, all the values.	a view you can loop over or wrap in list()
.items()	Key and value together.	pairs, used as for k, v in d.items()

FILE AND DATABASE	WHAT IT DOES	WHAT IT RETURNS
f.read()	The whole file.	one str, newlines included
f.readline()	The next line.	one str, ending in a newline; empty string at the end of the file
f.readlines()	Every line.	a LIST of str, each still carrying its newline
f.write(s)	Write text. It does not add a newline for you.	the number of characters written
f.close()	Flush and close. with does this for you.	None
sqlite3.connect(name)	Open or create the database file.	a connection
conn.cursor()	Get the object that runs SQL.	a cursor
cur.execute(sql, params)	Run one SQL statement. Use ? placeholders and pass the values.	the cursor
cur.executemany(sql, seq)	Run the SAME statement once per parameter tuple.	the cursor
cur.fetchone()	The next row.	a tuple, or None when the rows run out
cur.fetchall()	All remaining rows.	a list of tuples
conn.commit()	Make the changes permanent.	None. Skip it and the rows are lost with no error.
conn.close()	Close the connection.	None
TKINTER		WHAT IT DOES
tkinter.Tk()	Create the main window.	
Label, Entry, Button	Show text, take one line of typed text, run a command when clicked.	
.pack(), .grid(), .place()	The three geometry managers. Do not mix pack and grid in one container.	
.config(...)	Change a widget's settings after it has been made.	
entry.get()	Read what the user typed.	
root.mainloop()	Start the event loop. Without it the window appears and vanishes.	

RECALL AND CHECK

Name every call in this course that returns `None`, and say why that matters.

ANSWER

`print()`, `.append()`, `.remove()`, `.sort()`, `.write()` to a degree, `.close()`, `.commit()`, and any function with no return statement. It matters because assigning one of them to a variable stores `None`, which then fails much later and a long way from the mistake.

REFERENCE · R.2

The Must-Memorise List

The definitions and properties this paper asks for by name, in the words to give back. If you can write this page from memory you have the recall half of the paper already.

● MUST MEMORISE

THE PROPERTIES, BY TYPE

TYPE	ORDERED?	MUTABLE?	DUPLICATES?	WRITTEN
list	Yes	Yes	Yes	[1, 2, 3]
tuple	Yes	No	Yes	(1, 2, 3)
set	No	Yes	No	{1, 2, 3}
dict	Insertion order	Yes	Keys no, values yes	{"a": 1}
str	Yes	No	Yes	"abc"

That one table answers more marks on this paper than any other single thing in this book. Copy it out by hand until you can draw it blank.

- ▶ **Python is** an interpreted, high-level, general-purpose programming language, known for readable syntax and dynamic typing.
- ▶ **Dynamically typed** means the type belongs to the value, not to the name, so a variable may be rebound to a value of a different type and no declaration is needed.
- ▶ **Lists are ordered** means every item has a position, that position is fixed until you change it, and items can be reached and sliced by that position.
- ▶ **Strings are immutable** means a string object cannot be altered after it is created. Every string operation returns a NEW string, and `s[0] = "X"` raises `TypeError`. To change a string you rebind the name.
- ▶ **Sets are unique** means a set holds at most one copy of any value, so adding a value already present changes nothing and `len()` does not grow.
- ▶ **A KeyError** is raised when a dictionary is asked for a key it does not hold. `d.get(key)` returns `None` instead of raising.
- ▶ **A function** is a named block of code that runs when it is called, takes parameters and may return a value. It is the unit of reuse.
- ▶ **Modularising a program** gives reuse, independent testing, readability, easier maintenance, and division of work across a team. It does not make the program run faster.
- ▶ **An exception** is an error detected while the program runs. `try` and `except` let the program handle it instead of stopping.
- ▶ **A module** is a file of Python code that can be imported; a package is a directory of modules.

EXCEPTION	RAISED WHEN	STANDARD EXAMPLE
SyntaxError	Python cannot parse the line. Found BEFORE anything runs.	a missing colon
NameError	A name is used before it is assigned.	<code>print(x)</code> with no <code>x</code>
TypeError	The wrong KIND of thing.	<code>"5" + 5</code>
ValueError	The right kind, an impossible value.	<code>int("abc")</code>
IndexError	A sequence index out of range.	<code>[1,2,3][5]</code>
KeyError	A dictionary key that is not there.	<code>d["missing"]</code>
ZeroDivisionError	Dividing by zero.	<code>1 / 0</code>
IndentationError	The indentation does not line up.	a stray space before a line
FileNotFoundError	Opening a file for reading that is not there.	<code>open("no.txt")</code>

```
CREATE TABLE students (id INTEGER PRIMARY KEY, name TEXT, score REAL)
INSERT INTO students VALUES (?, ?, ?)
SELECT * FROM students WHERE score > 50 ORDER BY score DESC
UPDATE students SET score = 60 WHERE name = 'Ada'
DELETE FROM students WHERE score < 30
SELECT COUNT(*), MAX(score), MIN(score), AVG(score) FROM students
```

Six lines. WHERE is optional on UPDATE and DELETE, and leaving it out applies the statement to **every row in the table**.

REFERENCE · R.3

Exam Day Strategy

Three hours, one paper carrying an objective section and long questions, and no computer. How to spend the time, and how to write code that earns marks on paper.

TEACH

THE PLAN FOR THE THREE HOURS

- 1 First five minutes: read the whole paper.** Mark every question you can already do. That is not wasted time, it is the difference between answering the questions you know and running out of time before you reach them.
- 2 Do the objective section first, at pace.** Roughly 45 seconds a question. Any one you cannot settle in that time gets a mark in the margin and your best guess now, because there is no negative marking and a blank scores nothing.
- 3 Then the long question you marked as easiest.** Confidence is worth real marks, and an early complete answer stops you spending the whole paper on your worst topic.
- 4 Watch the marks, not the question numbers.** A part worth 2 marks wants two sentences. A part worth 8.5 wants a whole program. Writing an essay for 2 marks is how people run out of time.
- 5 Leave ten minutes.** Go back to the marked objective questions and to any part you left blank. Something written is always worth more than nothing.

The examiner cannot run your program, so what earns the marks is that it is readable and obviously correct. Five habits do most of that work.

- ▶ **Indent visibly.** Four clear spaces, or a ruled line down the block. An examiner who cannot see which lines are inside the loop cannot give you the loop mark.
- ▶ **Write the colons.** Every if, for, while, def and except ends its header with a colon, and it is the most commonly dropped character under time pressure.
- ▶ **Name things properly.** total, count, student_name. A program full of a, b and c reads as a guess even when it is right.
- ▶ **Convert your input.** Any number read with input() gets int() or float() around it. This single line appears in almost every program this paper asks for.
- ▶ **Trace your own answer once** with a small value before you move on. Three lines of a trace table in the margin catch an off-by-one that would otherwise cost the whole part.

SHAPE	WHAT EARNS THE MARKS
Explain a property (lists are ordered, strings are immutable)	Define it in one sentence, then give a two-line example that SHOWS it, then say what it means in practice. The example is usually worth as much as the sentence.
What is the output	Trace it line by line in the margin with a small table of variables. Write the output exactly as Python would print it, including quotes and brackets on a list.
Identify and correct the errors	Say WHERE (the line number), WHAT (the fault), and WHY it is wrong, then write the corrected line. Three parts, and most people give only the correction.
Write a program	Input, then process, then output, in that order, with a comment above each stage. Handle the obvious bad input if the marks are generous enough to imply it.

◆ KEY POINT

Marks are awarded for demonstrated understanding, not for a program that would compile. A partly written answer with correct structure and clear naming scores; a blank scores nothing at all.

BLUNT VERSION: never leave a part empty. Write the input line and the loop header even if you cannot finish.

The Trap Index and Cram Sheet

Every trap in this book on one page, then the whole course compressed into what fits on a single sheet. This is the section to read the night before and again in the last ten minutes.

THE TRAP	WHAT ACTUALLY HAPPENS
Adding 5 to what input() gave you	TypeError. input() returns a string; wrap it in int().
Expecting 7 / 2 to be 3	It is 3.5. A single slash always gives a float. Floor division is the double slash.
Expecting numbers[1:3] to include index 3	A slice stops BEFORE its stop index. range() does the same.
ys = xs.sort()	ys is None. sort() changes xs and returns nothing. Use sorted(xs) for a new list.
b = a on a list, then changing b	a changes too. Assignment does not copy. Use list(a) or a[:].
Looking a dictionary up by its value	KeyError. The subscript takes a KEY. Use .get() to avoid raising.
s[0] = "X" on a string	TypeError. Strings are immutable; build a new one.
Expecting "ab" * 3 to be an error	It is "ababab". Star repeats a string; it is "ab" + 3 that raises.
A second if where elif was meant	Both blocks run. An elif chain runs at most one.
Opening an existing file with "w"	It is emptied immediately. Append is "a".
Forgetting conn.commit()	The rows vanish and nothing is reported.
UPDATE or DELETE with no WHERE	It applies to every row in the table.
command=run() on a button	It runs once, immediately, and attaches the result. Pass command=run, with no brackets.
Forgetting mainloop()	The window appears and disappears at once.
Expecting int(3.9) to be 4	It is 3. int() truncates; round() rounds.
A recursion with no base case	RecursionError. Every recursion needs a stopping condition and a step towards it.

TOPIC	THE ONE THING	THE HOOK
Types	list ordered mutable, tuple ordered immutable, set unique unordered, dict key to value, str immutable	Square changeable, round fixed, curly unique
input()	Always a string	The keyboard has no number key that Python believes
Division	/ float, // floor, % remainder	One slash one point
Slices	Stop is excluded, negative counts from the end, [::-1] reverses	Up to, never including
In-place methods	sort, append, remove, reverse return None	If it changes the thing, it hands back nothing
Assignment	Binds a name, does not copy	One object, two labels
if chain	Only the first true branch runs	elif is a queue, if is a crowd
Loops	break leaves, continue skips, else runs if no break	Break out, continue on
Functions	print shows, return hands back; no return means None	Show or give
Scope	Assigning inside a function makes it local	What happens in the function stays in the function
Files	r read, w truncate, a append; with closes for you	w wipes
Exceptions	try, except by name, finally always	Finally means finally
Database	connect, cursor, execute, commit, close	No commit, no data
SQL	SELECT cols FROM table WHERE condition	WHERE or everywhere
tkinter	Tk, widgets, geometry manager, mainloop	No loop, no window

RECALL AND CHECK

Without looking: what does each of these return, and does it change what it was called on? `sorted(xs)`, `xs.sort()`, `s.upper()`, `xs.append(3)`, `d.get("k")`, `cur.fetchone()`.

ANSWER

`sorted(xs)` returns a new list, `xs` unchanged. `xs.sort()` returns `None`, `xs` sorted. `s.upper()` returns a new string, `s` unchanged, because strings are immutable. `xs.append(3)` returns `None`, `xs` one item longer. `d.get("k")` returns the value or `None`, `d` unchanged. `cur.fetchone()` returns the next row as a tuple, or `None` when they run out.

PAST PAPER 2025/2026

The Paper They Actually Set

The newest real examination, worked from end to end.

Every output here came from running the code. Read it after the modules and before the mocks: it is the clearest picture of what the question writer now wants.

Q1 and, strings, dict, BMI, interest

Q2 loops, control structures, grading

Q3 functions, lists, collections

Q4 immutability, a function, SQLite

Q5/Q6 files, nesting, void, math.pi



How the Paper Changed

The shape you drilled in the mocks loosened. The topics did not.

By the end of this section you can: say exactly how the 2025/2026 paper departs from the older template, and see where each of its parts is taught.

● MUST MEMORISE

FOUR, NOT SIX, STILL HOLDS

Course	CSC241, Python I
Session	2025/2026, Omega semester
Time	3 hours
Instruction	Answer any Four (4) questions
Marks	Six questions offered, four attempted, 70 in total.

The rubric is unchanged: six offered, choose four, and two whole topics can be ones you never touched.

⚠ TRAP

THREE HABITS THE OLD SHAPE GAVE YOU

The 2024/2025 paper was rigid: every question was five parts, each worth exactly 3.5 marks, and each question sat on one module. This paper breaks all three, and a student running on the old habits misreads it.

- ▶ **Marks vary per part.** Seen this paper: 1, 2, 2.5, 3, 3.5, 4.5, 5, 6, 6.5, 7, 7.5 and a single 10.5. A part is no longer worth a fixed 3.5, so **read the mark and spend your time to match it**. The 10.5-mark database program is worth three of the small parts.
- ▶ **The number of parts varies.** Question One has (a) to (e); Questions Four and Six have only (a) to (c). Do not hunt for a part (e) that is not there.
- ▶ **One question now mixes topics.** Question One alone touches the `and` operator, strings, a dictionary and two maths programs. Question Four mixes immutability, a function and a full database. You cannot skip a question just because its first part is a topic you dislike.

What did not change: the four underlying tasks are still define or describe, find and fix the errors, state the output, and write a program. Everything below is one of those four.

TEACH		WHERE EACH PART IS TAUGHT
QUESTION	PARTS	TAUGHT IN
One	and, strings, dict, BMI, interest	M2 U3/U4, M3 U4, M1
Two	count loop, control structures, grading	M3 U1/U2
Three	functions, list ops, collections	M4 U1, M3 U3/U4
Four	immutability, a function, SQLite	M2 U4, M4 U1, M5 U1
Five	file handling, modes, a records file	M4 U3
Six	nested lists, void, value-returning	M3 U3, M4 U1

Every part has a home in this book. Where the paper needs something the earlier edition skipped, the module now teaches it: the SQL UPDATE , DELETE and COUNT / MAX / AVG for Question Four, and lists inside lists for Question Six.

PAST PAPER 2025/2026

Questions One to Three, Worked

Every output was produced by running the code, not by reading it.

Question One

AS PRINTED	2025/2026 PAPER, QUESTION ONE (A)
Briefly describe how the and operator works in Python (use examples) [2.5 marks]	

BREAK IT DOWN	BEFORE YOU WRITE ANYTHING
WHAT I AM GIVEN Nothing but the operator name. This is a bookwork part: no data, no program, just describe and show. WHAT IS ACTUALLY ASKED Two things joined by "(use examples)". A description with no example cannot score full marks, and an example with no description cannot either. Answer both halves explicitly. WHAT THE MARKS SAY 2.5 marks is the smallest part on the paper. A truth table plus one worked line is the right size. Do not write an essay here and run out of time on the 10.5-mark database question. WHERE IT IS TAUGHT Module Two, Unit 3 (Operators).	

ANSWER

Q1 (A) THE AND OPERATOR, 2.5 MARKS

and is a boolean operator. It joins two conditions and is true **only when both are true**. If either side is false, the whole thing is false.

LEFT	RIGHT	LEFT AND RIGHT
True	True	True
True	False	False
False	True	False
False	False	False

The example the marker wants is a range check:

```
>>> score = 72
>>> score >= 0 and score <= 100
True
```

Both sides hold, so the score is valid. Change `score` to 150 and the right side fails, so the whole test is False. **and** also short-circuits: if the left is false, Python never even looks at the right, because the answer is already settled.

AS PRINTED

2025/2026 PAPER, QUESTION ONE (B)

What is a string in Python? List and briefly explain four common string methods used in Python. Provide an example for each method. [5 marks]

BREAK IT DOWN

BEFORE YOU WRITE ANYTHING

WHAT IS ACTUALLY ASKED

Three instructions in one part: **define**, **list** and **explain four**, and **give an example for each**. That is nine separate things to write.

WHAT THE MARKS SAY

5 marks over roughly nine items means each is worth about half a mark. Short and correct beats long and vague. A table is the fastest way to earn all of them.

THE TRAP

It says **four**. Writing six wastes time and earns nothing extra; writing three throws marks away. Count what you wrote.

WHERE IT IS TAUGHT

Module Two, Unit 4 (Working with Strings).

ANSWER

Q1 (B) STRINGS AND FOUR METHODS, 5 MARKS

A **string** is an ordered sequence of characters, written in quotes. It is **immutable**: its characters cannot be changed in place. Four common methods, one example each:

METHOD	DOES	EXAMPLE	GIVES
upper()	Capitals	"hi".upper()	'HI'
lower()	Lower case	"Hi".lower()	'hi'
strip()	Trims ends	" hi ".strip()	'hi'
replace(a,b)	Swaps text	"hi".replace("i","o")	'ho'

Each returns a **new** string, because the original cannot change. Others you know and could name instead: `split()`, `count()`, `find()`, `title()`. See Module Two, Unit Four for all of them.

AS PRINTED

2025/2026 PAPER, QUESTION ONE (C)

Identify the Errors in the following code segment [3 marks]

```
1 person = { "name": 'John Doe', 'age': 30, 'city':  
2     'New York', 'Email': 'john.doe@example.com' }  
3  
4 print("Name:", person['Name'])  
5 print("Age:", person['Age'])  
6 print("City:", person['City'])  
7 print("Email:", person['email'])  
8  
9 person['Age'] = "35"  
10 person['phone'] = '123-456-7890'  
11 del person['email']  
12  
13 print("Updated Person Dictionary:")  
14 print(person)  
15
```

WHAT IS ACTUALLY ASKED

Identify only. It does not ask you to rewrite the code, unlike Question Two (c), which does. Listing the faults earns the marks here; a full corrected listing earns no more.

WHAT I AM GIVEN

A dictionary whose keys are `'name'`, `'age'`, `'city'` and `'Email'`, then six accesses to it.

WHERE THE TRAP SITS

Every fault is the **same** fault: a dictionary key is case sensitive. Do not hunt for a syntax error, there is none. The line that raises nothing at all is the most dangerous one on the page.

WHAT THE MARKS SAY

3 marks and there are more than three faults, so name them in groups rather than one line at a time.

WHERE IT IS TAUGHT

Module Three, Unit 4 (Sets and Dictionaries).

The dictionary defines its keys as 'name', 'age', 'city' and 'Email'. Every access then uses the **wrong case**, and a dictionary key is case sensitive, so each is a `KeyError`. Python stops at the first one it reaches:

```
Traceback (most recent call last):
  File "person.py", line 4, in <module>
    print("Name:", person['Name'])
KeyError: 'Name'
```

The full list of faults, so you earn every mark:

- ▶ `person['Name']`, `person['Age']`, `person['City']`: the keys are lower case. Three `KeyError`s.
- ▶ `person['email']` on line 7 and `del person['email']` on line 11: the key is 'Email' with a capital E.
- ▶ `person['Age'] = "35"` does not raise, but it **creates a new key** 'Age' beside the real 'age', rather than updating the age. A silent logic error.

Corrected so the case matches throughout, this is what it prints:

PERSON.PY, CORRECTED

```
person = {"name": "John Doe", "age": 30,
          "city": "New York", "Email": "john.doe@example.com"}
print("Name:", person["name"])
print("Age:", person["age"])
print("City:", person["city"])
print("Email:", person["Email"])
person["age"] = "35"
person["phone"] = "123-456-7890"
del person["Email"]
print("Updated Person Dictionary:")
print(person)
```

```
Name: John Doe
Age: 30
City: New York
Email: john.doe@example.com
Updated Person Dictionary:
{'name': 'John Doe', 'age': '35', 'city': 'New York', 'phone': '123-456-7890'}
```

Note the final dictionary: 'age' is now the string '35', 'phone' was added, and 'Email' was deleted. That is `del` from Module Three doing its job.

Write a Python program that implements a mathematical model to calculate Body Mass Index (BMI). Use the formula: $BMI = W / H^2$, where: W = weight in kilograms, H = height in meters. [3.5 marks]

1. Classify the BMI:
 - $BMI < 18.5 \rightarrow$ Underweight
 - $18.5 \leq BMI < 25 \rightarrow$ Normal weight
 - $25 \leq BMI < 30 \rightarrow$ Overweight
 - $BMI \geq 30 \rightarrow$ Obese
2. Display the BMI value with the BMI class.

BREAK IT DOWN

BEFORE YOU WRITE ANYTHING

WHAT I AM GIVEN

The formula and four bands, with their boundaries written as $\leq$ and $<$. The paper has done the hard thinking; copy its boundaries exactly rather than inventing your own.

WHAT IS ACTUALLY ASKED

A **program**, not a function and not a formula. It must read input, compute, classify and print.

WHERE THE TRAP SITS

The bands touch. 25 is Overweight, not Normal weight, because the second band stops **below** 25. An `if/elif` chain in ascending order gets this right for free; a chain of separate `if`s does not.

WHERE IT IS TAUGHT

Module Three, Unit 1 (Decision Making).

H^2 in the question is `H ** 2` in Python: `**` is the power operator from Module Two, not `^`, which is something else entirely. Read the two values, compute, then classify with `if/elif`.

BMI.PY

```
weight = float(input("Weight (kg): "))
height = float(input("Height (m): "))

bmi = weight / height ** 2

if bmi < 18.5:
    category = "Underweight"
elif bmi < 25:
    category = "Normal weight"
elif bmi < 30:
    category = "Overweight"
else:
    category = "Obese"

print(f"BMI = {bmi:.2f}")
print("Class:", category)
```

For a 72 kg person 1.75 m tall:

```
BMI = 23.51
Class: Normal weight
```

The `elif` ladder needs no upper bound on each rung: by the time you reach `elif bmi < 25`, the first `if` has already ruled out everything below 18.5, so `< 25` alone means the 18.5 to 25 band. That is the whole economy of `elif`.

Write a Python program that calculates the future value of an investment using compound interest. The program should display A and the interest earned ($A - P$) to two decimal places. [3.5 marks]

- P = principal
- r = annual interest rate (decimal)
- n = number of times interest is compounded per year
- t = time in years

Use the formula: $A = P(1 + r/n)^{(nt)}$ where:

BREAK IT DOWN

BEFORE YOU WRITE ANYTHING

WHAT IS ACTUALLY ASKED

Two printed values, not one: **A** and **the interest earned**, both to two decimal places. Printing only A loses marks.

READING THE FORMULA

$\wedge$ in the paper means "to the power of". In Python that is `**`, never `^`, which is a different operator entirely.

WHERE THE TRAP SITS

"(decimal)" beside `r` means 0.05, not 5. And "to two decimal places" is a formatting instruction: use an f-string with `:.2f`, do not round the stored value.

WHERE IT IS TAUGHT

Module Two, Unit 3 (Operators) and Unit 4 (formatting).

ANSWER

Q1 (E) COMPOUND INTEREST, 3.5 MARKS

The formula $A = P(1 + r/n)^{nt}$ is one line once you use `**` for the power and brackets for the exponent, so the whole `n*t` is the power:

INTEREST.PY

```
P = float(input("Principal: "))
r = float(input("Annual rate (decimal): "))
n = int(input("Times compounded per year: "))
t = int(input("Years: "))

A = P * (1 + r / n) ** (n * t)

print(f"Future value (A): {A:.2f}")
print(f"Interest earned: {A - P:.2f}")
```

For 100000 at 0.05, compounded 4 times a year for 3 years:

```
Future value (A): 116075.45
Interest earned: 16075.45
```

The `:.2f` rounds to two decimal places for display only. The mark hinges on the brackets: `** (n * t)`, not `** n * t`, which would raise to the power `n` and only then multiply by `t`, a different and wrong number.

Question Two

AS PRINTED

2025/2026 PAPER, QUESTION TWO (A)

What is a count-controlled loop in Python? [3.5 mark]

BREAK IT DOWN**BEFORE YOU WRITE ANYTHING****WHAT IS ACTUALLY ASKED**

A definition. One clear sentence plus a short example is the whole answer.

NOTE THE PAPER

The mark text reads **3.5 mark**, singular, where every other part says "marks". That is the paper's own slip and it is quoted here as printed. It means 3.5 marks.

WHAT THE MARKS SAY

3.5 marks for a definition is generous, so do not stop at one line: name the loop, say what controls it, and show a `for` with `range()` .

WHERE IT IS TAUGHT

Module Three, Unit 2 (Loops).

ANSWER**Q2 (A) COUNT-CONTROLLED LOOP, 3.5 MARKS**

A **count-controlled loop** runs a known, fixed number of times, once per item in a sequence. In Python it is the `for` loop with `range` . You use it when you know how many passes you need before the loop starts.

```
for i in range(5):  
    print(i)
```

It contrasts with a **condition-controlled loop** (`while`), which runs an unknown number of times, until a condition turns false. Count-controlled: "do this five times". Condition-controlled: "do this until the user types done".

AS PRINTED**2025/2026 PAPER, QUESTION TWO (B)**

Explain the differences among the main types of control structures in Python. Give one example of each. [4.5 marks]

BREAK IT DOWN**BEFORE YOU WRITE ANYTHING****WHAT IS ACTUALLY ASKED**

"Differences **among**" means compare them against each other, not define each in isolation. Then one example each.

WHAT I SHOULD NAME

Sequence, selection and repetition. Naming only `if` and `for` misses a third of the answer.

WHAT THE MARKS SAY

4.5 marks across three structures, each needing a difference and an example, is 1.5 a structure. Keep each tight.

WHERE IT IS TAUGHT

Module Three, Units 1 and 2.

There are three kinds of control structure, and the answer names all three with an example each:

- ▶ **Sequence:** statements run top to bottom, in order. The default. `a = 2; b = 3; print(a + b)`
- ▶ **Selection:** a choice, with `if/elif/else`. `if score >= 50: print("Pass")`
- ▶ **Repetition:** doing something many times, with `for` or `while`. `for i in range(3): print(i)`

The difference is what decides the flow: sequence has no decision, selection runs a block or skips it once, repetition runs a block again and again.

Identify the Errors in this code segment and write the corrected version. [3 marks]

```
1 my_string = "Serve Jesus Now"
2 print("Original String:", mystring)
3
4 first_character = my_string[1]
5 last_character = my_string[-0]
6 print("First Character:", first_character)
7 print("Last Character:" last_character)
8
9 substring = my_string[17:]
10 print("Substring:", substring)
11
12 greeting = "Hi"
13 name = "Alice"
14 message = greeting + ", " + name + "!"
15 print("Message:", message)
16
17 length = len(mystring)
18 print("Length of the String:", length)
19
20 count_e = my_string.count('e')
21 print("Count of 'e':", count_e)
22
```

WHAT IS ACTUALLY ASKED

Two things, and this is the difference from Question One (c): identify the errors **and write the corrected version**. A list of faults alone is half an answer here.

WHERE THE TRAPS SIT

Three of the faults stop the program and two are silent. Line 1 never closes its string, so nothing runs at all until that is fixed. Line 5 uses `[-0]`, which is `[0]`, so it quietly returns the **first** character while claiming to be the last.

ORDER OF DISCOVERY

Fix the syntax first, then run it in your head again. A `SyntaxError` hides every fault after it, so the name errors on lines 2 and 17 cannot even appear until line 1 is closed.

WHERE IT IS TAUGHT

Module Two, Unit 4 (Strings) and Module Four, Unit 4 (Exceptions).

The variable is `my_string`, but lines 2 and 17 read `mystring`, which was never defined. Line 2 raises first:

```
NameError: name 'mystring' is not defined
```

Two more are logic faults that run without complaint but give the wrong thing:

- ▶ `my_string[1]` is the **second** character, not the first. The first is `[0]`.
- ▶ `my_string[-0]` is `my_string[0]`, because `-0` is `0`. It gives the first character, not the last. The last is `[-1]`.
- ▶ `my_string[17:1]` starts past the end and runs backwards, so it is the empty string. A real substring needs `start < stop` within the length.

Corrected, taking the first word "Serve" as the substring:

MY_STRING.PY, CORRECTED

```
my_string = "Serve Jesus Now"
print("Original String:", my_string)
first_character = my_string[0]
last_character = my_string[-1]
print("First Character:", first_character)
print("Last Character:", last_character)
substring = my_string[6:11]
print("Substring:", substring)
greeting = "Hi"
name = "Alice"
message = greeting + ", " + name + "!"
print("Message:", message)
length = len(my_string)
print("Length of the String:", length)
count_e = my_string.count("e")
print("Count of 'e':", count_e)
```

```
Original String: Serve Jesus Now
First Character: S
Last Character: w
Substring: Jesus
Message: Hi, Alice!
Length of the String: 15
Count of 'e': 3
```

A secondary school is developing a simple Python program to assist teachers in managing students' assessment activities. The school has provided the following grading scheme: [6.5 marks]

- 70 – 100: Grade A
- 60 – 69: Grade B
- 50 – 59: Grade C
- 45 – 49: Grade D
- 0 – 44: Grade F

Given that the correct password for accessing the system is: `admin123`, write a Python program that uses appropriate control structures to perform the required tasks.

BREAK IT DOWN

BEFORE YOU WRITE ANYTHING

WHAT IS ACTUALLY ASKED

One program doing **four** separate jobs: a password loop, a validated score with a grade, a multiplication table, and a sentinel loop that totals numbers. Read all eight requirements before you start, because they are really four programs in a row.

WHAT THE MARKS SAY

6.5 marks over eight requirements. Each requirement is worth under a mark, so a part-finished program still scores well: write the four blocks in order and do not stall on the first.

WHERE THE TRAPS SIT

Requirement i says **continue requesting until** correct, which is a `while`, not an `if`. Requirement viii excludes the terminating 0 from the sum **and** from the count.

WHERE IT IS TAUGHT

Module Three, Units 1 and 2. The sentinel loop is the idiom drilled in both mocks.

Six tasks in one program. Take them in order: a password loop, a validated score, a grade, a multiplication table, then a running sum. Each is a control structure from Module Three.

SYSTEM.PY

```
# i. password loop
while True:
    pw = input("Password: ")
    if pw == "admin123":
        print("Access Granted")
        break
    print("Access Denied")

# ii and iii. read a score and validate it
score = int(input("Score: "))
if score < 0 or score > 100:
    print("Error: score must be between 0 and 100")
else:
    # iv. grade it
    if score >= 70:
        print("Grade A")
    elif score >= 60:
        print("Grade B")
    elif score >= 50:
        print("Grade C")
    elif score >= 45:
        print("Grade D")
    else:
        print("Grade F")

# v. multiplication table 1 to 12
num = int(input("Number: "))
for i in range(1, 13):
    print(f"{num} x {i} = {num * i}")

# vi. sum until 0
total = 0
count = 0
while True:
    n = int(input("Number (0 to stop): "))
    if n == 0:
        break
    total += n
    count += 1
print("Sum:", total)
print("Count:", count)
```

The grade ladder is the same `elif` economy as the BMI: each rung needs only its lower bound, because the rungs above have already been ruled out. The order matters: put `>= 45` before `>= 70` and every score would grade D. Ranges must be tested high to low.

The two `while True` loops are the two loop patterns from Module Three, Unit Two: one broken by a right password, one by a sentinel of 0.

Question Three

AS PRINTED

2025/2026 PAPER, QUESTION THREE (A)

Mention five (5) benefits of modularizing a program with functions [2.5 marks]

BREAK IT DOWN

BEFORE YOU WRITE ANYTHING

WHAT IS ACTUALLY ASKED

"Mention" means list. No explanation is demanded, so five short named benefits is a complete answer.

WHAT THE MARKS SAY

2.5 marks over five items is half a mark each. Write exactly five: a sixth earns nothing, and four costs you half a mark.

WHERE IT IS TAUGHT

Module Four, Unit 1 (Functions).

ANSWER

Q3 (A) FIVE BENEFITS, 2.5 MARKS

Five benefits of modularizing a program with functions, which is building it out of named pieces:

1. **Reuse:** write it once, call it anywhere, no copying.
2. **Readability:** a name says what a block does.
3. **Maintenance:** fix a bug in one place and every caller is fixed.
4. **Decomposition:** a large problem splits into small, testable pieces.
5. **Abstraction:** a caller uses a function knowing what it does, not how.

Module Four, Unit One carries the same list under "why functions matter".

AS PRINTED

2025/2026 PAPER, QUESTION THREE (B)

What is a function in Python? Why are functions important in programming? Differentiate between built-in functions and user-defined functions. Distinguish between a Void and Value-returning function in Python (with examples). [6 marks]

BREAK IT DOWN**BEFORE YOU WRITE ANYTHING****WHAT IS ACTUALLY ASKED**

Four questions in one part: define, justify, differentiate built-in from user-defined, and distinguish void from value-returning. Answer them as four labelled paragraphs so the marker can find all four.

WHAT THE MARKS SAY

6 marks, so 1.5 per question. The "(with examples)" attaches to the last one and is not optional.

WHY THIS PART MATTERS

Void versus value-returning comes back twice more on this same paper, in Question Six (b) and (c), worth 14 marks between them. Getting the distinction crisp here pays three times.

WHERE IT IS TAUGHT

Module Four, Unit 1.

ANSWER**Q3 (B) FUNCTIONS, BUILT-IN VS USER-DEFINED, 6 MARKS**

A **function** is a named block of code that runs only when you call it. They matter for the five reasons above. The two kinds:

	BUILT-IN	USER-DEFINED
Who wrote it	Comes with Python	You did, with def
Defining needed?	No, always there	Yes, before you call it
Examples	print, len, input	cube, grade_of

`print("hi")` calls a built-in; after `def cube(n): return n*n*n` you call your own with `cube(4)`.

AS PRINTED**2025/2026 PAPER, QUESTION THREE (C)**

What will list1 and list2 contain after the following statements are executed? [3 marks]

- i. `list1 = [1, 2] * 2`
- ii. `list2 = [3]`
- iii. `list2 += list1`

BREAK IT DOWN

BEFORE YOU WRITE ANYTHING

WHAT IS ACTUALLY ASKED

The **contents of both lists** at the end, not a trace of each line. Write the two final values.

WHERE THE TRAP SITS

`*` on a list repeats it, it does not multiply the numbers inside. And `+=` on a list extends it in place rather than nesting, so the result is flat.

WHAT THE MARKS SAY

3 marks for two answers means the working counts. Show the intermediate value of `list1`.

WHERE IT IS TAUGHT

Module Three, Unit 3 (Lists and Tuples).

ANSWER

Q3 (C) LIST1 AND LIST2, 3 MARKS

```
list1 = [1, 2] * 2
list2 = [3]
list2 += list1
```

- ▶ `[1, 2] * 2` **repeats the list**, so `list1` is `[1, 2, 1, 2]`, not `[2, 4]`.
- ▶ `list2 += list1` extends `list2` in place, tacking on every item of `list1`.

```
list1 = [1, 2, 1, 2]
list2 = [3, 1, 2, 1, 2]
```

The trap is reading `*` as multiply-the-numbers. On a list it repeats. See Module Three, Unit Three.

AS PRINTED

2025/2026 PAPER, QUESTION THREE (D)

The Computer Science Department organized a beginner Python Programming Workshop for undergraduate students. You have been asked to write a simple Python program to manage the workshop information. The organizers provided the following records:

- *Registered Students: Esther, John, Paula, Paul, Peter, Ada, John, Mary*
- *Workshop Details: "Python Programming Workshop", "Room 204", "Monday", 30*
- *Students Who Attended: Esther, John, Mercy, Paul, Esther, John, Paula*
- *Student Scores:*
 - *Esther scored 60*
 - *John scored 55*
 - *Mercy scored 65*
 - *Paul scored 80*

Write a Python program that uses the most suitable Python collection for each type of data. [6 marks]

WHAT IS ACTUALLY ASKED

The phrase that decides the whole question is **the most suitable Python collection for each type of data**. The marks are for **choosing**, not just for printing.

READING THE DATA

The names repeat: John appears twice in the registered list, Esther and John twice in the attendance list. That repetition is not a typo, it is the question telling you which collection to reach for.

WHAT THE MARKS SAY

6 marks and ten requirements, so no single line is worth much. Requirements viii and ix are one operator each once the right collection is chosen.

WHERE IT IS TAUGHT

Module Three, Units 3 and 4.

Each kind of data wants a different collection, and choosing right **is** the question:

- ▶ Registered students, who may repeat and whose order does not matter, but which you want **unique**: a **set**.
- ▶ Workshop details, four fixed facts that never change: a **tuple**.
- ▶ Attendance, with duplicates to remove: a **set**.
- ▶ Scores, a name against a number: a **dictionary**.

The two "registered but did not attend" and "attended but not registered" questions are **set difference** both ways, from Module Three, Unit Four.

WORKSHOP.PY

```
registered = {"Esther", "John", "Paula", "Paul",
              "Peter", "Ada", "Mary"}
details = ("Python Programming Workshop", "Room 204", "Monday", 30)
attended = {"Esther", "John", "Mercy", "Paul", "Paula"}
scores = {"Esther": 72, "John": 55, "Paula": 68,
          "Paul": 40, "Peter": 91, "Ada": 59, "Mary": 83}

print("Workshop details:", details)
print("Registered:", sorted(registered))
print("Attended:", sorted(attended))
print("Registered, did not attend:", sorted(registered - attended))
print("Attended, not registered:", sorted(attended - registered))
print("Scored 60+:", sorted(n for n, s in scores.items() if s >= 60))
```

```
Workshop details: ('Python Programming Workshop', 'Room 204', 'Monday', 30)
Registered: ['Ada', 'Esther', 'John', 'Mary', 'Paul', 'Paula', 'Peter']
Attended: ['Esther', 'John', 'Mercy', 'Paul', 'Paula']
Registered, did not attend: ['Ada', 'Mary', 'Peter']
Attended, not registered: ['Mercy']
Scored 60+: ['Esther', 'Mary', 'Paula', 'Peter']
```

Why sorted is around every set: a set has no order, so its printed order is not reliable. Wrapping it in `sorted` gives a stable, markable list every run. Mercy attended without registering; Ada, Mary and Peter registered but did not come.

PAST PAPER 2025/2026

Questions Four to Six, Worked

The database question, the file question, and the one where the paper's own answer is wrong.

Question Four

AS PRINTED

2025/2026 PAPER, QUESTION FOUR (A)

Strings are Immutable. Explain and with reasons. [5 marks]

BREAK IT DOWN**BEFORE YOU WRITE ANYTHING****NOTE THE PAPER**

It reads "Explain and with reasons", which is not quite grammatical. It is the paper's own wording, quoted as printed. Read it as "explain, and give reasons".

WHAT IS ACTUALLY ASKED

Not just what immutable means, but **why**. Reasons carry the marks: an explanation with no reason is half an answer.

WHAT THE MARKS SAY

5 marks is a lot for a bookwork part, so show it as well as saying it: a two-line demonstration that reassignment makes a new string is worth more than another paragraph.

WHERE IT IS TAUGHT

Module Two, Unit 4.

ANSWER**Q4 (A) STRINGS ARE IMMUTABLE**

Immutable means a string cannot be changed after it is made. Trying to change one character is an error:

```
>>> s = "cat"
>>> s[0] = "b"
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
TypeError: 'str' object does not support item assignment
```

The reason: every method that "changes" a string, like `upper()` or `replace()`, actually returns a **new** string and leaves the original untouched. To "edit" a string you build a new one and reassign it: `s = s.replace("c", "b")`. A list, by contrast, is mutable and does allow `list[0] = "b"`.

AS PRINTED**2025/2026 PAPER, QUESTION FOUR (B)**

A program contains the following function definition:

```
2
3 def cube(num):
4     return num * num * num
5
```

- i. *Write a statement that passes the value 4 to this function and assigns its return value to the variable result. [1 mark]*
- ii. *What will be the value in result [1 mark]*

BREAK IT DOWN**BEFORE YOU WRITE ANYTHING****WHAT IS ACTUALLY ASKED**

One line of code, then one number. Nothing more: part i is a single statement and part ii is a single value.

NOTE THE PAPER

Part ii is printed without a question mark. Quoted as printed; it is still a question.

WHAT THE MARKS SAY

1 mark each, the smallest on the paper. Write the line, write the number, move on.

WHERE IT IS TAUGHT

Module Four, Unit 1.

ANSWER**Q4 (B) THE CUBE FUNCTION, 1 + 1 MARKS**

(i) A statement that passes 4 and stores the result:

```
result = cube(4)
```

(ii) `cube(4)` returns `4 * 4 * 4`, so:

```
result = 64
```

The call `cube(4)` **is** a value, because `cube` returns one, so it can sit on the right of an `=`. That is a value-returning function from Module Four.

AS PRINTED**2025/2026 PAPER, QUESTION FOUR (C)**

A secondary school has decided to computerize the management of students' academic records. You have been asked to develop a simple Python program that uses an SQLite database to store, retrieve, update, and manage students' records. The school requires that all records be stored in a database named `school.db` in a table named `student`. The table should contain the following fields: `StudentID` (INTEGER), `StudentName` (TEXT), `Class` (TEXT), and `Score` (REAL). The school has provided the following student records: [10.5 marks]

StudentID | StudentName | Class | Score

101 | David James | SS1 | 78

102 | Mary Johnson | SS2 | 85

103 | Peter Okoro | SS1 | 67

104 | Grace Bello | SS3 | 92

105 | Esther Ade | SS2 | 74

Write a Python program that uses SQLite to manage the students' records.

WHAT IS ACTUALLY ASKED

The single biggest part on the paper. Seven requirements: create, insert, read all, read filtered, aggregate, update, delete. That is the whole of SQL in one question.

WHAT THE MARKS SAY

10.5 marks is three of the small parts put together, and it is 15 per cent of the whole examination. If you can do this question, attempt it first: nothing else on the paper pays as well per line written.

WHERE THE TRAPS SIT

The names are exact: database `school.db`, table `student` (singular), and those four field names with those four types. Marks go for matching them, not for a working program with your own names. Requirement i says **if they do not already exist**.

WHERE IT IS TAUGHT

Module Five, Unit 1 (Python and Databases).

The biggest single part on the paper, and it walks through the whole of Module Five, Unit One plus the new WHERE, UPDATE, DELETE and summary functions. Take the seven requirements in order.

SCHOOL.PY

```
import sqlite3

conn = sqlite3.connect("school.db")
cursor = conn.cursor()

# i. create the table if it is not there
cursor.execute("""CREATE TABLE IF NOT EXISTS student
                  (StudentID INTEGER PRIMARY KEY, StudentName TEXT,
                   Class TEXT, Score REAL)""")

# ii. insert the given records
records = [(101, "David James", "SS1", 78),
           (102, "Mary Johnson", "SS2", 85),
           (103, "Peter Okoro", "SS1", 67),
           (104, "Grace Bello", "SS3", 92),
           (105, "Esther Ade", "SS2", 74)]
cursor.executemany("INSERT INTO student VALUES (?, ?, ?, ?)", records)
conn.commit()

# iii. all records
print("All students:")
cursor.execute("SELECT * FROM student")
for row in cursor.fetchall():
    print(row)

# iv. SS2 only
print("SS2 students:")
cursor.execute("SELECT * FROM student WHERE Class = ?", ("SS2",))
for row in cursor.fetchall():
    print(row)

# v. statistics in one query
cursor.execute("SELECT COUNT(*), MAX(Score), AVG(Score) FROM student")
total, highest, average = cursor.fetchone()
print("Total:", total, " Highest:", highest, " Average:", average)

# vi. update Peter's score
cursor.execute("UPDATE student SET Score = ? WHERE StudentName = ?",
              (75, "Peter Okoro"))
conn.commit()

# vii. delete student 105
cursor.execute("DELETE FROM student WHERE StudentID = ?", (105,))
conn.commit()

conn.close()
```

The output of the two SELECT s and the statistics:

```

All students:
(101, 'David James', 'SS1', 78.0)
(102, 'Mary Johnson', 'SS2', 85.0)
(103, 'Peter Okoro', 'SS1', 67.0)
(104, 'Grace Bello', 'SS3', 92.0)
(105, 'Esther Ade', 'SS2', 74.0)
SS2 students:
(102, 'Mary Johnson', 'SS2', 85.0)
(105, 'Esther Ade', 'SS2', 74.0)
Total: 5 Highest: 92.0 Average: 79.2

```

Five marks are in the details a rushed answer drops:

- ▶ **Scores print as 78.0**, with a decimal, because the column is `REAL`.
- ▶ **Placeholders, not glued strings**, for every value. `("SS2",)` keeps its comma.
- ▶ **UPDATE and DELETE each commit**. Forget the commit and the change is thrown away silently.
- ▶ **The statistics are one query**. `COUNT`, `MAX` and `AVG` come back as a single row, unpacked into three names.

Question Five

AS PRINTED

2025/2026 PAPER, QUESTION FIVE (A)

What is file handling in Python? [2 marks]

BREAK IT DOWN

BEFORE YOU WRITE ANYTHING

WHAT IS ACTUALLY ASKED

A definition. Two marks, two sentences.

WHERE IT IS TAUGHT

Module Four, Unit 3 (File Handling).

ANSWER

Q5 (A) FILE HANDLING, 2 MARKS

File handling is reading data from, and writing data to, a file on disk, so the data lasts after the program ends. Python does it with `open()`, the read and write methods, and `close()` (or a `with` block that closes for you). See Module Four, Unit Three.

AS PRINTED

2025/2026 PAPER, QUESTION FIVE (B)

List and explain four (4) file access modes used in the `open()` function. [6 marks]

BREAK IT DOWN

BEFORE YOU WRITE ANYTHING

WHAT IS ACTUALLY ASKED

Four modes, each **listed and explained**. The mode letter alone is not an explanation.

WHAT THE MARKS SAY

6 marks over four modes is 1.5 each, so each needs the letter, what it does, and what happens if the file does not exist.

WHERE THE TRAP SITS

The difference that earns the marks is what each mode does to an **existing** file. One of them silently destroys it.

WHERE IT IS TAUGHT

Module Four, Unit 3.

ANSWER

Q5 (B) FOUR FILE MODES, 6 MARKS

MODE	MEANS	IF THE FILE EXISTS	IF IT DOES NOT
"r"	Read	Reads it	FileNotFoundError
"w"	Write	Erased first	Created
"a"	Append	Adds at the end	Created
"r+"	Read and write	Both, no erase	FileNotFoundError

The two that catch people: "w" **destroys** the existing contents the moment you open it, and "r" is the only common one that fails on a missing file.

AS PRINTED

2025/2026 PAPER, QUESTION FIVE (C)

Mention the difference between reading from a file and appending to a file. [2 marks]

BREAK IT DOWN

BEFORE YOU WRITE ANYTHING

WHAT IS ACTUALLY ASKED

One difference, stated. "Mention" signals a short answer, and 2 marks confirms it.

WHERE IT IS TAUGHT

Module Four, Unit 3.

ANSWER

Q5 (C) READING VS APPENDING, 2 MARKS

Reading ("r") takes data **out** of a file and changes nothing. **Appending** ("a") adds new data to the **end**, keeping what is already there. Reading needs the file to exist; appending creates it if it does not. In short: one looks, the other adds without erasing.

The Computer Science Department conducted a Continuous Assessment (CA) for CSC241 students. You have been asked to develop a Python program to manage the students' assessment records. The department requires that all student records be stored in a single text file throughout the program. The file created at the beginning of the program must be used for all subsequent operations. The assessment records should contain the following information for each student: Name, Matric Number, Department, and Score. The records should be stored in a text file named: `your_matric_number_CSC241.txt`. (Replace `your_matric_number` with your actual matric number. For example: `23CG012345_CSC241.txt`). Write a Python program that uses file handling operations to manage the students' assessment records. A sample record is shown below:

- Name: John Doe, Matric No: 23CG012345, Department: Computer Science, Score: 78
- Name: Mary James, Matric No: 23CG012346, Department: MIS, Score: 85

[7.5 marks]

BREAK IT DOWN

BEFORE YOU WRITE ANYTHING

WHAT IS ACTUALLY ASKED

Four operations on **one** file: write ten records, append five more without destroying them, read all, read the first five. The words "the same file" appear in three of the four requirements.

WHERE THE TRAP SITS

Requirement ii is the whole question. Opening the file the same way you did in i would erase the ten records you just wrote. The mode you choose here is worth more than the rest of the program.

THE INSTRUCTION PEOPLE MISS

The file name must be built from **your own matric number**. Writing the literal example name is a mark lost for not reading.

WHERE IT IS TAUGHT

Module Four, Unit 3.

Write the first records with "w", add more with "a" so the first are kept, then read the whole file and just the first five. The same file name throughout, as the question insists.

CA_RECORDS.PY

```
filename = "23CG012345_CSC241.txt"

# i. first 10 records, "w" starts the file fresh
with open(filename, "w") as f:
    for i in range(10):
        name = input("Name: ")
        matric = input("Matric No: ")
        dept = input("Department: ")
        score = input("Score: ")
        f.write(f"Name: {name}, Matric No: {matric}, "
              f"Department: {dept}, Score: {score}\n")

# ii. 5 more, "a" keeps the first ten
with open(filename, "a") as f:
    for i in range(5):
        name = input("Name: ")
        matric = input("Matric No: ")
        dept = input("Department: ")
        score = input("Score: ")
        f.write(f"Name: {name}, Matric No: {matric}, "
              f"Department: {dept}, Score: {score}\n")

# iii. read and show all
print("All records:")
with open(filename, "r") as f:
    for line in f:
        print(line.strip())

# iv. read and show the first five
print("First five:")
with open(filename, "r") as f:
    for i, line in enumerate(f):
        if i >= 5:
            break
        print(line.strip())
```

The marks live in the modes: "w" once at the start, then "a" so the append does not wipe the ten already written. If both used "w", the second open would erase the first ten before adding five, and you would end with five records, not fifteen. Reading the first five uses `enumerate` and `break` to stop early, from Module Three.

Question Six

AS PRINTED

2025/2026 PAPER, QUESTION SIX (A)

Given the following code segment [3 marks]

```
1
2 nested_list = [
3     [1, 2, 3],
4     [4, 5, 6],
5     [7, 8, 9],
6     [10, 11, 12, 13],
7     [14, [15, 16, 17], 18],
8     [19, [20, [21, 22], 23], 24]
9 ]
10
```

- What is the value at index [3][2]?
- `print (len (nested_list[4]))`
- What is the value at index [5][1][1][0]?

BREAK IT DOWN

BEFORE YOU WRITE ANYTHING

WHAT IS ACTUALLY ASKED

Three values read out of one nested list. No program: just the answers.

HOW TO READ AN INDEX CHAIN

Left to right, one bracket at a time, writing down what you are holding after each step. Trying to see [5][1][1][0] in one go is how people lose this mark.

WHERE THE TRAP SITS

The rows are ragged on purpose: row 3 has four items and rows 4 and 5 contain lists inside lists. Counting from 1 instead of 0 gets every one of the three wrong.

WHERE IT IS TAUGHT

Module Three, Unit 3.

ANSWER

Q6 (A) NESTED INDEXING, 3 MARKS

Follow the brackets left to right, one layer at a time. The list is six rows, some of them holding lists of their own.

- ▶ **[3][2]** : row 3 is [10, 11, 12, 13] , item 2 of it is 12 .
- ▶ **len(nested_list[4])** : row 4 is [14, [15, 16, 17], 18] , which has three items, so 3 . The inner list counts as one item.
- ▶ **[5][1][1][0]** : row 5 is [19, [20, [21, 22, 23], 24]] . [1] is [20, [21, 22, 23], 24] , its [1] is [21, 22, 23] , and [0] of that is 21 .

```
12
3
21
```

Lists inside lists, from Module Three, Unit Three. Each bracket peels one layer; write down what each step gives and the chain is harmless.

A fitness club organized a Weekly Step Challenge to encourage members to maintain an active lifestyle. You have been asked to develop a Python program that tracks a participant's daily step count over a specified number of days and determines whether the participant achieved the overall fitness milestone. The club requires that the final performance report be displayed using a void function named `display_achievement_summary()`, which receives the participant's name, total steps taken, and overall target as arguments. Write a Python program that uses a void function to evaluate and display the participant's fitness achievement. [7 marks]

BREAK IT DOWN

BEFORE YOU WRITE ANYTHING

WHAT IS ACTUALLY ASKED

A program built around **a void function with a named signature**. The question names the function and its three parameters for you, so use those names exactly.

VOID IS THE POINT

A void function **prints** and returns nothing. Returning the report string instead of printing it answers a different question and loses the marks this part exists to test.

WHAT THE MARKS SAY

7 marks, and the paper prints a Sample Interaction. Match its wording and layout: it is the marking scheme in disguise.

WHERE IT IS TAUGHT

Module Four, Unit 1, and Question Three (b) above.

A **void** function does the displaying and returns nothing. Read the inputs, sum the daily steps, work out the target, then hand three values to the function.

STEPS.PY

```
def display_achievement_summary(user_name, actual_total, target_total):
    print(f"--- Progress Report for {user_name} ---")
    print("Total Steps:", actual_total)
    if actual_total >= target_total:
        print("Milestone Achieved!!")
    else:
        print("Milestone Not Achieved.")

name = input("Enter your name: ")
days = int(input("Enter the number of days to track: "))
goal = int(input("Enter your daily step goal: "))

total_steps = 0
for day in range(1, days + 1):
    steps = int(input(f"Enter steps for day {day}: "))
    total_steps += steps

target = goal * days
display_achievement_summary(name, total_steps, target)
```

For Alice, 3 days, a 10000 goal, and steps 12000, 9000, 11000:

```
--- Progress Report for Alice ---
Total Steps: 32000
Milestone Achieved!!
```

The target is $10000 \times 3 = 30000$, and 32000 clears it, so the milestone is achieved. The function **prints and returns nothing**, which is exactly what "void" means: it is called for its effect, not for a value.

A commercial renovation company specializes in installing flooring in circular meeting rooms for corporate clients. You have been asked to develop a Python program that estimates the total cost of a flooring project. To promote code reusability and modular programming, the company requires that the program be developed using value-returning functions. The area of the meeting room is calculated using the formula: $\text{Area} = \pi r^2$ where π should be obtained from the math module. The total project cost is calculated as: $\text{Total Cost} = (\text{Area} \times \text{Flooring Price}) + \text{Labor Fee}$. Write a Python program that uses value-returning functions to estimate the cost of the flooring project. [7 marks]

WHAT IS ACTUALLY ASKED

The mirror of (b): three **value-returning** functions, named for you, that hand results back to a main program which does the printing.

THE INSTRUCTION THAT CARRIES MARKS

" π should be obtained from the math module" is not a suggestion. Typing 3.14 answers a different question and gives a different number.

READ THE SAMPLE OUTPUT TWICE

The paper prints Total Calculated Cost: 10546.90 for radius 12, price 20, fee 1500. Work it through with `math.pi` before you trust it. The answer below shows what the run actually gives.

WHERE IT IS TAUGHT

Module Four, Unit 1 and Unit 2 (Modules).

ANSWER**Q6 (C) VALUE-RETURNING FUNCTIONS AND MATH.PI, 7 MARKS**

Three value-returning functions, each handing a value back, and `math.pi` for the area of the circle. This is the modular style the question demands.

FLOORING.PY

```
import math

def get_data():
    radius = float(input("Enter room radius (meters): "))
    price = float(input("Enter flooring price per square meter: "))
    labor = float(input("Enter labor fee: "))
    return radius, price, labor

def calculate_area(radius):
    return math.pi * radius ** 2

def calculate_total(area, material_price, labor_fee):
    return area * material_price + labor_fee

radius, price, labor = get_data()
area = calculate_area(radius)
total_cost = calculate_total(area, price, labor)

print("--- Project Estimate ---")
print(f"Total Calculated Cost: {total_cost:.2f}")
if total_cost > 10000:
    print("Status: Requires Senior Management Approval.")
else:
    print("Status: Approval not required.")
```

For radius 12, price 20, labor 1500:

```
--- Project Estimate ---
Total Calculated Cost: 10547.79
Status: Requires Senior Management Approval.
```

The question sheet prints the sample cost as **10546.90**. Run the arithmetic it asks for and you get **10547.79**. The paper is wrong by 0.89, and it is worth seeing why, because the method is sound and only the printed figure is off.

```
>>> import math
>>> area = math.pi * 12 ** 2
>>> area
452.3893421169302
>>> area * 20 + 1500
10547.786842338604
```

Rounded to two places that is 10547.79 . There is no radius, price or fee near the sheet's own sample that reaches 10546.90 with `math.pi` . **Write the working, show the run, and state your figure.** A marker who has run it will credit the correct number; one who has not still sees a fully justified method. This is the same discipline the whole book is built on: trust the run, not the printout, even when the printout is the examiner's.

OBJECTIVE SECTION

Seventy Questions, Every Option Explained

The other half of the paper. Seventy objective questions across the five modules, every code snippet run on a real interpreter, and for every question a reason why each of the four options is right or wrong.

```
>>> numbers[-3:] * 2
```

~~[6, 8, 10]~~

☒ [3, 4, 5, 3, 4, 5]

~~[2, 3, 4]~~

~~TypeError~~

How to Attack an Objective Question

The paper carries an objective section as well as the long questions, and it is sat under more time pressure than anything else on it. This part is where you practise that half.

TEACH

WHERE THESE QUESTIONS COME FROM

Say this plainly, because it decides how you should read them. This course has no computer-based test among its papers, so unlike the long questions in this book, which are quoted from the examiner word for word, **these 70 questions were written for this manual**. They are not the examiner's and they do not pretend to be.

What they are grounded in is real. Every one is drawn from something this manual teaches and something the ten past papers actually ask about, and the paper's own habits are copied on purpose: it leans hard on asking for the output of a code snippet, on asking you to identify and correct an error, and on asking for a property by name, such as lists being ordered, sets being unique, or strings being immutable. Those are descriptions of what the paper does, not quotations of its wording, which is why none of them is set in quotation marks here. So what you are drilling is this course's own material in objective form, even though the wording is ours.

Every code snippet below was **run on a real interpreter** and the answer is what came back, not what it ought to have been. If your answer differs from the book's, the interpreter settled it, not an opinion.

◆ KEY POINT

In a four-option question, three options are wrong for a reason you can state. Finding that reason is faster and safer than recognising the right answer, because recognition is exactly what a well-built distractor attacks.

BLUNT VERSION: do not hunt for the right one. Kill the wrong ones.

TEACH

THE FOUR MOVES, IN ORDER

- 1 **Underline the instruction word.** NOT, TRUE, BEST, PRIMARILY and EXCEPT each flip or narrow the question. A question containing NOT wants the odd one out, and read as an ordinary question every wrong answer starts to look right.
- 2 **For a code question, trace it on paper before you look at the options.** Write the variables down the margin and update them line by line. The options are built from the answers you get by tracing it WRONG, so reading them first plants the mistake.
- 3 **Strike the impossible.** Two of four usually die at once: an option that contradicts a definition, or that is true but answers a different question.
- 4 **Split the last two on the exact wording.** It is nearly always one word: only, always, all, entirely, every. An option saying a function "eliminates complexity entirely" is wrong because of "entirely".

Five facts answer a large share of every output-tracing question this course can set. Know them cold and the tracing gets much shorter.

1. **input() returns a string**, always, however numeric it looks.
2. **A slice excludes its stop**, and range() excludes its stop too.
3. **A single slash always gives a float**; double slash floors.
4. **Methods that change a list in place return None**: sort, append, reverse, extend.
5. **Assignment does not copy**. `b = a` gives one object two names.

HOW TO USE THIS PART

Sit one module at a time, timed at 45 seconds a question, with the answers covered. Then read the whole answer block for that module, including the questions you got right: on an objective paper a right answer for the wrong reason is a mark you will not get twice.

OBJECTIVE SECTION

Introduction to Python Programming

8 questions on Module One. Cover the answers, sit the whole set, then read every line of the answer block.

1. Python is best described as which kind of language?
 - (a) A compiled language, translated to machine code before it runs
 - (b) An interpreted, high-level, general-purpose language
 - (c) A markup language for describing documents
 - (d) An assembly language for a specific processor family
2. A Python script contains a syntax error on its last line. What happens when you run it?
 - (a) The first lines run, then the error is reported when that line is reached
 - (b) Nothing runs: the whole file is checked before any of it executes
 - (c) The error is ignored and the script finishes normally
 - (d) The line is skipped and the rest of the script continues
3. Which of these is NOT a stated advantage of Python?
 - (a) It is easy to read, with a syntax close to English
 - (b) It is portable: the same code runs on different platforms
 - (c) It is the fastest executing language available
 - (d) It has a large standard library and a large community
4. At the interactive prompt you type `2 + 3` and press enter. Why does 5 appear even though you never wrote `print`?
 - (a) The prompt echoes the value of any expression you type
 - (b) Python adds a `print` automatically to every line of every program
 - (c) The plus operator prints its result
 - (d) It does not appear; the prompt shows nothing without `print`

5. What is the correct way to run a saved file called `grades.py` from the command line?
- (a) `run grades.py`
 - (b) `python grades.py`
 - (c) `grades.py run`
 - (d) `execute python grades.py`
6. Which line is a valid single-line comment in Python?
- (a) `// this is a comment`
 - (b) `/* this is a comment */`
 - (c) `# this is a comment`
 - (d) `-- this is a comment`
7. In Python, what does indentation do?
- (a) Nothing: it is style only, as in C or Java
 - (b) It marks which statements belong to a block, and is part of the syntax
 - (c) It sets how far output is printed from the left edge
 - (d) It is required only inside functions
8. Which of these is a VALID Python variable name?
- (a) `2nd_score`
 - (b) `class`
 - (c) `total score`
 - (d) `_total2`

1. The answer is An interpreted, high-level, general-purpose language.

Interpreted, high-level, general-purpose. Those three words are the definition the paper wants back.

- ✗ **A compiled language, translated to machine code before it runs.** Wrong. Compilation to machine code ahead of running is the C and Java model. Python translates and runs statement by statement through an interpreter, which is why an error can appear only when that line runs.
- ✓ **An interpreted, high-level, general-purpose language.** Correct. Interpreted means a program runs it line by line; high-level means it hides the machine; general-purpose means it is not tied to one kind of task.
- ✗ **A markup language for describing documents.** Wrong. A markup language describes how content is structured, like HTML. Python computes, it does not mark up.
- ✗ **An assembly language for a specific processor family.** Wrong. Assembly is tied to one processor and its instruction set. Python code runs unchanged wherever the interpreter runs.

2. The answer is Nothing runs: the whole file is checked before any of it executes.

Syntax errors are found at parse time, before execution. Runtime errors are found when their line runs. Knowing which is which explains why some faulty scripts print nothing at all.

- ✗ **The first lines run, then the error is reported when that line is reached.** Wrong, and it is the trap. This is true of a NAME error or a type error, which are found only when the line runs, but a SYNTAX error is found while the file is being parsed, before anything executes.
- ✓ **Nothing runs: the whole file is checked before any of it executes.** Correct. Python parses the whole file first. A syntax error anywhere stops it before the first line runs, which is why a stray colon can produce no output at all.
- ✗ **The error is ignored and the script finishes normally.** Wrong. Python never ignores an error. Something is always reported.
- ✗ **The line is skipped and the rest of the script continues.** Wrong. Python does not skip statements it cannot parse. There is no "carry on regardless" mode.

3. The answer is It is the fastest executing language available. Speed of execution is a DISadvantage of Python, not an advantage. Everything else on that list is real.

- ✗ **It is easy to read, with a syntax close to English.** Wrong choice, because readability IS an advantage. The whitespace rules exist to force it.
- ✗ **It is portable: the same code runs on different platforms.** Wrong choice, because portability IS an advantage. The interpreter hides the platform.
- ✓ **It is the fastest executing language available.** Correct, and this is the one to remember. Interpretation costs speed. Python is chosen for programmer time, not for execution time, and a paper listing slow execution as a disadvantage of Python is common.
- ✗ **It has a large standard library and a large community.** Wrong choice, because the standard library IS an advantage. It is the "batteries included" claim.

4. The answer is The prompt echoes the value of any expression you type. The prompt echoes; a script does not. This is why the manual marks its three kinds of block differently, and why copying prompt examples into a file makes the output vanish.

- ✓ **The prompt echoes the value of any expression you type.** Correct. The interactive prompt evaluates what you type and displays the result. That is a feature of the prompt, not of the language.
- ✗ **Python adds a print automatically to every line of every program.** Wrong, and this is the trap. Inside a SCRIPT, `2 + 3` on its own line computes 5 and throws it away silently. Only the prompt echoes.
- ✗ **The plus operator prints its result.** Wrong. The plus operator produces a value and does nothing else with it.

✗ **It does not appear; the prompt shows nothing without print.** Wrong. The prompt does echo. Try it: this is the difference between the three kinds of code block this manual sets out.

5. The answer is **python grades.py**. `python <filename>`. The interpreter is the program; your file is the input to it.

✗ **run grades.py.** Wrong. There is no run command. The interpreter is the program you launch, and the script is what you hand it.

✓ **python grades.py.** Correct. You launch the interpreter, `python`, and give it the file to run as its argument.

✗ **grades.py run.** Wrong. The order is wrong and `run` is not a Python word.

✗ **execute python grades.py.** Wrong. There is no execute command wrapping it.

6. The answer is **# this is a comment**. Hash to end of line. Python has no block comment: a run of hashes, or a string used as a docstring, is how several lines are done.

✗ **// this is a comment.** Wrong. Two slashes are the C, Java and JavaScript comment. In Python two slashes are the floor division operator, so this line is an error.

✗ **/* this is a comment */.** Wrong. Slash-star is the C block comment. Python has no block comment syntax at all.

✓ **# this is a comment.** Correct. The hash starts a comment that runs to the end of the line.

✗ **-- this is a comment.** Wrong. Two hyphens are the SQL comment, which matters here because this course also writes SQL through Python. The two languages do not share it.

7. The answer is **It marks which statements belong to a block, and is part of the syntax**. Indentation is syntax, not decoration. It replaces the braces that other languages use to group statements.

✗ **Nothing: it is style only, as in C or Java.** Wrong, and it is the single biggest habit to break coming from another language. In Python the indentation IS the block structure, so getting it wrong changes the meaning of the program or stops it running.

✓ **It marks which statements belong to a block, and is part of the syntax.** Correct. Where C uses braces, Python uses the indentation itself. An inconsistent indent raises `IndentationError`.

✗ **It sets how far output is printed from the left edge.** Wrong. Output position is decided by what you print, not by how the source is laid out.

✗ **It is required only inside functions.** Wrong. Every block needs it: `if`, `else`, `for`, `while`, `def`, `class`, `try`, `with`.

8. The answer is **_total2**. Start with a letter or underscore, then letters, digits or underscores, and never a keyword. Names are case sensitive, so `Total` and `total` are two different variables.

✗ **2nd_score.** Wrong. A name may not start with a digit. Letters and the underscore only, then digits are allowed after the first character.

✗ **class.** Wrong. `class` is a reserved keyword, so it cannot be used as a name. The same applies to `if`, `for`, `def`, and the rest.

✗ **total score.** Wrong. A space separates two names, so Python reads this as two things and reports a syntax error. Use an underscore.

✓ **_total2.** Correct. It starts with an underscore, which is allowed, and the digit is not in first position.

OBJECTIVE SECTION

Python Basics, Syntax, Operators and Strings

18 questions on Module Two. Cover the answers, sit the whole set, then read every line of the answer block.

1. What does it mean to say Python is dynamically typed?

- (a) A variable must be declared with its type before use
- (b) The type belongs to the value, and a name may be rebound to a value of another type
- (c) Types are checked by the compiler before the program runs
- (d) Every value in Python is a string until converted

2. A program does `age = input()` and the user types 20. What is printed?

```
age = "20"          # this is exactly what input() hands back
print(type(age).__name__)
print(age * 2)
```

- (a) `int`
`40`
- (b) `str`
`2020`
- (c) `str`
`40`
- (d) `int`
`2020`

3. What is printed?

```
print(7 / 2)
print(7 // 2)
print(7 % 2)
```

- (a) `3.5`
`3`
`1`
- (b) `3`
`3`
`1`
- (c) `3.5`
`3.5`
`1`
- (d) `3.5`
`3`
`3.5`

4. What is printed?

```
print(2 + 3 * 4 ** 2)
```

- (a) 50
- (b) 80
- (c) 400
- (d) 100

5. What is printed?

```
print(2 ** 3 ** 2)
```

- (a) 64
- (b) 512
- (c) 18
- (d) 12

6. What is printed?

```
x = 10  
x += 5  
x -= 3  
x *= 2  
print(x)
```

- (a) 24
- (b) 12
- (c) 30
- (d) 17

7. What is printed?

```
x = 5  
print(1 < x < 10)  
print(x == 5 and x != 5)
```

- (a) True
False
- (b) True
True
- (c) False
False
- (d) TypeError

8. What is printed?

```
print(True and False or True)  
print(not True and False)
```

- (a) True
False
- (b) False
False
- (c) True
True
- (d) False
True

9. What is printed?

```
print(int("7") + int(3.9))  
print(float("2.5") * 2)  
print(str(4) + "2")
```

- (a) 10
5.0
42
- (b) 10.9
5.0
6
- (c) 11
5.0
42
- (d) 10
5
42

10. What is meant by saying that Python strings are immutable?

- (a) A string cannot be printed twice
- (b) Once created, a string object cannot be changed; operations return a new string
- (c) A string cannot be assigned to a variable more than once
- (d) A string cannot contain numbers

11. What is printed?

```
s = "PYTHON"  
print(s[0], s[-1], s[1:4], s[::-1])
```

- (a) P N YTH NOHTYP
- (b) P N YTHO NOHTYP
- (c) Y N YTH NOHTYP
- (d) P O YTH NOHTYP

12. What is printed?

```
s = " Hello World "  
print(s.strip().upper())  
print(s.strip().split())  
print("-".join(["a", "b", "c"]))
```

- (a) HELLO WORLD
['Hello', 'World']
a-b-c
- (b) HELLO WORLD
['Hello World']
a-b-c
- (c) HELLO WORLD
['Hello', 'World']
a-b-c
- (d) Hello World
['Hello', 'World']
abc

13. What is printed?

```
print("ab" * 3)
print("ab" + "3")
```

- (a) ababab
ab3
- (b) ab3
ababab
- (c) abababab
ab3
- (d) TypeError
ab3

14. What is printed?

```
name = "Ada"
score = 87.456
print(f"{name} scored {score:.1f}")
```

- (a) Ada scored 87.456
- (b) Ada scored 87.5
- (c) Ada scored 87.4
- (d) {name} scored {score:.1f}

15. What is printed?

```
print("a\tb")
print("c\nd")
print("e\\f")
```

- (a) a b
c
d
e\f
- (b) a\tb
c\nd
e\\f
- (c) ab
cd
ef
- (d) a b
cd
ef

16. What is printed?

```
a = [1, 2]
b = [1, 2]
print(a == b)
print(a is b)
```

- (a) True
True
- (b) True
False
- (c) False
False
- (d) False
True

17. What is printed?

```
a, b = 1, 2
a, b = b, a
print(a, b)
```

- (a) 1 2
- (b) 2 1
- (c) 2 2
- (d) 1 1

18. Which error does Python raise for `int("abc")`?

- (a) `TypeError`
- (b) `ValueError`
- (c) `SyntaxError`
- (d) `NameError`

1. The answer is **The type belongs to the value, and a name may be rebound to a value of another type**. The value carries the type; the name is only a label. That is why `type()` reports on the value and why a name can change type mid-program.

- ✗ **A variable must be declared with its type before use.** Wrong, and this is the static-typing model of C and Java. Python needs no declaration at all: assignment creates the name.
- ✓ **The type belongs to the value, and a name may be rebound to a value of another type.** Correct. `x = 5` then `x = "five"` is legal, because the type travels with the value, not with the name.
- ✗ **Types are checked by the compiler before the program runs.** Wrong. There is no compiler pass checking types ahead of time. A type error surfaces when the offending line runs.
- ✗ **Every value in Python is a string until converted.** Wrong. This describes what `input()` returns, not what Python does generally. `5` is an `int` the moment you write it.

2. The answer is **str then 2020**. `input()` always returns a string. That is why `age + 5` raises `TypeError` and `age * 2` gives `2020`, and it is the single most common first-program bug in this course. Wrap it in `int()` or `float()`.

- ✗ **int then 40.** Wrong on both lines. `input()` never returns an `int`, and star on a string repeats rather than multiplies.
- ✓ **str then 2020.** Correct. `input()` always returns a `str`, so `type` reports `str`, and `"20" * 2` repeats the characters to give `2020`.
- ✗ **str then 40.** Wrong on the second line. `40` would need `age` to be a number; it is a string, so star repeats it.
- ✗ **int then 2020.** Wrong on the first line. The value came from `input()`, so it is a string however numeric it looks.

3. The answer is **3.5 then 3 then 1**. Slash gives a float, double slash floors, percent is the remainder. Note `7 // 2` is `3` but `-7 // 2` is `-4`, because flooring goes DOWN, not towards zero.

- ✓ **3.5 then 3 then 1.** Correct. A single slash is true division and always gives a float, even when it divides exactly. Double slash floors the result to an integer. The percent sign gives the remainder.
- ✗ **3 then 3 then 1.** Wrong. A single slash never truncates in Python 3. `7 / 2` is `3.5`, not `3`. That was Python 2 behaviour, and the change is a favourite exam point.
- ✗ **3.5 then 3.5 then 1.** Wrong. Double slash is floor division, so it produces `3`, not `3.5`.
- ✗ **3.5 then 3 then 3.5.** Wrong. The percent sign is remainder, not division, so `7 % 2` is `1`.

4. The answer is **50**. Exponent, then multiply and divide, then add and subtract. When in doubt in the exam, write the brackets in yourself: they cost nothing and remove the question.

- ✓ **50.** Correct. Exponent first: `4 ** 2` is `16`. Then multiply: `3 * 16` is `48`. Then add: `2 + 48` is `50`.
- ✗ **80.** Wrong. That is `(2 + 3) * 4 ** 2`, which adds before multiplying. Addition is last, not first.
- ✗ **400.** Wrong. That is `((2 + 3) * 4) ** 2`, which applies the exponent to everything.
- ✗ **100.** Wrong. That is `(2 + 3 * 4) ** 2`, exponent applied last.

5. The answer is **512**. Exponent is right-associative, and it is the only common operator that is. Everything else in this course groups left to right.

- ✗ **64.** Wrong. `64` is `(2 ** 3) ** 2`, grouping from the left. The exponent operator is the exception: it groups from the RIGHT.
- ✓ **512.** Correct. It reads as `2 ** (3 ** 2)`, which is `2 ** 9`, which is `512`.
- ✗ **18.** Wrong. `18` would be `2 * 3 * 3`, which is not what any part of this expression does.
- ✗ **12.** Wrong. There is no reading of this expression that gives `12`.

6. The answer is **24**. `x += 5` is exactly `x = x + 5`. Work these line by line and write the running value in the margin; the exam asks this shape often.

- ✓ **24.** Correct. 10 plus 5 is 15, minus 3 is 12, times 2 is 24. Each line uses the value the line before it left behind.
- ✗ **12.** Wrong. 12 is the value after the subtraction. The multiplication still has to happen.
- ✗ **30.** Wrong. 30 would be 15 times 2, which skips the subtraction.
- ✗ **17.** Wrong. 17 would be 10 plus 5 plus 2, which misreads two of the three operators.

7. The answer is **True then False**. Chained comparison is real Python and reads exactly as it does in mathematics. Most languages do not have it, so it is worth a mark when the paper shows one.

- ✓ **True then False.** Correct. Python allows a chained comparison, and $1 < 5 < 10$ is True. The second line asks for something and its own opposite at once, which can never be True.
- ✗ **True then True.** Wrong. The second line cannot be True: $x == 5$ and $x != 5$ contradict each other by construction.
- ✗ **False then False.** Wrong. The first line is True, because chaining is legal in Python and 5 does lie between 1 and 10.
- ✗ **TypeError.** Wrong. Nothing here is a type error; all the operands are numbers.

8. The answer is **True then False**. Precedence among the three: not, then and, then or. Read them in that order and these questions become mechanical.

- ✓ **True then False.** Correct. and binds tighter than or, so the first line is (True and False) or True, which is False or True, which is True. On the second, not binds tightest, so it is (not True) and False, which is False and False.
- ✗ **False then False.** Wrong. The first line is True, because the or rescues it: only one side of an or needs to be true.
- ✗ **True then True.** Wrong. The second line is False. not True is False, and False and anything is False.
- ✗ **False then True.** Wrong on both counts, and it has the precedence backwards.

9. The answer is **10 then 5.0 then 42**. int() truncates towards zero and round() rounds. Mixing an int with a float gives a float. Plus on two strings joins them.

- ✓ **10 then 5.0 then 42.** Correct. int("7") is 7 and int(3.9) TRUNCATES to 3, so the sum is 10. float("2.5") * 2 is 5.0, a float because one operand is. str(4) + "2" joins two strings into "42".
- ✗ **10.9 then 5.0 then 6.** Wrong. int() applied to a float does not keep the fraction. It truncates towards zero.
- ✗ **11 then 5.0 then 42.** Wrong. int(3.9) is 3, not 4. int() truncates, it does not round; round() is the function that rounds.
- ✗ **10 then 5 then 42.** Wrong. Multiplying a float by an int gives a float, so it prints 5.0, with the point.

10. The answer is **Once created, a string object cannot be changed; operations return a new string**. Immutable means the object cannot be modified in place. This is a question the paper asks in words ("Strings are Immutable. Explain and with reasons."), so have the sentence ready and an example with it.

- ✗ **A string cannot be printed twice.** Wrong. Printing does not change anything, and there is no limit on how often you may print.
- ✓ **Once created, a string object cannot be changed; operations return a new string.** Correct. s.upper() hands back a NEW string and leaves s untouched, and s[0] = "X" raises TypeError. To "change" a string you must rebind the name to a new one.
- ✗ **A string cannot be assigned to a variable more than once.** Wrong. Rebinding a name is always allowed. That changes what the name points to, not the string it used to point at.
- ✗ **A string cannot contain numbers.** Wrong. "abc123" is a perfectly ordinary string. The characters in it are irrelevant to mutability.

11. The answer is **P N YTH NOHTYP**. Index from 0, negative counts from the end, a slice excludes its stop, and [::-1] reverses. Those four facts answer most string ques-

tions on this paper.

- ✓ **P N YTH NOHTYP.** Correct. Index 0 is the first character, minus 1 is the last, a slice stops BEFORE its end index so 1:4 gives characters 1, 2 and 3, and a step of minus 1 walks the string backwards.
- ✗ **P N YTHO NOHTYP.** Wrong. The slice s[1:4] does not include index 4, so it is three characters, not four.
- ✗ **Y N YTH NOHTYP.** Wrong. s[0] is P. Python indexes from 0, not from 1.
- ✗ **P O YTH NOHTYP.** Wrong. s[-1] is the last character, N, not the second to last.

12. The answer is **HELLO WORLD then ['Hello', 'World'] then a-b-c.** strip, upper, lower, split, join, replace, find. Those seven cover nearly every string question this course has ever set.

- ✓ **HELLO WORLD then ['Hello', 'World'] then a-b-c.** Correct. strip() removes the leading and trailing spaces, upper() uppercases what is left, split() with no argument splits on whitespace into a list of words, and join() glues a list together with the string it is called on as the separator.
- ✗ **HELLO WORLD then ['Hello World'] then a-b-c.** Wrong. split() with no argument splits on EVERY run of whitespace, so two words come back as two items.
- ✗ **HELLO WORLD then ['Hello', 'World'] then a-b-c.** Wrong. strip() really does remove the outer spaces. That is its only job.
- ✗ **Hello World then ['Hello', 'World'] then abc.** Wrong on two counts: upper() does uppercase, and join() really does put the hyphen between the items.

13. The answer is **ababab then ab3.** Star repeats, plus joins, and "ab" + 3 is the error. The operator means different things for numbers and for strings, which is what operator overloading is.

- ✓ **ababab then ab3.** Correct. A string times an integer repeats it. A string plus a string joins them. Both operators are overloaded for strings.
- ✗ **ab3 then ababab.** Wrong, and it has the two operators swapped. Star repeats, plus joins.
- ✗ **abababab then ab3.** Wrong. Times 3 gives three copies, not four.
- ✗ **TypeError then ab3.** Wrong. String times int is perfectly legal. It is string PLUS int that raises TypeError.

14. The answer is **Ada scored 87.5.** f"..." substitutes; :.2f fixes the decimal places. Forgetting the f is the classic error and the output tells you at once.

- ✗ **Ada scored 87.456.** Wrong. The .1f asks for one decimal place, so the full value is not printed.
- ✓ **Ada scored 87.5.** Correct. The f before the quote turns it into a formatted string, the braces are replaced by the values, and .1f rounds to one decimal place, giving 87.5.
- ✗ **Ada scored 87.4.** Wrong. Formatting ROUNDS, it does not truncate, and 87.456 rounds up to 87.5.
- ✗ **{name} scored {score:.1f}.** Wrong. That is what you get WITHOUT the f prefix, which is a real and common slip: the braces then print literally.

15. The answer is **a b then c then d then e\f.** Backslash n newline, backslash t tab, double backslash for a real backslash. This matters for file paths, where a Windows path needs doubling or a raw string.

- ✓ **a b then c then d then e\f.** Correct. Backslash t is a tab, backslash n is a newline, and two backslashes print one literal backslash.
- ✗ **a\tb then c\nd then e\\f.** Wrong. That is what a RAW string would print, written r"a\tb". Without the r, the escapes are interpreted.
- ✗ **ab then cd then ef.** Wrong. The escapes produce real whitespace: a tab and a line break, not nothing.
- ✗ **a b then cd then ef.** Wrong. Backslash t is a tab character, which is wider than a single space, and the last line keeps its backslash.

16. The answer is **True then False**. `==` is equal in value, `is` is the very same object. Use `is` only for `None`, and use `==` for everything else.

- ✗ **True then True.** Wrong. `is` asks whether they are the SAME object, and two separately written lists are two objects however alike they look.
- ✓ **True then False.** Correct. `==` compares contents, so it is `True`. `is` compares identity, so it is `False`: `a` and `b` name two different list objects.
- ✗ **False then False.** Wrong. `==` on two lists compares them item by item, and these match.
- ✗ **False then True.** Wrong, and it has the two operators exactly backwards.

17. The answer is **2 1**. `a, b = b, a` swaps in one line because the right side is built completely before anything is assigned.

- ✗ **1 2.** Wrong. That is the value before the swap.
- ✓ **2 1.** Correct. The right-hand side is evaluated first, into the pair `(2, 1)`, and only then unpacked into `a` and `b`. So the swap needs no temporary variable.
- ✗ **2 2.** Wrong. That is what happens in a language where `a = b` runs first and destroys the old `a`. Python does not: it builds the whole right side first.
- ✗ **1 1.** Wrong, for the same reason: nothing is overwritten before it is read.

18. The answer is **ValueError**. `TypeError` is the wrong type, `ValueError` is the right type carrying an impossible value. `int("abc")` is the standard example of the second.

- ✗ **TypeError.** Wrong. `TypeError` is for the wrong KIND of thing, such as `"5" + 5`. Here the kind is right, a string, and `int()` does accept strings.
- ✓ **ValueError.** Correct. The type is acceptable but the VALUE is not: `"abc"` is a string that does not spell a number. That is exactly what `ValueError` means.
- ✗ **SyntaxError.** Wrong. The line parses perfectly. A syntax error would mean Python could not read the line at all.
- ✗ **NameError.** Wrong. `NameError` means a name was used before it was assigned. Every name here exists.

OBJECTIVE SECTION

Control Flow, Lists, Tuples, Sets and Dictionaries

18 questions on Module Three. Cover the answers, sit the whole set, then read every line of the answer block.

1. What is printed?

```
score = 75
if score >= 70:
    print("A")
elif score >= 60:
    print("B")
else:
    print("C")
```

- (a) A
- (b) B
- (c) A
- B
- (d) C

2. What is printed?

```
for i in range(1, 6, 2):  
    print(i, end=" ")
```

- (a) 1 2 3 4 5
- (b) 1 3 5
- (c) 1 3 5 7
- (d) 2 4

3. What is printed?

```
n = 0  
while True:  
    n += 1  
    if n == 3:  
        break  
print(n)
```

- (a) 0
- (b) 2
- (c) 3
- (d) the loop never ends

4. What is printed?

```
for i in range(5):  
    if i % 2 == 0:  
        continue  
    print(i, end=" ")
```

- (a) 0 2 4
- (b) 1 3
- (c) 0 1 2 3 4
- (d) 1 3 5

5. What is printed?

```
for i in range(3):  
    print(i, end=" ")  
else:  
    print("done")
```

- (a) 0 1 2 done
- (b) 0 1 2
- (c) done
- (d) 0 1 2 3 done

6. What is printed?

```
xs = [3, 1, 2]
xs.append(4)
xs.sort()
print(xs)
print(xs[0], xs[-1])
```

- (a) [1, 2, 3, 4]
1 4
- (b) [3, 1, 2, 4]
3 4
- (c) [4, 3, 2, 1]
4 1
- (d) [1, 2, 3]
1 3

7. What is printed?

```
xs = [3, 1, 2]
ys = xs.sort()
print(ys)
```

- (a) [1, 2, 3]
- (b) None
- (c) [3, 1, 2]
- (d) TypeError

8. What is printed?

```
numbers = [1, 2, 3, 4, 5]
print(numbers[-3:])
print(numbers[:2])
print(numbers[-3:] * 2)
```

- (a) [3, 4, 5]
[1, 2]
[3, 4, 5, 3, 4, 5]
- (b) [2, 3, 4]
[1, 2]
[6, 8, 10]
- (c) [3, 4, 5]
[1, 2, 3]
[6, 8, 10]
- (d) [3, 4, 5]
[1, 2]
[6, 8, 10]

9. What is printed?

```
m = [[1, 2], [3, 4, 5], [6]]  
print(m[1][2])  
print(len(m))
```

- (a) 5
3
- (b) 4
3
- (c) 5
6
- (d) 3
3

10. Which statement about tuples is TRUE?

- (a) A tuple can be changed after creation, like a list
- (b) A tuple is ordered and immutable, and can be used as a dictionary key
- (c) A tuple cannot contain more than two items
- (d) A tuple is written with square brackets

11. What is printed?

```
s = {3, 1, 2, 3, 1}  
print(len(s))  
print(sorted(s))
```

- (a) 5
[1, 1, 2, 3, 3]
- (b) 3
[1, 2, 3]
- (c) 3
[3, 1, 2]
- (d) 5
[1, 2, 3]

12. What is printed?

```
a = {1, 2, 3, 4}
b = {3, 4, 5}
print(sorted(a & b))
print(sorted(a | b))
print(sorted(a - b))
```

- (a) [3, 4]
[1, 2, 3, 4, 5]
[1, 2]
- (b) [1, 2]
[1, 2, 3, 4, 5]
[3, 4]
- (c) [3, 4]
[1, 2, 5]
[1, 2]
- (d) [3, 4]
[1, 2, 3, 3, 4, 4, 5]
[1, 2]

13. What is printed?

```
d = {"a": 1, "b": 2}
d["c"] = 3
d["a"] = 9
print(d)
print(len(d))
```

- (a) {'a': 9, 'b': 2, 'c': 3}
3
- (b) {'a': 1, 'b': 2, 'c': 3}
3
- (c) {'a': 9, 'b': 2, 'c': 3}
4
- (d) {'a': 1, 'a': 9, 'b': 2, 'c': 3}
4

14. What is printed?

```
months = {"Jan": 1, "Feb": 2}
try:
    print(months[2])
except KeyError as e:
    print("KeyError", e)
```

- (a) 2
- (b) KeyError 2
- (c) Feb
- (d) None

15. What is printed?

```
d = {"x": 1, "y": 2}
for k in d:
    print(k, end=" ")
print()
for k, v in d.items():
    print(k, v, end=" ")
```

- (a) x y
x 1 y 2
- (b) 1 2
x 1 y 2
- (c) ('x', 1) ('y', 2)
x 1 y 2
- (d) x y
1 2

16. What is printed?

```
a = [1, 2, 3]
b = a
b.append(4)
print(a)
```

- (a) [1, 2, 3]
- (b) [1, 2, 3, 4]
- (c) [4]
- (d) [[1, 2, 3], 4]

17. What is printed?

```
print("a" in "cat")
print(2 in [1, 2, 3])
print("x" in {"x": 1})
```

- (a) True
True
True
- (b) True
True
False
- (c) False
True
True
- (d) True
False
True

18. What is printed?

```
xs = [10, 20, 30, 40]
del xs[1]
print(xs)
d = {"a": 1, "b": 2}
del d["a"]
print(d)
```

- (a) [10, 30, 40]
{'b': 2}
- (b) [20, 30, 40]
{'b': 2}
- (c) [10, 20, 30]
{'a': 1}
- (d) [10, 30, 40]
{'a': 1}

ANSWERS, AND WHY EVERY OPTION IS WHAT IT IS

1. The answer is **A**. `elif` tests only if everything above it failed. Replacing `elif` with a second `if` changes the answer, and the paper has asked for exactly that comparison.

- ✓ **A**. Correct. The first condition that is true wins, its block runs, and the whole chain is then finished.
- ✗ **B**. Wrong. 75 is at or above 60, but the chain never reaches that test: it stopped at the first true one.
- ✗ **A then B**. Wrong. An `if/elif` chain runs AT MOST one block. Separate `if` statements would print both, which is exactly the trap the paper sets.
- ✗ **C**. Wrong. `else` runs only when every test above it failed.

2. The answer is **1 3 5**. `range(stop)`, `range(start, stop)`, `range(start, stop, step)`, and the `stop` is never included. That last part is the one people lose marks on.

- ✗ **1 2 3 4 5**. Wrong. That ignores the third argument, the step of 2.
- ✓ **1 3 5**. Correct. `range(start, stop, step)` begins at 1, adds 2 each time, and stops BEFORE 6. So 1, 3, 5.
- ✗ **1 3 5 7**. Wrong. 7 is past the stop value, so it is never produced.
- ✗ **2 4**. Wrong. The start is 1, not 2. `range` starts where you tell it to.

3. The answer is **3**. `break` leaves the loop at once; `continue` jumps to the next iteration. `while True` with a `break` inside is the standard way to write a menu loop.

- ✗ **0**. Wrong. `n` is incremented before anything else in the body, so it can never still be 0 at the end.
- ✗ **2**. Wrong. `break` happens AFTER the increment that made `n` equal to 3, so 3 is the value that survives.
- ✓ **3**. Correct. `n` reaches 3, the `if` fires, `break` leaves the loop immediately, and `print` sees 3.
- ✗ **the loop never ends**. Wrong. `while True` does loop forever unless something leaves it, and `break` is exactly that something.

4. The answer is **1 3**. `continue` abandons this pass and starts the next. Reading it as "skip the rest of the body" is the reliable way to trace it.

- ✗ **0 2 4**. Wrong, and it is backwards: `continue` SKIPS the even numbers, so they are the ones that never print.
- ✓ **1 3**. Correct. When `i` is even the `continue` skips the print, so only the odd values 1 and 3 appear. `range(5)` stops at 4.
- ✗ **0 1 2 3 4**. Wrong. The `continue` means not every value reaches the print.
- ✗ **1 3 5**. Wrong. `range(5)` produces 0 to 4, so 5 is never reached.

5. The answer is **0 1 2 done**. A loop `else` means "no break happened". It is Python-specific and a favourite for a "what is the output" question because it looks like an `if else` and is not.

- ✓ **0 1 2 done**. Correct. A loop `else` runs when the loop finished normally, that is without a break. Nothing breaks here, so it runs.
- ✗ **0 1 2**. Wrong. The `else` does run, because there is no break in this loop.
- ✗ **done**. Wrong. The loop body runs first, three times.
- ✗ **0 1 2 3 done**. Wrong. `range(3)` gives 0, 1 and 2. It never reaches 3.

6. The answer is **[1, 2, 3, 4] then 1 4**. Lists are ORDERED, so position has meaning, and MUTABLE, so they can be changed in place. The paper asks for both properties by name.

- ✓ **[1, 2, 3, 4] then 1 4**. Correct. `append` adds to the end, `sort` orders the list IN PLACE, and the first and last items are then 1 and 4.
- ✗ **[3, 1, 2, 4] then 3 4**. Wrong. `sort()` really does reorder the list. It changes it and returns `None`.
- ✗ **[4, 3, 2, 1] then 4 1**. Wrong. `sort()` ascends by default. Descending needs `reverse=True`.

✗ **[1, 2, 3] then 1 3.** Wrong. The 4 that append added is still there. append modifies the list itself.

7. The answer is **None**. Methods that change a list in place return None: sort, append, reverse, extend. The functions that return a NEW value are sorted() and reversed().

✗ **[1, 2, 3].** Wrong, and this is the trap. sort() changes xs and hands back nothing, so the sorted list is in xs, not in ys.

✓ **None.** Correct. sort() returns None. If you want a new sorted list without touching the original, use sorted(xs).

✗ **[3, 1, 2].** Wrong. xs itself IS sorted now; it is ys that is empty of meaning.

✗ **TypeError.** Wrong. Nothing raises here. It runs and quietly gives you None, which is what makes the bug hard to see.

8. The answer is **[3, 4, 5] then [1, 2] then [3, 4, 5, 3, 4, 5]**. This exact shape, numbers[-3:] * 2, is on a past paper. A slice excludes its stop, and star repeats rather than scales.

✓ **[3, 4, 5] then [1, 2] then [3, 4, 5, 3, 4, 5].** Correct. Minus 3 to the end is the last three items. Up to 2 stops before index 2, so two items. A list times 2 REPEATS the list, it does not multiply the numbers in it.

✗ **[2, 3, 4] then [1, 2] then [6, 8, 10].** Wrong. numbers[-3:] counts three back from the end, giving 3, 4 and 5.

✗ **[3, 4, 5] then [1, 2, 3] then [6, 8, 10].** Wrong. numbers[:2] excludes index 2, so it is two items, not three.

✗ **[3, 4, 5] then [1, 2] then [6, 8, 10].** Wrong on the last line. Star on a list repeats the list; to multiply each number you would need a loop or a comprehension.

9. The answer is **5 then 3**. Read a nested index left to right: first pick the sublist, then index into it. The paper prints a three-deep version of this, so do it one bracket at a time on paper.

✓ **5 then 3.** Correct. m[1] is the middle list, [3, 4, 5], and its index 2 is 5. len(m) counts the OUTER items, of which there are three.

✗ **4 then 3.** Wrong. Index 2 of [3, 4, 5] is 5. Index 1 would be 4.

✗ **5 then 6.** Wrong. len() of a nested list counts the sublists, not the numbers inside them.

✗ **3 then 3.** Wrong. m[1][2] reaches into the second list and takes its third item.

10. The answer is **A tuple is ordered and immutable, and can be used as a dictionary key**. List against tuple is a guaranteed question. Ordered both, mutable only the list, and only the tuple can be a dictionary key.

✗ **A tuple can be changed after creation, like a list.** Wrong. That is a list. Assigning to t[0] on a tuple raises TypeError.

✓ **A tuple is ordered and immutable, and can be used as a dictionary key.** Correct. Ordered like a list, immutable unlike one, and because it cannot change it can be hashed, so it may be a dictionary key where a list may not.

✗ **A tuple cannot contain more than two items.** Wrong. The name comes from the general mathematical term, not from two. A tuple can hold any number of items, including none.

✗ **A tuple is written with square brackets.** Wrong. Round brackets, or in fact just commas: 1, 2 is already a tuple. Square brackets make a list.

11. The answer is **3 then [1, 2, 3]**. Sets are unique and unordered. That is why the manual compares set answers as sets, never by printed order: Python randomises string hashing per run.

✗ **5 then [1, 1, 2, 3, 3].** Wrong. A set discards duplicates the moment it is built, so it never held five items.

✓ **3 then [1, 2, 3].** Correct. The duplicates collapse, leaving three items, and sorted() puts them in order for printing, which is how you show a set reliably.

✗ **3 then [3, 1, 2].** Wrong. `sorted()` returns an ordered LIST, so it prints in ascending order whatever order the set happened to be in.

✗ **5 then [1, 2, 3].** Wrong. `len()` of that set is 3, not 5, because uniqueness is enforced.

12. The answer is **[3, 4] then [1, 2, 3, 4, 5] then [1, 2]**. Ampersand intersection, bar union, minus difference, caret symmetric difference. The named methods are `intersection()`, `union()`, `difference()` and they mean the same thing.

✓ **[3, 4] then [1, 2, 3, 4, 5] then [1, 2].** Correct. The ampersand is intersection, what is in both. The bar is union, everything once. The minus is difference, what is in a and not in b.

✗ **[1, 2] then [1, 2, 3, 4, 5] then [3, 4].** Wrong, and it has intersection and difference swapped.

✗ **[3, 4] then [1, 2, 5] then [1, 2].** Wrong. Union is EVERYTHING from both, not only the items unique to one.

✗ **[3, 4] then [1, 2, 3, 3, 4, 4, 5] then [1, 2].** Wrong. A union is still a set, so nothing appears twice in it.

13. The answer is **{'a': 9, 'b': 2, 'c': 3} then 3**. Keys are unique; assigning to an existing key replaces its value. Since Python 3.7 a dictionary also keeps insertion order, which is why the printed order is predictable.

✓ **{'a': 9, 'b': 2, 'c': 3} then 3.** Correct. Assigning to a key that does not exist ADDS it; assigning to one that does REPLACES its value. Three keys remain.

✗ **{'a': 1, 'b': 2, 'c': 3} then 3.** Wrong. `d["a"] = 9` really does overwrite the 1.

✗ **{'a': 9, 'b': 2, 'c': 3} then 4.** Wrong. There are three keys, not four. Overwriting a key does not add one.

✗ **{'a': 1, 'a': 9, 'b': 2, 'c': 3} then 4.** Wrong. A dictionary cannot hold the same key twice. That is what makes it a mapping.

14. The answer is **KeyError 2**. `months[2]` asks for the KEY 2. Use `months.get(2)` to get None instead of a `KeyError`. This exact confusion is on a past paper.

✗ **2.** Wrong. Looking up 2 asks for a KEY equal to 2, and there is no such key. The values are 1 and 2, but you cannot look a dictionary up by its value.

✓ **KeyError 2.** Correct. The subscript is a KEY lookup, the key 2 is absent, `KeyError` is raised and caught, and the missing key is what the exception carries.

✗ **Feb.** Wrong. Nothing here searches the values to find the key that maps to 2. That would need a loop or a reversed dictionary.

✗ **None.** Wrong. A missing key raises rather than returning None. It is `get()` that returns None instead of raising.

15. The answer is **x y then x 1 y 2**. `for k in d` gives keys. `d.keys()`, `d.values()`, `d.items()` give what they say. Forgetting `items()` and then trying to unpack a key is a common exam slip.

✓ **x y then x 1 y 2**. Correct. Iterating a dictionary directly gives its KEYS. `items()` gives key and value pairs, which unpack into the two loop variables.

✗ **1 2 then x 1 y 2**. Wrong. The plain loop gives keys, not values. For values you would write `d.values()`.

✗ **('x', 1) ('y', 2) then x 1 y 2**. Wrong. The plain loop yields keys alone, not pairs. Pairs come from `items()`.

✗ **x y then 1 2**. Wrong on the second line: `items()` gives both parts, and both are printed.

16. The answer is **[1, 2, 3, 4]**. Assignment binds a name, it does not copy. To copy, use `a[:]` or `list(a)` or `a.copy()`. This is the most expensive misunderstanding in the whole course.

✗ **[1, 2, 3].** Wrong. `b = a` does not copy anything. It gives the same list a second name, so a change through one name is visible through the other.

✓ **[1, 2, 3, 4].** Correct. There is only ONE list here with two names, so appending through b shows up in a.

- ✗ **[4]**. Wrong. `append` adds to the existing contents; it does not replace them.
- ✗ **[[1, 2, 3], 4]**. Wrong. `append` adds an item to the list, it does not nest the old list inside a new one.

17. The answer is **True then True then True**. `in` means substring on a string, membership on a list or set, and KEY membership on a dictionary. That last one is the trap.

✓ **True then True then True**. Correct. On a string, `in` tests for a substring. On a list, for an item. On a dictionary, for a KEY.

✗ **True then True then False**. Wrong on the third line. `in` on a dictionary searches the keys, and "x" is a key here.

✗ **False then True then True**. Wrong on the first line. "a" is one of the characters of "cat", so it is found.

✗ **True then False then True**. Wrong on the second line. 2 is an item of that list.

18. The answer is **[10, 30, 40] then {'b': 2}**. `del` removes by index on a list and by key on a dictionary. `remove()` removes by VALUE on a list, and `pop()` removes and gives the item back.

✓ **[10, 30, 40] then {'b': 2}**. Correct. `del` removes by POSITION on a list, so index 1 (the 20) goes, and by KEY on a dictionary.

✗ **[20, 30, 40] then {'b': 2}**. Wrong. Index 1 is the second item. Index 0 is the 10.

✗ **[10, 20, 30] then {'a': 1}**. Wrong. `del xs[1]` takes out the second item, not the last, and `del d["a"]` takes out the "a" entry.

✗ **[10, 30, 40] then {'a': 1}**. Wrong on the dictionary: `del d["a"]` removes the key it names.

OBJECTIVE SECTION

Functions, Modules, Files and Exceptions

14 questions on Module Four. Cover the answers, sit the whole set, then read every line of the answer block.

1. What is printed?

```
def add(a, b=10):  
    return a + b  
  
print(add(5))  
print(add(5, 1))  
print(add(b=2, a=3))
```

(a) 15

6

5

(b) 15

6

6

(c) 5

6

5

(d) TypeError

6

5

2. What is printed?

```
def f(x):  
    print(x * 2)  
  
def g(x):  
    return x * 2  
  
a = f(3)  
b = g(3)  
print(a, b)
```

- (a) 6
None 6
- (b) 6
6 6
- (c) 6 6
None 6
- (d) None 6

3. What is printed?

```
x = 1  
  
def f():  
    x = 2  
    print(x)  
  
f()  
print(x)
```

- (a) 2
1
- (b) 2
2
- (c) 1
1
- (d) 1
2

4. Which of these is NOT a benefit of breaking a program into functions?

- (a) The same code can be reused instead of repeated
- (b) The program runs measurably faster because it is shorter
- (c) Each piece can be tested on its own
- (d) The program is easier to read and to maintain

5. What is printed?

```
import math  
from math import sqrt  
print(math.sqrt(16), sqrt(16))
```

- (a) 4.0 4.0
- (b) 4 4
- (c) 4.0 4
- (d) NameError

6. A file is opened with `open("data.txt", "w")` when `data.txt` already exists and contains text. What happens?
- (a) The new text is added after the existing text
 - (b) The existing contents are erased
 - (c) Python raises `FileExistsError`
 - (d) The file is opened read-only
7. Why is `with open(...)` as `f` preferred to `f = open(...)`?
- (a) It is faster to type
 - (b) It closes the file automatically, even if an error is raised inside the block
 - (c) It is the only way to read a file line by line
 - (d) It opens the file in a special faster mode
8. A text file holds three lines. Which call gives a LIST of those three lines?
- (a) `f.read()`
 - (b) `f.readline()`
 - (c) `f.readlines()`
 - (d) `f.line()`
9. What is printed?

```
try:
    x = 1 / 0
except ZeroDivisionError:
    print("caught")
finally:
    print("always")
```

- (a) caught
always
 - (b) always
 - (c) caught
 - (d) the program crashes
10. Which exception does this raise?
- ```
xs = [1, 2, 3]
print(xs[5])
```
- (a) `KeyError`
  - (b) `IndexError`
  - (c) `ValueError`
  - (d) `TypeError`
11. Why is `except:` on its own considered bad practice?
- (a) It is slower than naming the exception
  - (b) It catches everything, including errors you did not anticipate, and hides real bugs
  - (c) It is not valid Python
  - (d) It can only be used once per program

12. What is printed?

```
def check(age):
 if age < 0:
 raise ValueError("negative age")
 return age

try:
 check(-1)
except ValueError as e:
 print(e)
```

- (a) negative age
- (b) ValueError
- (c) -1
- (d) nothing is printed

13. What is a docstring?

- (a) A comment starting with a hash inside a function
- (b) A string literal as the first statement of a function, module or class, describing what it does
- (c) A special variable holding the function name
- (d) A file that documents the whole program

14. What is printed?

```
def fact(n):
 if n <= 1:
 return 1
 return n * fact(n - 1)

print(fact(5))
```

- (a) 15
- (b) 120
- (c) 5
- (d) RecursionError

1. The answer is **15 then 6 then 5**. A parameter with a default may be omitted. Keyword arguments name their target, so order stops mattering. Defaults must come after non-defaults in the definition.

- ✓ **15 then 6 then 5**. Correct. The default fills in for b when only one argument is given. Positional arguments fill left to right. Keyword arguments may be given in any order because they name their parameter.
- ✗ **15 then 6 then 6**. Wrong on the last line:  $3 + 2$  is 5, not 6.
- ✗ **5 then 6 then 5**. Wrong on the first line. b defaults to 10, so `add(5)` is 15.
- ✗ **TypeError then 6 then 5**. Wrong. `add(5)` is legal precisely BECAUSE b has a default.

2. The answer is **6 then None 6**. `print` SHOWS a value, `return` HANDS IT BACK. A function that only prints returns None, and using that None later is a standard exam bug.

- ✓ **6 then None 6**. Correct. f prints 6 and returns nothing, so a is None. g returns 6 silently, so b is 6. The last line then prints None 6.
- ✗ **6 then 6 6**. Wrong. a is None. A function with no return statement gives back None.
- ✗ **6 6 then None 6**. Wrong. f prints once, when it is called; the value it gives back is still None.
- ✗ **None 6**. Wrong. The 6 printed inside f appears first, on its own line.

3. The answer is **2 then 1**. Assigning to a name inside a function makes it local. To change a global you must say `global x`, and this pair of prints is exactly how the paper tests it.

- ✓ **2 then 1**. Correct. The assignment inside f creates a LOCAL x. The global x is untouched, so it is still 1 afterwards.
- ✗ **2 then 2**. Wrong. Assigning inside a function does not change the global unless the function declares global x.
- ✗ **1 then 1**. Wrong. Inside f, the local x shadows the global one, so 2 prints first.
- ✗ **1 then 2**. Wrong, and it has the two the wrong way round.

4. The answer is **The program runs measurably faster because it is shorter**. Every benefit of modularising is a benefit to the person reading or changing the code. Speed is not one, and a question offering it is offering the distractor.

- ✗ **The same code can be reused instead of repeated**. Wrong choice, because reuse IS a benefit, and usually the first one listed.
- ✓ **The program runs measurably faster because it is shorter**. Correct. Calling a function costs a little time rather than saving it. The benefits of modularising are all about the PROGRAMMER: reuse, testing, readability, maintainability, teamwork. Speed is not among them.
- ✗ **Each piece can be tested on its own**. Wrong choice, because testing a piece in isolation IS a benefit.
- ✗ **The program is easier to read and to maintain**. Wrong choice, because readability and maintenance ARE benefits, and the paper lists them.

5. The answer is **4.0 4.0**. `import math` needs the `math.` prefix; `from math import sqrt` puts the name in directly. Both are examinable and the difference is which name you must then write.

- ✓ **4.0 4.0**. Correct. Both forms reach the same function. `sqrt` always returns a float, so both print 4.0.
- ✗ **4 4**. Wrong. `math.sqrt` returns a float even for a perfect square, so the point and zero are printed.
- ✗ **4.0 4**. Wrong. Both calls are the same function and give the same value.
- ✗ **NameError**. Wrong. Both imports are valid, and each makes its name available in its own way.

6. The answer is **The existing contents are erased**. "r" read, "w" write and TRUNCATE, "a" append, "x" create only if absent. Choosing "w" where you meant "a" destroys the file, and the paper asks about this.

- ✗ **The new text is added after the existing text.** Wrong, and this is the mode people mean when they choose "w" by mistake. Appending is "a".
- ✓ **The existing contents are erased.** Correct. "w" truncates the file to nothing the moment it is opened, before you write a single character. Data is lost even if the program then crashes.
- ✗ **Python raises FileExistsError.** Wrong. "w" is perfectly happy for the file to exist. It just empties it. The mode that refuses to overwrite is "x".
- ✗ **The file is opened read-only.** Wrong. "w" is write-only. Reading is "r", which is also the default.

7. The answer is **It closes the file automatically, even if an error is raised inside the block**. with guarantees the close. It is the same guarantee finally gives, written more briefly, and saying that in the exam earns the mark.

- ✗ **It is faster to type.** Wrong. Brevity is not the reason, and it is barely shorter anyway.
- ✓ **It closes the file automatically, even if an error is raised inside the block.** Correct. The with block guarantees f.close() runs on the way out, whether the block finished normally or raised. Without it, an exception can leave the file open and its buffer unwritten.
- ✗ **It is the only way to read a file line by line.** Wrong. A plain open() can be iterated line by line just as well. The difference is only about closing.
- ✗ **It opens the file in a special faster mode.** Wrong. The mode is whatever you pass to open. with changes nothing about how the file is opened.

8. The answer is **f.readlines()**. read whole, readline one, readlines list. Note the new-line stays on each item, so print() of one adds a blank line unless you strip.

- ✗ **f.read()**. Wrong. read() gives the WHOLE file as one string, newlines and all.
- ✗ **f.readline()**. Wrong. readline() gives ONE line, the next one, as a string.
- ✓ **f.readlines()**. Correct. readlines() gives a list with one string per line, each still carrying its trailing newline unless you strip it.
- ✗ **f.line()**. Wrong. There is no such method.

9. The answer is **caught then always**. try, except by exception type, else if nothing was raised, finally always. finally is where you put the cleanup that must happen either way.

- ✓ **caught then always.** Correct. The division raises, the matching except handles it, and finally runs afterwards as it always does.
- ✗ **always.** Wrong. The except block ran too, and its print came first.
- ✗ **caught.** Wrong. finally runs whether or not anything was raised. That is its whole point.
- ✗ **the program crashes.** Wrong. The exception was caught, so nothing propagates and nothing crashes.

10. The answer is **IndexError**. IndexError for a sequence, KeyError for a mapping. Naming the right exception is worth marks on its own in this paper.

- ✗ **KeyError.** Wrong. KeyError is for a missing DICTIONARY key. A list is indexed by position, not by key.
- ✓ **IndexError.** Correct. The index is outside the range of the list, which is exactly what IndexError means.
- ✗ **ValueError.** Wrong. ValueError is for a value of the right type that cannot be used, such as int("abc").
- ✗ **TypeError.** Wrong. TypeError would be xs["a"], where the index is the wrong KIND of thing entirely.

11. The answer is **It catches everything, including errors you did not anticipate, and hides real bugs**. Catch the exception you expect and no more. A bare except turns a loud failure into a silent wrong answer.

- ✗ **It is slower than naming the exception.** Wrong. Speed is not the issue and the difference is immeasurable.
- ✓ **It catches everything, including errors you did not anticipate, and hides real bugs.** Correct. A bare except swallows every exception, including a typo raising NameError, so the program keeps going in a broken state and the real fault is invisible.
- ✗ **It is not valid Python.** Wrong. It is valid, which is exactly why it is dangerous.
- ✗ **It can only be used once per program.** Wrong. There is no such limit.

**12.** The answer is **negative age**. raise makes your own error, and validating input this way is what the paper means by defensive programming.

- ✓ **negative age.** Correct. raise creates the exception with that message, the except catches it and binds it to e, and printing an exception prints its message.
- ✗ **ValueError.** Wrong. Printing the exception object shows its MESSAGE, not the name of its class. For the class name you would print type(e).\_\_name\_\_.
- ✗ **-1.** Wrong. The argument is never returned, because the raise happens first.
- ✗ **nothing is printed.** Wrong. The except block does run and does print.

**13.** The answer is **A string literal as the first statement of a function, module or class, describing what it does**. First statement, a string, becomes \_\_doc\_\_. It is the "self-documenting" part of the modularisation answer.

- ✗ **A comment starting with a hash inside a function.** Wrong. A hash comment is stripped and cannot be read back at run time. A docstring survives into the object.
- ✓ **A string literal as the first statement of a function, module or class, describing what it does.** Correct. It sits directly under the def line, usually in triple quotes, and becomes the object's \_\_doc\_\_, which is what help() shows you.
- ✗ **A special variable holding the function name.** Wrong. That is \_\_name\_\_, a different attribute entirely.
- ✗ **A file that documents the whole program.** Wrong. A docstring lives inside the code, not beside it.

**14.** The answer is **120**. Every recursion needs a base case and a step that moves towards it. Missing the base case gives RecursionError, and the paper has asked for exactly that fault to be identified.

- ✗ **15.** Wrong. 15 is 1+2+3+4+5, the SUM. This function multiplies.
- ✓ **120.** Correct. 5 \* 4 \* 3 \* 2 \* 1 is 120. The base case at n <= 1 stops the recursion.
- ✗ **5.** Wrong. The function does not simply return its argument; it multiplies down to the base case.
- ✗ **RecursionError.** Wrong. There IS a base case, so the recursion terminates. Without one, RecursionError would be right.

## OBJECTIVE SECTION

# Databases and GUI Development

*12 questions on Module Five. Cover the answers, sit the whole set, then read every line of the answer block.*

1. What is the correct order for working with a database in Python?
  - (a) connect, cursor, execute, commit, close
  - (b) cursor, connect, commit, execute, close
  - (c) connect, execute, cursor, close, commit
  - (d) open, read, write, close

2. A program inserts three rows and then closes the connection without calling commit. What is in the table?
  - (a) All three rows
  - (b) The first row only
  - (c) No rows
  - (d) An error is raised
3. Which SQL statement retrieves only the students whose score is above 50?
  - (a) `SELECT * FROM students WHERE score > 50`
  - (b) `SELECT * FROM students IF score > 50`
  - (c) `GET * FROM students WHERE score > 50`
  - (d) `SELECT students WHERE score > 50`
4. Why is `cur.execute("INSERT INTO t VALUES (?, ?)", (a, b))` preferred to building the SQL string with the values in it?
  - (a) It runs faster on small tables
  - (b) It keeps the values separate from the SQL, so a value cannot be read as SQL
  - (c) It is the only form `sqlite3` accepts
  - (d) It commits automatically
5. What does `cur.executemany()` do that `execute()` does not?
  - (a) It runs several DIFFERENT statements in one call
  - (b) It runs the same statement once for each item in a sequence of parameter tuples
  - (c) It commits after every row
  - (d) It returns all the rows of a table
6. After executing a `SELECT` that matches five rows, what does `fetchone()` return?
  - (a) All five rows as a list
  - (b) The first row, as a tuple
  - (c) The number 5
  - (d) None
7. Which SQL function returns the number of rows matching a query?
  - (a) `SUM()`
  - (b) `COUNT()`
  - (c) `TOTAL()`
  - (d) `LEN()`
8. What does `UPDATE students SET score = 0` do, with no `WHERE` clause?
  - (a) Nothing: a `WHERE` clause is required
  - (b) It sets score to 0 for EVERY row in the table
  - (c) It sets score to 0 for the first row only
  - (d) It deletes the score column
9. In `tkinter`, what does the `mainloop()` call do?
  - (a) It draws the window once and returns
  - (b) It starts the event loop, which waits for and dispatches user actions
  - (c) It creates the main window object
  - (d) It closes the application

10. Which tkinter widget is used to collect a single line of typed text?
- (a) Label
  - (b) Entry
  - (c) Button
  - (d) Canvas
11. In `Button(root, text="Go", command=run)`, when is `run` called?
- (a) Immediately, as the button is created
  - (b) When the button is clicked
  - (c) When `mainloop()` starts
  - (d) Never, unless `bind()` is also used
12. Which of these is NOT a tkinter geometry manager?
- (a) pack
  - (b) grid
  - (c) place
  - (d) align

1. The answer is **connect, cursor, execute, commit, close**. connect, cursor, execute, commit, close. Forget the commit and the rows vanish when the program ends, which is the classic lost mark.

- ✓ **connect, cursor, execute, commit, close**. Correct. Connect to the database, get a cursor from the connection, execute SQL through the cursor, commit so the change is saved, then close the connection.
- ✗ **cursor, connect, commit, execute, close**. Wrong. A cursor is obtained FROM a connection, so the connection must exist first.
- ✗ **connect, execute, cursor, close, commit**. Wrong. You cannot execute before you have a cursor, and committing after closing is too late.
- ✗ **open, read, write, close**. Wrong. That is the file-handling pattern, not the database one.

2. The answer is **No rows**. Commit is what makes a change permanent. No commit, no data, and no error message to tell you.

- ✗ **All three rows**. Wrong. Uncommitted work is not saved. It lived only in the transaction.
- ✗ **The first row only**. Wrong. There is nothing special about the first row; all three are in the same uncommitted transaction.
- ✓ **No rows**. Correct. Without a commit the transaction is rolled back when the connection closes, so the table is unchanged.
- ✗ **An error is raised**. Wrong. Nothing raises. The program appears to work, which is what makes this bug expensive.

3. The answer is **SELECT \* FROM students WHERE score > 50**. SELECT columns FROM table WHERE condition. That one line is most of what this course asks of SQL.

- ✓ **SELECT \* FROM students WHERE score > 50**. Correct. SELECT names the columns, FROM names the table, WHERE filters the rows.
- ✗ **SELECT \* FROM students IF score > 50**. Wrong. SQL filters with WHERE. IF is not part of a SELECT statement.
- ✗ **GET \* FROM students WHERE score > 50**. Wrong. There is no GET in SQL. Retrieval is SELECT.
- ✗ **SELECT students WHERE score > 50**. Wrong. The table must be named after FROM, and the columns before it.

4. The answer is **It keeps the values separate from the SQL, so a value cannot be read as SQL**. The question mark placeholder separates data from code. It is the standard answer to "how do you prevent SQL injection".

- ✗ **It runs faster on small tables**. Wrong. Speed is not the reason, though prepared statements can help on repeated inserts.
- ✓ **It keeps the values separate from the SQL, so a value cannot be read as SQL**. Correct. The placeholder keeps data as data. Pasting a value into the text of a statement lets a value containing a quote change the statement, which is SQL injection.
- ✗ **It is the only form sqlite3 accepts**. Wrong. String building is accepted. It is accepted and unsafe, which is the point.
- ✗ **It commits automatically**. Wrong. You still have to commit yourself.

5. The answer is **It runs the same statement once for each item in a sequence of parameter tuples**. One statement, many parameter sets. It is the tidy way to insert a list of records and this manual teaches it because the paper reached for it.

- ✗ **It runs several DIFFERENT statements in one call**. Wrong. The statement is the same every time. Only the parameters change.

- ✓ **It runs the same statement once for each item in a sequence of parameter tuples.**

Correct. You give it one statement and a list of parameter tuples, and it runs the statement once per tuple, which is how a batch insert is done.

- ✗ **It commits after every row.** Wrong. It does not commit at all. You still call commit yourself.
- ✗ **It returns all the rows of a table.** Wrong. Retrieving rows is fetchall(), after a SELECT.

**6.** The answer is **The first row, as a tuple.** fetchone next row or None, fetchall the rest as a list of tuples. A row is a tuple, so index it by position.

- ✗ **All five rows as a list.** Wrong. That is fetchall(). fetchone gets one.
- ✓ **The first row, as a tuple.** Correct. fetchone() hands back the next row as a tuple, and returns None once the rows run out.
- ✗ **The number 5.** Wrong. Counting rows is done in SQL, with COUNT(\*).
- ✗ **None.** Wrong. None comes back only when there are no rows left, and here there are five.

**7.** The answer is **COUNT()**. COUNT how many, SUM the total, AVG the mean, MAX and MIN the extremes. The paper has asked for all five.

- ✗ **SUM().** Wrong. SUM() adds up the values in a column. It answers how much, not how many.
- ✓ **COUNT().** Correct. COUNT(\*) counts rows; COUNT(column) counts the rows where that column is not null.
- ✗ **TOTAL().** Wrong. There is no standard TOTAL() in this course.
- ✗ **LEN().** Wrong. LEN() is not SQL. Python has len(); SQL has COUNT().

**8.** The answer is **It sets score to 0 for EVERY row in the table.** No WHERE means every row. Write the WHERE first and the SET second if that helps you never forget it.

- ✗ **Nothing: a WHERE clause is required.** Wrong. WHERE is optional, and that is precisely the danger.
- ✓ **It sets score to 0 for EVERY row in the table.** Correct. With no WHERE, the statement applies to the whole table. The same is true of DELETE FROM students, which empties it.
- ✗ **It sets score to 0 for the first row only.** Wrong. SQL is set-based: it acts on every row that matches, and with no WHERE every row matches.
- ✗ **It deletes the score column.** Wrong. Removing a column is ALTER TABLE. UPDATE changes values.

**9.** The answer is **It starts the event loop, which waits for and dispatches user actions.** Tk() makes the window, widgets are created and placed, mainloop() runs the event loop. Without the last one nothing stays on screen.

- ✗ **It draws the window once and returns.** Wrong. Drawing once and returning would make the window appear and vanish immediately, which is exactly what happens if you forget mainloop.
- ✓ **It starts the event loop, which waits for and dispatches user actions.** Correct. mainloop() keeps the program running, watching for clicks and keystrokes and calling the handlers bound to them, until the window is closed.
- ✗ **It creates the main window object.** Wrong. Creating the window is Tk(). mainloop is what makes it live.
- ✗ **It closes the application.** Wrong. It is what keeps the application OPEN.

**10.** The answer is **Entry.** Label shows, Entry takes one line, Text takes many, Button acts. Read what was typed with entry.get().

- ✗ **Label.** Wrong. A Label DISPLAYS text. The user cannot type into it.
- ✓ **Entry.** Correct. Entry is the one-line text box, and its get() method reads what was typed.
- ✗ **Button.** Wrong. A Button is clicked; it does not accept typing.
- ✗ **Canvas.** Wrong. A Canvas is a drawing surface for shapes and images.

**11.** The answer is **When the button is clicked.** Pass the function, not a call to it. command=run attaches it; command=run() runs it once immediately and attaches its result.

- ✗ **Immediately, as the button is created.** Wrong, and this is the trap. It WOULD be immediate if you wrote `command=run()`, with brackets: that calls the function and hands its return value to `command`.
- ✓ **When the button is clicked.** Correct. `command` is given the function ITSELF, without brackets, and `tkinter` calls it later, each time the button is pressed.
- ✗ **When `mainloop()` starts.** Wrong. `mainloop` starts the waiting; it does not fire the handlers by itself.
- ✗ **Never, unless `bind()` is also used.** Wrong. `command=` is a complete way to attach a handler to a button. `bind()` is for other events.

**12.** The answer is **align**. `pack`, `grid`, `place`. Three of them, and mixing `pack` with `grid` in one container makes the window hang while they argue.

- ✗ **pack.** Wrong choice, because `pack` IS one: it stacks widgets in the space available.
- ✗ **grid.** Wrong choice, because `grid` IS one: it places widgets by row and column.
- ✗ **place.** Wrong choice, because `place` IS one: it positions by exact coordinates.
- ✓ **align.** Correct. There is no align manager. The three are `pack`, `grid` and `place`, and you should not mix `pack` and `grid` in the same container.

## MOCK EXAMINATION ONE

# Sit This Paper Under Exam Conditions

*Same shape as the real thing, question for question: six questions, attempt any four, five parts each at three and a half marks. Different problems. Three hours, no computer, no notes. The answers start after the paper, and looking at them early is the only way to waste this.*

|    |                     |      |                                                                                          |
|----|---------------------|------|------------------------------------------------------------------------------------------|
| Q1 | (a) (b) (c) (d) (e) | 17.5 |                                                                                          |
| Q2 | (a) (b) (c) (d) (e) | 17.5 |                                                                                          |
| Q3 | (a) (b) (c) (d) (e) | 17.5 |                                                                                          |
| Q4 | (a) (b) (c) (d) (e) | 17.5 | = 70  |

# Question Paper

## ● MUST MEMORISE

## THE RUBRIC ITSELF

|             |                                                                                          |
|-------------|------------------------------------------------------------------------------------------|
| Course      | CSC241, Python Programming Language I                                                    |
| Credit unit | 3                                                                                        |
| Time        | 3 hours                                                                                  |
| Instruction | <b>Answer Four (4) questions only</b>                                                    |
| Marks       | Six questions offered. Each is five parts at 3.5 marks, so 17.5 a question, 70 in total. |

**Read the rubric before you write anything.** Four questions, not six. Answering all six wastes an hour and earns nothing extra: only four are marked. Spend the first five minutes choosing your four, and pick them by part (d) and (e), which is where 7 of the 17.5 marks sit.

## Question One

### QUESTION ONE

17.5 MARKS

(a) Explain the differences among the three main types of control structures in Python. Give one example of each. **[3.5mks]**

(b) The following code is intended to print all odd numbers from 1 to 9, skipping the number 5. Identify and correct the error(s): **[3.5mks]**

```
n = 1
while n <= 9:
 if n % 2 != 0:
 if n == 5:
 continue
 print(n)
 n += 1
```

(c) Explain what the following code snippet does. What is the output? **[3.5mks]**

```
for i in range(1, 6):
 if i % 4 == 0:
 break
 print(i)
else:
 print("All done")
```

(d) You are writing the ticketing system for a cinema. The price depends on the customer's age. Your program should use decision making structures to implement the logic. **[3.5mks]**

#### Program requirements:

1. Prompt the user to enter an integer for the age.
2. Check that the age is positive.
3. If the age is under 13, print "Child - N500".
4. Else if the age is under 18, print "Teen - N800".
5. Else if the age is 60 or over, print "Senior - N600".
6. Else, print "Adult - N1200".
7. If the age is not positive, print "Please enter a valid age".

(e) Develop a new ticketing system as an extension of the one you did in (d) above. This updated version should allow the user to enter multiple ages in one input, separated by spaces. **[3.5mks]**

#### Program requirements:

1. Prompt the user to enter multiple ages separated by spaces.
2. Split the input and convert each item to an integer.
3. For each age entered, check it is positive, apply the pricing logic, and display the category.
4. After all ages, display the total number of tickets priced.

## Question Two

### QUESTION TWO

17.5 MARKS

(a) List and briefly describe four fundamental built-in data structures in Python. For each one, mention a real-world example where it can be appropriately used. **[3.5mks]**

(b) The following code is supposed to store the courses a student is taking in a list, remove any duplicates, and print the number of distinct courses. Identify and fix all the issues: **[3.5mks]**

```
courses = ["CSC241", "MAT121", "CSC241", "PHY121", "MAT121"]
courses = list(courses)
distinct = courses
print("Distinct courses:", len(distinct))
```

(c) What will be the output of the following code? Explain what happens at each stage of execution: **[3.5mks]**

```
student = {
 "name": "Bola",
 "level": 200,
 "courses": ["CSC241", "MAT121"]
}
student["courses"].append("PHY121")
student["level"] = 300
print(student["courses"])
print(len(student))
```

(d) You are building a stock checking tool for a shop. The program should allow the user to enter the quantities of four products, store them in a list, sort them in ascending order, and display the sorted quantities, the highest and the lowest. **[3.5mks]**

#### Program requirements:

1. Initialize an empty list to hold the quantities.
2. Use a loop to input the quantities of four products (as integers).
3. Store each quantity in the list.
4. Sort the list in ascending order.
5. Print the sorted list.
6. Print the maximum and minimum using appropriate built-in functions.

(e) Develop a new stock checking tool as an extension of (d). This version should allow any number of quantities (not just four), separated by spaces. **[3.5mks]**

#### Program requirements:

1. Prompt the user to enter all quantities, separated by spaces.
2. Convert the input to a list of integers.
3. Sort the quantities in descending order.
4. Display the sorted list, the highest and lowest, the average, and the number of products that need reordering. **A product needs reordering if its quantity is below 10.**

## Question Three

### QUESTION THREE

17.5 MARKS

**(a)** What is a function in Python? Why are functions important in programming? Differentiate between built-in functions and user-defined functions. Explain the two types of user-defined functions in Python. **[3.5mks]**

**(b)** The following code is meant to define a function that returns the cube of a number and then calls it. However, it does not work as expected. Identify and fix the errors: **[3.5mks]**

```
def cube(n)
 return n ** 3
print(Cube(3))
```

**(c)** Consider the following Python code. What will be the output? Explain how the function works, especially with regard to default parameters. **[3.5mks]**

```
def total(price, tax=0.1):
 return price + price * tax

print(total(100))
print(total(100, 0.2))
```

**(d)** You have been asked to build a results tool for a department. The program should define a function `grade(score)` that returns "A" for 70 and above, "B" for 60 to 69, "C" for 50 to 59, and "F" below 50. The program should prompt for the names and scores of three students, call the function for each, and display the result in the format: **[3.5mks]**

```
Student: Ada | Score: 72 | Grade: A
Student: Bola | Score: 45 | Grade: F
...
```

#### Program requirements:

1. Define `grade(score)` returning the letter as described.
2. In the main program, use a loop to collect each student's name and score.
3. For each student, call the function and determine the grade.
4. Display the result in a clean, formatted output.

**(e)** Develop a new results tool as an extension of (d). This version should allow any number of students (not just three), all in one session. **[3.5mks]**

#### Program requirements:

1. Define `grade(score)` as before.
2. Use a loop to repeatedly accept student names and scores.
3. Stop the loop when the user enters 'done' as the name.
4. For each student, call the function and display the grade.
5. After the loop, display summary statistics: how many students, and how many passed. **A pass is 50 and above.**

## Question Four

### QUESTION FOUR

17.5 MARKS

**(a)** What is a string in Python? List and briefly explain four common string methods used in Python. Provide an example for each method. **[3.5mks]**

**(b)** The following code is intended to extract the surname from a full name and convert it to lowercase. However, it contains errors. Identify and fix them: **[3.5mks]**

```
full_name = "Grace Hopper"
surname = full_name[full_name.index():]
print("Surname:", surname.lowercase())
```

**(c)** What will be the output of the following code? Explain the result of each line: **[3.5mks]**

```
tag = " CSC241 Python "
print(tag.strip())
print(tag.upper().replace("PYTHON", "CODE"))
print(len(tag.strip()))
```

**(d)** Write a Python program for a tool that processes a list of student registration numbers and prints the entry year. The registration numbers have the form YY/DEPT/NNN , and the entry year is the part before the first slash. **[3.5mks]**

#### Program Requirements:

1. Create a list containing at least five registration numbers.
2. Loop through the list.
3. For each one, split the string at the slash and extract the year.
4. Print the result in a clean format, such as:

```
Reg: 24/CSC/001 | Year: 24
```

**(e)** Develop a new text tool as an extension of (d). This version should help a user analyse multiple sentences entered in a single paragraph. **[3.5mks]**

#### Program Requirements:

1. Prompt the user to enter a paragraph of text.
2. Split the paragraph into sentences using "." as the delimiter.
3. For each sentence: remove leading and trailing whitespace, skip empty sentences, count the number of words, convert to lowercase, and count the number of consonants (letters that are not a, e, i, o, u).
4. Display the results for that sentence.

## Question Five

### QUESTION FIVE

17.5 MARKS

(a) What is file handling in Python? List and explain four file access modes used in the `open()` function. Also mention the difference between reading from a file and appending to a file. **[3.5mks]**

(b) The following code is intended to write a heading to a file named `report.txt`, but it contains errors. Identify and fix them: **[3.5mks]**

```
f = open("report.txt", "w")
f.write("Weekly Report")
f.close
```

(c) What will be the result of executing the following code? Explain what each line does. **[3.5mks]**

```
with open("log.txt", "a") as f:
 f.write("Entry recorded.\n")

print("Log updated.")
```

(d) Develop a Python program for a library book log. The program should allow the librarian to enter the titles of books borrowed today and save the list to a file called `borrowed.txt`. **[3.5mks]**

#### Program Requirement:

1. Ask the librarian how many books were borrowed.
2. Use a loop to input each title.
3. Open `borrowed.txt` in write mode.
4. Write each title to the file on a new line.
5. After saving, print "Log saved successfully."

(e) Develop a new library record system as an extension of (d). This version should allow the user to enter multiple records (title and days overdue) until 'done' is entered as the title. Each entry should be stored in a text file called `fines.txt` in the format:

**[3.5mks]**

```
Things Fall Apart, 3
Purple Hibiscus, 0
```

#### Program Requirement:

1. Open `fines.txt` in write mode.
2. Use a loop to collect titles and days overdue until 'done' is entered.
3. Write each entry in the format: Title, Days.
4. Close the file, then reopen it in read mode.
5. For each line: split the title and days, print the record, and total the fine. **The fine is N50 per day overdue.**
6. Display the total fine owed.

## Question Six

### QUESTION SIX

17.5 MARKS

(a) i. What is a database, and why is it used in programming? ii. What role does the `sqlite3` module play in Python? iii. Mention three common SQL commands used when working with databases and their functions. **[3.5mks]**

(b) The following code is meant to create a database table named `books`, but it contains errors. Identify and fix the issues: **[3.5mks]**

```
import sqlite3
conn = sqlite3.connect("library.db")
cursor = conn.cursor()
cursor.execute("CREATE TABLE books (id INTEGER PRIMARY KEY, title TEXT)")
conn.commit()
conn.close()
```

(c) i. Examine the following code and explain what it does. ii. What will happen when this code runs? iii. What assumptions must be true for it to execute successfully? **[3.5mks]**

```
import sqlite3
conn = sqlite3.connect("library.db")
cursor = conn.cursor()
cursor.execute("INSERT INTO books (title) VALUES ('Things Fall Apart')")
conn.commit()
conn.close()
```

(d) Develop a simple Python-based student registry using `sqlite3`. The program should create a table `students` with fields: `id`, `name` and `matric`. It prompts the user to enter the name and matric number of three students and inserts each into the database. **[3.5mks]**

#### Program requirements:

1. Import `sqlite3` and connect to a database named `registry.db`.
2. Create a table `students` if it does not already exist.
3. Use a loop to collect name and matric number for three students.
4. Insert each student into the table.
5. Commit the changes and close the connection.

(e) Develop a new student registry as an extension of (d). This version should allow the user to enter multiple students until they type 'done' as the name, store the records in `registry.db`, and after all entries are added, fetch and display all students stored in the table. **[3.5mks]**

The answers follow. **Do not read them until you have written yours.**

Reading a worked answer produces a strong, false feeling of competence: everything looks obvious once somebody else has done it. The only way to find out whether you can write a sentinel loop from nothing is to sit with a blank page and fail to, which is unpleasant and is the entire point. Three hours now is worth more than three days of rereading.

# Answers

Every output below was produced by running the code, not by reading it. Where your answer differs in wording but not in substance, you are right.

## Question One

### ANSWER

### QUESTION ONE (A)

#### THREE CONTROL STRUCTURES, ONE EXAMPLE EACH

- ▶ **Sequence.** Statements run one after another, top to bottom, with no choices.  
Example: `a = 2` then `b = a + 1`.
- ▶ **Selection** (decision). A block runs only if a condition is true. Written with `if`, `elif`, `else`. Example: `if n % 2 == 0: print("even")`.
- ▶ **Iteration** (repetition). A block runs many times. Written with `while` or `for`.  
Example: `for i in range(3): print(i)`.

The question says "give one example of each", so a sentence without code will not score full marks. Write the three lines.

### ANSWER

### QUESTION ONE (B)

#### THE ERROR

The code **does not just print the wrong numbers. It never stops.** It prints 1, then 3, and then hangs forever.

When `n` reaches 5, `continue` jumps straight back to the top of the `while`, **skipping** `n += 1` on the way. So `n` stays 5 for ever and the condition `n <= 9` never turns false.

#### THE FIX

```
n = 1
while n <= 9:
 if n % 2 != 0 and n != 5:
 print(n)
 n += 1
```

```
1
3
7
9
```

Removing `continue` removes the bug at its root. If you keep `continue`, `n += 1` must happen before it, or the loop hangs again.

**Say the word "infinite" in your answer.** A candidate who writes "it prints 5 by mistake" has not spotted the actual error and will not get the marks.

## OUTPUT

```
1
2
3
```

## EXPLANATION

The loop counts 1, 2, 3, 4, 5. It prints each until `i` is 4, where `4 % 4 == 0` is true and `break` leaves the loop at once, before the `print`. So 4 and 5 never print.

**"All done" never prints.** A loop's `else` runs only when the loop finishes *normally*. A `break` skips it. Read `else` here as "if no break happened".

This is the marked difference between a good answer and a lucky one. Most candidates add "All done" to the output.

```
age = int(input("Enter age: "))

if age > 0:
 if age < 13:
 print("Child - N500")
 elif age < 18:
 print("Teen - N800")
 elif age >= 60:
 print("Senior - N600")
 else:
 print("Adult - N1200")
else:
 print("Please enter a valid age")
```

Run with 8, 15, 70, 30 and 0:

```
Child - N500
Teen - N800
Senior - N600
Adult - N1200
Please enter a valid age
```

## WHY THE ORDER IS WHAT IT IS

**The tests must run narrowest first.** `age < 13` before `age < 18`, because a 10 year old satisfies both and the first true test wins. Swap them and no child could ever be priced.

`elif age >= 60` is reached only by people 18 or over, because anything younger was caught above. That is why it needs no upper bound.

```
line = input("Enter ages separated by spaces: ")
count = 0

for part in line.split():
 age = int(part)
 if age > 0:
 if age < 13:
 print(age, "Child - N500")
 elif age < 18:
 print(age, "Teen - N800")
 elif age >= 60:
 print(age, "Senior - N600")
 else:
 print(age, "Adult - N1200")
 count += 1
 else:
 print(age, "Please enter a valid age")

print("Tickets priced:", count)
```

Entering 8 15 70 30 :

```
Enter ages separated by spaces: 8 15 70 30
8 Child - N500
15 Teen - N800
70 Senior - N600
30 Adult - N1200
Tickets priced: 4
```

#### WHAT EARNED THE MARKS

The decision logic is **identical** to part (d). Only two things changed: `split()` replaced the single `input`, and a `for` loop wrapped the whole thing. That is what part (e) is testing, every single year.

`count` is incremented inside the positive branch only, because an invalid age is not a ticket.

## Question Two

### ANSWER

### QUESTION TWO (A)

| STRUCTURE                | DESCRIPTION                            | REAL WORLD EXAMPLE                           |
|--------------------------|----------------------------------------|----------------------------------------------|
| <b>List</b> [ ]          | Ordered, changeable, allows duplicates | The register of a class, in order            |
| <b>Tuple</b> ( )         | Ordered, cannot be changed             | The latitude and longitude of a place        |
| <b>Set</b> { }           | Unordered, no duplicates               | The distinct courses a student is taking     |
| <b>Dictionary</b> {k: v} | Pairs looked up by key                 | A student's matric number against their name |

The question asks for a real-world example for each, so a bare definition scores half. Give the example.

### ANSWER

### QUESTION TWO (B)

#### THE ISSUES

**The code runs perfectly and prints the wrong answer.** There is no error message to find, which is the difficulty.

```
Distinct courses: 5
```

- ▶ `list(courses)` copies a list that is already a list. It does nothing useful and removes no duplicates.
- ▶ `distinct = courses` merely gives the same list a second name. Nothing was deduplicated at all.

#### THE FIX

```
courses = ["CSC241", "MAT121", "CSC241", "PHY121", "MAT121"]
distinct = set(courses)
print("Distinct courses:", len(distinct))
```

```
Distinct courses: 3
```

The distinct courses are CSC241, MAT121 and PHY121. The question allows a `for` loop instead of a set, which is worth the same marks but more writing:

```
distinct = []
for c in courses:
 if c not in distinct:
 distinct.append(c)
```

## OUTPUT

```
['CSC241', 'MAT121', 'PHY121']
3
```

## EACH STAGE

- ▶ `student["courses"]` looks up the key and returns **the list itself**, not a copy. `.append("PHY121")` then changes that list, so the dictionary's value is changed. No reassignment was needed.
- ▶ `student["level"] = 300` replaces an existing value. The key already existed, so nothing is added.
- ▶ `print(student["courses"])` shows the three courses.
- ▶ `len(student)` is **3**, because **len on a dictionary counts its keys**: name, level and courses. It does not count the courses inside it, and it did not change when level was reassigned.

**The trap is the last line.** Answering 4 means you counted something else. There were three keys at the start and three at the end.

```
quantities = []

for i in range(4):
 q = int(input("Enter quantity: "))
 quantities.append(q)

quantities.sort()
print("Sorted:", quantities)
print("Highest:", max(quantities))
print("Lowest:", min(quantities))
```

Entering 25, 4, 60, 12:

```
Sorted: [4, 12, 25, 60]
Highest: 60
Lowest: 4
```

## TWO WAYS TO LOSE MARKS HERE

- ▶ `quantities = quantities.sort()` would leave `None` in the variable and everything after would fail. `sort()` sorts in place and returns nothing. Call it on its own line.
- ▶ Forgetting `int()` means you sort strings, so `"12"` would come before `"4"` alphabetically.

```
line = input("Enter quantities separated by spaces: ")

quantities = []
for part in line.split():
 quantities.append(int(part))

quantities.sort(reverse=True)

reorder = 0
for q in quantities:
 if q < 10:
 reorder += 1

average = sum(quantities) / len(quantities)

print("Sorted:", quantities)
print("Highest:", max(quantities))
print("Lowest:", min(quantities))
print(f"Average: {average:.2f}")
print("Need reordering:", reorder)
```

Entering 25 4 60 12 8:

```
Enter quantities separated by spaces: 25 4 60 12 8
Sorted: [60, 25, 12, 8, 4]
Highest: 60
Lowest: 4
Average: 21.80
Need reordering: 2
```

#### THE DETAILS THAT EARN THE MARKS

- ▶ `sort(reverse=True)` for descending. The question says descending.
- ▶ `{average:.2f}`. The average is a float and will often be a long ugly one. Format it. This is the difference between a professional answer and a puzzling one.
- ▶ `q < 10`, strictly below, because the question says "below 10". A quantity of exactly 10 does **not** need reordering. Here 4 and 8 do.

## Question Three

### ANSWER

### QUESTION THREE (A)

**What a function is:** a named block of code that runs only when it is called. It may take values in and may hand a value back.

**Why they matter:** reuse (write once, call anywhere); readability (a name says what a block does); maintenance (fix a bug in one place); decomposition (a large problem splits into small testable pieces).

**Built in against user defined:** a built in function comes with Python and is always available ( `print` , `len` , `int` , `max` ). A user defined function is one you write with `def` , and it must be defined before it is called.

**The two types of user defined function:**

- ▶ **Void:** does something and returns nothing. It has no `return` , so calling it gives `None` . Example: a function that only prints.
- ▶ **Value returning:** works something out and hands it back with `return` , so the call can be used as a value.

Four things are asked here and each is worth roughly a quarter. Answer all four explicitly, with a heading for each if it helps you not to forget one.

### ANSWER

### QUESTION THREE (B)

#### THE ERRORS

- ▶ **No colon** after `def cube(n)` . This is a `SyntaxError`, found while reading the file, so **nothing runs at all**.
- ▶ **Cube with a capital C** does not exist. The function is `cube` , and names are case sensitive.

Only the first is reported. Fix it, run again, and only then do you see:

```
NameError: name 'Cube' is not defined. Did you mean: 'cube'?
```

#### THE FIX

```
def cube(n):
 return n ** 3

print(cube(3))
```

```
27
```

Say **both** errors. The question says "errors", plural, and the marks are split. Say also that the first hides the second: it shows you understand why a `SyntaxError` stops everything.

## ANSWER

## QUESTION THREE (C)

## OUTPUT

```
110.0
120.0
```

## EXPLANATION

- ▶ `tax=0.1` is a **default parameter**: a value used when the caller does not supply one.
- ▶ `total(100)` passes only `price`, so `tax` falls back to `0.1`. The result is  $100 + 10.0 = 110.0$ .
- ▶ `total(100, 0.2)` supplies `tax`, and the argument **overrides** the default. The result is  $100 + 20.0 = 120.0$ .

**Both answers are floats**, printed as `110.0` and `120.0`, not `110` and `120`. The multiplication by a float makes the whole expression a float. Writing `110` without the decimal point is not the output.

Worth stating in your answer: a parameter with a default must come after any without one, so `def total(tax=0.1, price)` would be a `SyntaxError`.

## ANSWER

## QUESTION THREE (D)

```
def grade(score):
 if score >= 70:
 return "A"
 elif score >= 60:
 return "B"
 elif score >= 50:
 return "C"
 else:
 return "F"

for i in range(3):
 name = input("Student name: ")
 score = int(input("Score: "))
 print(f"Student: {name} | Score: {score} | Grade: {grade(score)}")
```

Entering Ada 72, Bola 45, John 65:

```
Student: Ada | Score: 72 | Grade: A
Student: Bola | Score: 45 | Grade: F
Student: John | Score: 65 | Grade: B
```

## TWO THINGS THE MARKER IS LOOKING FOR

- ▶ **return, not print**. The question says the function "returns" the letter. A function that prints instead hands back `None`, and the caller has nothing to display.
- ▶ **Descending order of tests**. 70 first. Test 50 first and every score above 50 would be a C.

`grade(score)` is called **inside the f-string**, which is neat and legal. Assigning it to a variable first is equally correct.

```
def grade(score):
 if score >= 70:
 return "A"
 elif score >= 60:
 return "B"
 elif score >= 50:
 return "C"
 else:
 return "F"

total = 0
passed = 0

while True:
 name = input("Student name (or 'done'): ")
 if name == "done":
 break
 score = int(input("Score: "))
 print(f"Student: {name} | Score: {score} | Grade: {grade(score)}")
 total += 1
 if score >= 50:
 passed += 1

print(f"Students: {total}, Passed: {passed}, Failed: {total - passed}")
```

Entering Ada 72, Bola 45, John 65, then done:

```
Student: Ada | Score: 72 | Grade: A
Student: Bola | Score: 45 | Grade: F
Student: John | Score: 65 | Grade: B
Students: 3, Passed: 2, Failed: 1
```

#### THE SHAPE

**The function is untouched.** Only the loop changed, from `for i in range(3)` to a sentinel `while True` with a `break`. That is the whole of part (e), and it is why a function is worth writing in part (d) in the first place.

**Test for 'done' before asking for the score**, or the program will demand a score for a student who does not exist.

A pass is 50 and above, so the test is `>=`. Bola's 45 fails.

## Question Four

### ANSWER

### QUESTION FOUR (A)

**What a string is:** a sequence of characters, written in single or double quotes. It is **immutable**, so it cannot be changed in place.

| METHOD                     | EXPLANATION                                           | EXAMPLE                              | GIVES              |
|----------------------------|-------------------------------------------------------|--------------------------------------|--------------------|
| <code>upper()</code>       | Returns a copy in capitals                            | <code>"ada".upper()</code>           | <code>'ADA'</code> |
| <code>lower()</code>       | Returns a copy in small letters                       | <code>"ADA".lower()</code>           | <code>'ada'</code> |
| <code>strip()</code>       | Returns a copy with whitespace removed from both ends | <code>" hi ".strip()</code>          | <code>'hi'</code>  |
| <code>replace(a, b)</code> | Returns a copy with every a changed to b              | <code>"a-b".replace("-", " ")</code> | <code>'a b'</code> |

Notice every explanation says **"returns a copy"**. That single word shows the marker you understand immutability, and it costs you nothing to write.

### ANSWER

### QUESTION FOUR (B)

#### THE ERRORS

- ▶ `full_name.index()` is called with **nothing to search for**. It needs the text whose position you want, here the space. `TypeError: index expected at least 1 argument, got 0`
- ▶ `lowercase()` **does not exist**. The method is `lower()`.

#### THE SUBTLETY THAT COSTS HALF THE MARKS

Fixing `index()` to `index(" ")` is not enough. That gives the position **of the space**, which is 5, and slicing from 5 **includes the space**:

```
>>> full_name = "Grace Hopper"
>>> full_name[full_name.index(" "):]
' Hopper'
```

Note the leading space inside the quotes. You must start **one past** it:

#### THE FIX

```
full_name = "Grace Hopper"
surname = full_name[full_name.index(" ") + 1:]
print("Surname:", surname.lower())
```

```
Surname: hopper
```

A cleaner answer, worth full marks and less error prone:

```
surname = full_name.split()[1]
print("Surname:", surname.lower())
```

## OUTPUT

```
CSC241 Python
CSC241 CODE
13
```

## EACH LINE

- ▶ `tag.strip()` removes the three spaces at each end and returns `CSC241 Python`. **tag itself is unchanged.**
- ▶ `tag.upper()` gives `" CSC241 PYTHON "`. **Now** `replace("PYTHON", "CODE")` matches, because the text is in capitals by the time `replace` sees it. **The spaces are still there**, because `strip` on the line before returned a new string and changed nothing.
- ▶ `len(tag.strip())` is **13**: six for `CSC241`, one space, six for `Python`. Note it is `len` of the *stripped* string, not of `tag`, which is 19.

**Two traps in three lines.** The middle line only matches because of the `upper()` before it, and the last line only gives 13 because of the `strip()` inside it.

```
regs = [
 "24/CSC/001",
 "23/CSC/117",
 "24/MAT/042",
 "22/PHY/008",
 "24/CSC/236",
]

for reg in regs:
 year = reg.split("/")[0]
 print(f"Reg: {reg} | Year: {year}")
```

```
Reg: 24/CSC/001 | Year: 24
Reg: 23/CSC/117 | Year: 23
Reg: 24/MAT/042 | Year: 24
Reg: 22/PHY/008 | Year: 22
Reg: 24/CSC/236 | Year: 24
```

## WHY INDEX 0

`"24/CSC/001".split("/")` gives `['24', 'CSC', '001']`. The year is the part before the first slash, so it is item **0**. The separator itself is thrown away.

The year stays a **string**, which is correct here: it is a label, not a number, and `"001"` would lose its zeros as an integer.

```
paragraph = input("Enter a paragraph: ")

for sentence in paragraph.split("."):
 sentence = sentence.strip()
 if sentence == "":
 continue

 words = len(sentence.split())

 consonants = 0
 for ch in sentence.lower():
 if ch.isalpha() and ch not in "aeiou":
 consonants += 1

 print(f"Sentence: {sentence.lower()}")
 print(f" Words: {words}, Consonants: {consonants}")
```

Entering Python is fun. I like code.

```
Enter a paragraph: Python is fun. I like code.
Sentence: python is fun
Words: 3, Consonants: 8
Sentence: i like code
Words: 3, Consonants: 4
```

#### THE ONE THAT CATCHES EVERYBODY

**ch.isalpha() is not optional.** A consonant is a *letter* that is not a vowel. Without that test the **spaces** get counted, since a space is certainly not in "aeiou", and the answer comes out as 10 instead of 8.

The eight consonants in "python is fun" are p, y, t, h and n from *python*, s from *is*, then f and n from *fun*. The vowels o, i and u are excluded, and the two spaces are excluded by `isalpha()`.

**y counts as a consonant here**, because the question defines vowels as a, e, i, o, u and y is not among them. Follow the question's definition, not your instinct about English.

The `continue` is needed because the paragraph ends in a full stop, so splitting leaves an empty final piece.

## Question Five

### ANSWER

### QUESTION FIVE (A)

**File handling** is working with files on disk from a program, so that data survives after the program ends. It is always three steps: open, process, close.

| MODE | NAME             | BEHAVIOUR                                                              |
|------|------------------|------------------------------------------------------------------------|
| "r"  | Read             | Opens for reading. <b>Error if the file does not exist.</b>            |
| "w"  | Write            | <b>Erases the file completely</b> , then writes. Creates it if absent. |
| "a"  | Append           | Adds to the end, keeping what is there. Creates it if absent.          |
| "x"  | Exclusive create | Creates it. <b>Error if it already exists.</b>                         |

**Reading against appending:** reading takes data *out* of the file and changes nothing, and the file must already exist. Appending puts data *into* the file at the end, keeps everything already there, and creates the file if needed.

Worth adding, and it shows real understanding: the dangerous distinction is "w" against "a". Opening with "w" destroys the contents immediately, before you write anything.

### ANSWER

### QUESTION FIVE (B)

#### THE ERRORS

- ▶ The write line is **missing its closing bracket**. This is a `SyntaxError`, so **nothing in the file runs**.
- ▶ `f.close` has **no brackets**, so it is not a call. It names the method and discards it. The file is never closed.

**The first hides the second**, and the second is the more dangerous, because it reports nothing at all: no error, no message, and possibly an empty file.

#### THE FIX

```
f = open("report.txt", "w")
f.write("Weekly Report")
f.close()
```

Better still, and worth saying:

```
with open("report.txt", "w") as f:
 f.write("Weekly Report")
```

`with` closes the file for you, so the second bug becomes impossible.

## RESULT

```
Log updated.
```

## EACH LINE

- ▶ `with open("log.txt", "a") as f` opens `log.txt` in **append** mode, **creating it if it does not exist**, and calls it `f`.
- ▶ `f.write("Entry recorded.\n")` adds that line **to the end** of the file, keeping everything already in it. The `\n` is needed because `write` adds no newline of its own.
- ▶ The block ends, so `with` **closes the file automatically**. There is no `close()` to forget.
- ▶ `print("Log updated.")` displays that text on screen.

**Only "Log updated." appears on screen.** The written line went into the file, not to the display. Saying so is the mark. Run it three times and the file has three entries, while the screen still shows one line each time.

```
count = int(input("How many books were borrowed? "))

with open("borrowed.txt", "w") as f:
 for i in range(count):
 title = input("Title: ")
 f.write(title + "\n")

print("Log saved successfully.")
```

Entering 2, then Things Fall Apart, then Purple Hibiscus:

```
Log saved successfully.
```

and `borrowed.txt` now holds:

```
Things Fall Apart
Purple Hibiscus
```

## THE DETAIL

+ `"\n"` **is essential**. The requirement says "on a new line", and `write` adds none. Without it the file would hold `Things Fall ApartPurple Hibiscus` on one line. An `f`-string works equally well: `f.write(f"{title}\n")`.

```
with open("fines.txt", "w") as f:
 while True:
 title = input("Title (or 'done'): ")
 if title == "done":
 break
 days = int(input("Days overdue: "))
 f.write(f"{title}, {days}\n")

total_fine = 0
with open("fines.txt", "r") as f:
 for line in f:
 title, days = line.strip().split(", ")
 days = int(days)
 fine = days * 50
 total_fine += fine
 print(f"{title}: {days} days, N{fine}")

print("Total fine: N" + str(total_fine))
```

Entering Things Fall Apart 3, Purple Hibiscus 0, Half of a Yellow Sun 2, then done:

```
Things Fall Apart: 3 days, N150
Purple Hibiscus: 0 days, N0
Half of a Yellow Sun: 2 days, N100
Total fine: N250
```

#### THE DETAILS THAT DECIDE THIS

- ▶ **line.strip() before split(", ")**. Every line read from a file keeps its `\n`. Without the strip, `days` would be `"3\n"`, and `int("3\n")` happens to work, but the habit will bite you elsewhere. Strip first, always.
- ▶ **split(", ") with the space**, matching exactly the format you wrote. Splitting on `","` alone leaves a leading space on the number.
- ▶ The fine is N50 per day, so a book 0 days overdue owes N0 and still appears in the list. Do not skip it.

## Question Six

### ANSWER

### QUESTION SIX (A)

**i. What a database is, and why:** an organised collection of data held in tables of rows and columns, stored so it can be searched, changed and retrieved efficiently. It is used because the data lasts beyond the program; it can be searched without reading everything; it enforces structure, so a column keeps its type; it serves many users at once; and it keeps data consistent through rules such as a primary key.

**ii. The role of `sqlite3`:** it is the built in Python module that connects a Python program to an SQLite database file, letting you create tables, insert rows and query them from Python. It ships with Python, so nothing needs installing.

**iii. Three SQL commands:**

| COMMAND      | FUNCTION                                  |
|--------------|-------------------------------------------|
| CREATE TABLE | Makes a new table and defines its columns |
| INSERT INTO  | Adds a new row of data to a table         |
| SELECT       | Retrieves rows from a table               |

### ANSWER

### QUESTION SIX (B)

#### THE ISSUES

**Both are missing brackets**, and they behave very differently.

- ▶ `conn.cursor` does not call the method, so `cursor` holds the method itself rather than a cursor. The next line then fails with `AttributeError`:  
'builtin\_function\_or\_method' object has no attribute 'execute'.
- ▶ `conn.commit` does nothing at all, **silently**. The table would never be saved.

#### THE FIX

```
import sqlite3
conn = sqlite3.connect("library.db")
cursor = conn.cursor()
cursor.execute("CREATE TABLE IF NOT EXISTS books (id INTEGER PRIMARY KEY, title TEXT)")
conn.commit()
conn.close()
```

**Worth the extra mark:** add `IF NOT EXISTS`. Without it the fixed program works once and then fails on every later run with `sqlite3.OperationalError: table books already exists`, because `connect` keeps the file between runs.

**i. What it does:** connects to `library.db`, creating the file if it does not exist; gets a cursor; runs an `INSERT` adding one row to the `books` table with the title 'Things Fall Apart', letting `id` fill itself; commits, making the change permanent; and closes the connection.

**ii. What happens when it runs: nothing is displayed at all.** There is no `print` and no `SELECT`, so a successful run produces no output. One row is silently added to the table.

**iii. The assumptions:**

- ▶ The `books` table **must already exist**, with a `title` column. This code never creates it, so without it you get `sqlite3.OperationalError: no such table: books`.
- ▶ `id` must fill itself, that is, be declared `INTEGER PRIMARY KEY`, since no `id` is supplied.
- ▶ The file must be writable, and the disk must have room.

Part ii is the easy mark most candidates miss. "No output" is the answer, and it is worth saying in those words.

```
import sqlite3

conn = sqlite3.connect("registry.db")
cursor = conn.cursor()

cursor.execute("""CREATE TABLE IF NOT EXISTS students
 (id INTEGER PRIMARY KEY, name TEXT, matric TEXT)""")

for i in range(3):
 name = input("Name: ")
 matric = input("Matric: ")
 cursor.execute("INSERT INTO students (name, matric) VALUES (?, ?)", (name, matric))

conn.commit()
conn.close()
print("Students saved.")
```

Entering Ada 24/CSC/001, Bola 24/CSC/117, John 24/MAT/042:

Students saved.

#### WHAT EARNS THE MARKS

- ▶ **IF NOT EXISTS**, because the question says "if it does not already exist" in as many words.
- ▶ **The ? placeholders**, not string joining. A matric number contains slashes and a name may contain an apostrophe, which would break joined SQL outright.
- ▶ **matric is TEXT**, not `INTEGER`. It contains slashes, so it is not a number at all.
- ▶ `commit()` before `close()`, or nothing is saved.

```
import sqlite3

conn = sqlite3.connect("registry.db")
cursor = conn.cursor()
cursor.execute("""CREATE TABLE IF NOT EXISTS students
 (id INTEGER PRIMARY KEY, name TEXT, matric TEXT)""")

while True:
 name = input("Name (or 'done'): ")
 if name == "done":
 break
 matric = input("Matric: ")
 cursor.execute("INSERT INTO students (name, matric) VALUES (?, ?)", (name, matric))

conn.commit()

cursor.execute("SELECT * FROM students")
for row in cursor.fetchall():
 print(f"{row[0]} | {row[1]} | {row[2]}")

conn.close()
```

Entering Ada 24/CSC/001, Bola 24/CSC/117, then done:

```
1 | Ada | 24/CSC/001
2 | Bola | 24/CSC/117
```

#### THE SHAPE, ONE LAST TIME

The sentinel loop is **exactly** the one from Questions Three and Five. The only new thing in this question is the `SELECT` and `fetchall()` at the end, which the question asks for in its last line.

**Each row is a tuple**, so `row[0]`, `row[1]` and `row[2]` are id, name and matric, in the order the table declares them.

**Commit before you select.**

#### ● MUST MEMORISE

#### WHAT THIS PAPER WAS REALLY TESTING

Look back at the six questions you just answered. Underneath the different scenarios, **every part (e) was the same two lines of thinking:**

- ▶ **The sentinel loop:** `while True`, read, test for 'done', `break`, otherwise process. Questions Three, Five and Six.
- ▶ **Split and convert:** read one line, `split()`, loop, convert each with `int()`. Questions One and Two.

That is 17.5 marks of the 70 you can attempt, and it is two patterns. If you can write both from a blank page, without thinking, you have banked a quarter of the paper before you read a single question.

And **every part (b) was a hidden second error:** a `SyntaxError` masking a `NameError`, an unclosed bracket masking a missing pair of brackets. Look for two.

## MOCK EXAMINATION TWO

# Sit This One After the First

*The same six questions, the same five parts, the same three hours. New problems throughout, and the parts that caught you in Mock One are asked here from the other side. Sit it cold, on paper, with the book shut.*

|    |                     |      |          |                                                                                          |
|----|---------------------|------|----------|------------------------------------------------------------------------------------------|
| Q1 | (a) (b) (c) (d) (e) | 17.5 | answer   |                                                                                          |
| Q2 | (a) (b) (c) (d) (e) | 17.5 | answer   |                                                                                          |
| Q3 | (a) (b) (c) (d) (e) | 17.5 | answer   |                                                                                          |
| Q4 | (a) (b) (c) (d) (e) | 17.5 | answer   | = 70  |
| Q5 | (a) (b) (c) (d) (e) | 17.5 | leave it |                                                                                          |
| Q6 | (a) (b) (c) (d) (e) | 17.5 | leave it |                                                                                          |

# Question Paper

## ● MUST MEMORISE

FOUR, NOT SIX

|             |                                                                                          |
|-------------|------------------------------------------------------------------------------------------|
| Course      | CSC241, Python Programming Language I                                                    |
| Credit unit | 3                                                                                        |
| Time        | 3 hours                                                                                  |
| Instruction | <b>Answer Four (4) questions only</b>                                                    |
| Marks       | Six questions offered. Each is five parts at 3.5 marks, so 17.5 a question, 70 in total. |

You chose four questions in Mock One. Choose four different ones here. The rubric is generous in a way students miss: **two whole questions can be subjects you never mastered**, and you still leave with full marks. The skill is picking, and you pick on parts (d) and (e).

## Question One

### QUESTION ONE

17.5 MARKS

(a) What is a control structure? Explain `break`, `continue` and the `else` branch of a loop, and give one short example of each. **[3.5mks]**

(b) The following code is intended to add up the numbers 1 to 4 and print the total. Identify and correct the error(s): **[3.5mks]**

```
total = 0
for i in range(1, 5)
 total += i
print("Total:", Total)
```

(c) Explain what the following code does. What is the output, and why does the last line print at all? **[3.5mks]**

```
for i in range(3, 0, -1):
 print(i)
else:
 print("Liftoff")
```

(d) You are writing the billing program for an electricity company. The charge depends on the number of units consumed. Your program should use decision making structures to implement the logic. **[3.5mks]**

#### Program requirements:

1. Prompt the user to enter an integer for the units consumed.
2. Check that the reading is positive.
3. If the units are under 50, the band is "Lifeline" at N10 per unit.
4. Else if the units are under 200, the band is "Domestic" at N20 per unit.
5. Else, the band is "Commercial" at N30 per unit.
6. Print the band and the bill, which is the units times the rate.
7. If the reading is not positive, print "Please enter a valid reading".

(e) Develop a new billing program as an extension of the one you did in (d) above. This updated version should allow the user to enter several meter readings in one input, separated by spaces. **[3.5mks]**

#### Program requirements:

1. Prompt the user to enter all the readings separated by spaces.
2. Split the input and convert each item to an integer.
3. For each reading, check it is positive, apply the banding, and display the band and the bill.
4. After all readings, display how many were billed and the total of all the bills.

## Question Two

### QUESTION TWO

17.5 MARKS

(a) Differentiate between a list and a tuple, and say when each is the right choice. What is a set for, and what does a dictionary give you that a list cannot? Give a real-world example of each of the four. **[3.5mks]**

(b) The following code is supposed to add a parcel weight to the collection and print the weights in order. It fails, and fixing the first fault reveals a second. Identify and fix both: **[3.5mks]**

```
weights = (12, 5, 20, 8)
weights.append(15)
print("Sorted:", weights.sort())
```

(c) What will be the output of the following code? Explain what happens at each stage of execution: **[3.5mks]**

```
stock = {"pen": 12, "book": 5}
stock["pen"] = stock["pen"] - 3
stock["bag"] = 7
print(stock["pen"])
print(len(stock))
print("book" in stock)
```

(d) You are building a weighing tool for a courier. The program should allow the user to enter the weights of four parcels, store them in a list, sort them in ascending order, and display the sorted weights, the heaviest and the lightest. **[3.5mks]**

#### Program requirements:

1. Initialize an empty list to hold the weights.
2. Use a loop to input the weights of four parcels (as integers).
3. Store each weight in the list.
4. Sort the list in ascending order.
5. Print the sorted list.
6. Print the heaviest and the lightest using appropriate built-in functions.

(e) Develop a new weighing tool as an extension of (d). This version should accept any number of parcels, not just four, entered separated by spaces. **[3.5mks]**

#### Program requirements:

1. Prompt the user to enter all the weights, separated by spaces.
2. Convert the input to a list of integers.
3. Sort the weights in descending order.
4. Display the sorted list, the heaviest and lightest, the average weight, and the number of parcels that attract a surcharge. **A parcel attracts a surcharge if it is above 20 kg.**

## Question Three

### QUESTION THREE

17.5 MARKS

(a) What is the difference between a parameter and an argument? What does `return` do, and what does a function give back if you leave it out? What is a default parameter, and why must it come last? **[3.5mks]**

(b) The following code is meant to define a function that gives the parking fee for a number of hours and then print it. Correcting the first error exposes a second. Identify and fix both: **[3.5mks]**

```
def fee(hours)
 total = 200 + (hours - 1) * 150
print(fee(3))
```

(c) Consider the following Python code. What will be the output? Explain how the function works, especially with regard to default parameters. **[3.5mks]**

```
def pay(hours, rate=500):
 return hours * rate

print(pay(8))
print(pay(8, 700))
```

(d) You have been asked to build a car park till. The program should define a function `fee(hours)` that charges N200 for the first hour and N150 for each hour after it. The program should prompt for the plate number and hours of three vehicles, call the function for each, and display the result in the format: **[3.5mks]**

```
Plate: ABC123 | Hours: 3 | Fee: N500
Plate: XYZ789 | Hours: 1 | Fee: N200
...
```

#### Program requirements:

1. Define `fee(hours)` returning the charge as described.
2. In the main program, use a loop to collect each vehicle's plate and hours.
3. For each vehicle, call the function to work out the fee.
4. Display the result in a clean, formatted output.

(e) Develop a new car park till as an extension of (d). This version should handle any number of vehicles in one session. **[3.5mks]**

#### Program requirements:

1. Define `fee(hours)` as before.
2. Use a loop to repeatedly accept a plate number and hours.
3. Stop the loop when the user enters 'done' as the plate number.
4. For each vehicle, call the function and display the fee.
5. After the loop, display how many vehicles were charged and the total takings.

## Question Four

### QUESTION FOUR

17.5 MARKS

(a) What is a string in Python, and what does it mean to say a string is immutable? What follows from that when you call a method on one? List four string methods with an example of each, **at least one of which asks a question about the string rather than changing it.** [3.5mks]

(b) The following code is intended to print the domain of an email address in lower case. It contains two errors, and the second is hidden behind the first. Identify and fix them: [3.5mks]

```
email = "Ada@CU.edu.ng"
domain = email.split("@")
print("Domain:", domain.lower)
```

(c) What will be the output of the following code? Explain the result of each line: [3.5mks]

```
code = " csc241-python "
print(code.strip().upper())
print(code.strip().split("-"))
print(len(code.strip()))
```

(d) Write a Python program for a tool that processes a list of staff email addresses and prints the domain of each. An address has the form `name@domain`, and the domain is the part after the `@`. [3.5mks]

#### Program Requirements:

1. Create a list containing at least five email addresses.
2. Loop through the list.
3. For each one, split the string at the `@` and take the domain.
4. Print the result in a clean format, such as:

```
Email: ada@cu.edu.ng | Domain: cu.edu.ng
```

(e) Develop a new text tool as an extension of (d). This version should check a batch of product codes typed in a single line. [3.5mks]

#### Program Requirements:

1. Prompt the user to enter several product codes separated by commas.
2. Split the line at the comma.
3. For each code: remove leading and trailing whitespace, skip it if it is empty, and put it in capitals.
4. For each code, display how many of its characters are letters, how many are not letters, and whether it begins with "CSC".

## Question Five

### QUESTION FIVE

17.5 MARKS

(a) What does `with open(...)` give you that a plain `open()` does not? Explain the modes `"r"`, `"w"` and `"a"`, and say what happens in each if the file does not yet exist. **[3.5mks]**

(b) The following code is intended to add a name to a file. It contains two errors. Identify and fix them: **[3.5mks]**

```
f = open("names.txt", "r")
f.write("Ada\n")
f.close
```

(c) What will the following code print? Explain what happens to the first line that was written, and why. **[3.5mks]**

```
with open("data.txt", "w") as f:
 f.write("first\n")

with open("data.txt", "w") as f:
 f.write("second\n")

with open("data.txt") as f:
 print(f.read())
```

(d) Develop a Python program for a lecturer's attendance sheet. The program should allow the lecturer to enter the names of the students present today and save the list to a file called `attendance.txt`. **[3.5mks]**

#### Program Requirement:

1. Ask the lecturer how many students are present.
2. Use a loop to input each name.
3. Open `attendance.txt` in write mode.
4. Write each name to the file on a new line.
5. After saving, print "Attendance saved successfully."

(e) Develop a new results recorder as an extension of (d). This version should allow the user to enter multiple records (name and score) until 'done' is entered as the name. Each entry should be stored in a text file called `scores.txt` in the format: **[3.5mks]**

```
Ada, 72
Bola, 45
```

#### Program Requirement:

1. Open `scores.txt` in write mode.
2. Use a loop to collect names and scores until 'done' is entered.
3. Write each entry in the format: Name, Score.
4. Close the file, then reopen it in read mode.
5. For each line: split the name and the score, print the record, and keep a running total. **A pass is 50 and above.**
6. Display the class average and how many students passed.

## Question Six

### QUESTION SIX

17.5 MARKS

**(a)** i. In a database, what is a table, a row and a column? ii. What does `conn.commit()` do, and what happens if you forget it? iii. What is the difference between `fetchone()` and `fetchall()`? **[3.5mks]**

**(b)** The following code is meant to create a `products` table and put one row in it, but it contains errors. Identify and fix the issues: **[3.5mks]**

```
import sqlite3
conn = sqlite3.connect("shop.db")
cursor = conn.cursor()
fields = "(id INTEGER PRIMARY KEY, name TEXT, price INTEGER)"
cursor.execute("CREATE TABLE products " + fields)
cursor.execute("INSERT INTO products (name, price) VALUE ('Pen', 150)")
conn.commit()
conn.close
```

**(c)** i. Examine the following code and explain what it does. ii. What will it print? iii. What assumptions must be true for it to execute successfully? **[3.5mks]**

```
import sqlite3
conn = sqlite3.connect("shop.db")
cursor = conn.cursor()
cursor.execute("SELECT name FROM products")
print(cursor.fetchall())
conn.close()
```

**(d)** Develop a simple Python-based shop inventory using `sqlite3`. The program should create a table `products` with fields: `id`, `name` and `price`. It prompts the user to enter the name and price of three products and inserts each into the database. **[3.5mks]**

#### Program requirements:

1. Import `sqlite3` and connect to a database named `inventory.db`.
2. Create a table `products` if it does not already exist.
3. Use a loop to collect the name and price of three products.
4. Insert each product into the table.
5. Commit the changes and close the connection.

**(e)** Develop a new inventory as an extension of (d). This version should allow the user to enter products until they type 'done' as the name, store the records in `inventory.db`, and after all entries are added, fetch and display every product stored in the table. **[3.5mks]**

The answers follow. **Do not read them until you have written yours.**

You have seen one set of answers already, and that makes this page more dangerous, not less: the shapes are familiar now, and familiarity reads exactly like competence until you are asked to produce one from nothing. If you can write Q1(e) without looking, you have learned the pattern. If you cannot, you have only recognised it, and recognition earns no marks.

# Answers

Every output below was produced by running the code, not by reading it. Where your answer differs from the marking, the difference is information: find out which of you is right before you move on.

## Question One

### ANSWER

Q1 (A)

A **control structure** decides which statements run, and how often. Without one a program is a list that runs top to bottom, once.

- ▶ **break** leaves the loop immediately. Nothing after it in the loop body runs, and the loop does not continue. `for i in range(5): if i == 2: break`
- ▶ **continue** abandons *this pass only* and jumps to the next. The loop carries on. `for i in range(5): if i % 2: continue`
- ▶ **else on a loop** runs when the loop finishes **without being broken**. It is the part everyone forgets: it is not "otherwise", it is "and we got all the way through".

The three-way answer the examiner wants: sequence, selection ( `if` ), repetition ( `for` , `while` ). `break` and `continue` steer the third.

### ANSWER

Q1 (B)

**Two errors, and the second only appears once the first is gone.**

- 1 The `for` line has **no colon**. Python stops before running anything:  
*SyntaxError: expected ':'*
- 2 Add the colon and it runs, and now fails on the last line:  
*NameError: name 'Total' is not defined*  
`Total` is not `total` . Python is case sensitive, and a name with a capital letter is a different name.

FIXED

```
total = 0
for i in range(1, 5):
 total += i
print("Total:", total)
```

Total: 10

`range(1, 5)` gives 1, 2, 3, 4. The 5 is the stop and never arrives, so the total is 10 and not 15. If you answered 15 you have the off by one that this part exists to catch.

## ANSWER

## Q1 (C)

```
3
2
1
Liftoff
```

`range(3, 0, -1)` counts **down**: start at 3, stop before 0, step by -1. So 3, 2, 1, and never 0.

**Why the last line prints at all** is the whole question. A loop's `else` runs when the loop **completes without a break**. There is no `break` here, so the loop finishes, and the `else` fires.

This is the mirror of Mock One Q1(c), where a `break` did fire and the `else` was skipped. Same rule, both sides of it. If you can state that rule in one sentence you have both marks.

## ANSWER

## Q1 (D)

BILL.PY

```
units = int(input("Enter units consumed: "))

if units > 0:
 if units < 50:
 band, rate = "Lifeline", 10
 elif units < 200:
 band, rate = "Domestic", 20
 else:
 band, rate = "Commercial", 30
 print(f"Band: {band} | Bill: N{units * rate}")
else:
 print("Please enter a valid reading")
```

Entering 120:

```
Band: Domestic | Bill: N2400
```

The `elif` chain is doing work you get for free: by the time Python reaches `elif units < 200` it already knows the units are **not** under 50, so you never write `units >= 50` and `units < 200`. Writing that instead is not wrong, but it is the long way round and it is where people put the boundary in the wrong place.

**The boundaries decide the mark:**

| UNITS | BAND       | BILL  |
|-------|------------|-------|
| 49    | Lifeline   | N490  |
| 50    | Domestic   | N1000 |
| 199   | Domestic   | N3980 |
| 200   | Commercial | N6000 |

50 is **not** under 50, so it is Domestic. 200 is not under 200, so it is Commercial. Read `<` as "strictly less than" every time, and these four fall out on their own.

Same banding, wrapped in **split and convert**. Nothing about the pricing changes. That is the point of "as an extension of (d)": do not rewrite it, wrap it.

BILLS.PY

```
line = input("Enter the readings separated by spaces: ")

count = 0
grand = 0
for part in line.split():
 units = int(part)
 if units > 0:
 if units < 50:
 band, rate = "Lifeline", 10
 elif units < 200:
 band, rate = "Domestic", 20
 else:
 band, rate = "Commercial", 30
 bill = units * rate
 grand += bill
 count += 1
 print(f"{units} units | {band} | N{bill}")
 else:
 print(units, "Please enter a valid reading")

print("Readings billed:", count)
print("Total:", grand)
```

Entering 30 120 250 :

```
30 units | Lifeline | N300
120 units | Domestic | N2400
250 units | Commercial | N7500
Readings billed: 3
Total: 10200
```

Three things earn the marks: `split()` with no argument (any run of spaces), `int(part)` on every piece because `input()` hands you text, and `count` incremented **inside** the valid branch, so a rejected reading is not counted as billed.

## Question Two

### ANSWER

Q2 (A)

| STRUCTURE                | WHAT IT IS                              | USE IT FOR                                                         |
|--------------------------|-----------------------------------------|--------------------------------------------------------------------|
| <b>List</b> []           | Ordered, changeable, duplicates allowed | The parcels weighed today: the set of them changes all day.        |
| <b>Tuple</b> ()          | Ordered, <b>unchangeable</b>            | A fixed pair like (latitude, longitude), or the days of the week.  |
| <b>Set</b> {}            | Unordered, <b>no duplicates</b>         | The distinct courses a student takes: ask it "how many different?" |
| <b>Dictionary</b> {k: v} | Keyed pairs                             | Stock levels: look up "pen" directly, without searching.           |

**List against tuple:** use a tuple when the collection must not change, and the fact that it cannot is the protection. A list is the default; reach for a tuple deliberately.

**What a dictionary gives that a list cannot:** it finds a value **by name**. In a list you must know the position, or search the whole thing. In a dictionary `stock["pen"]` goes straight there, however many items there are.

### ANSWER

Q2 (B)

**Two errors, the second hidden behind the first.**

- 1 `weights` is a **tuple**, and a tuple cannot be changed:  
*AttributeError: 'tuple' object has no attribute 'append'*  
The collection is being added to, so it should have been a list: `[12, 5, 20, 8]`.
- 2 Make it a list and the crash stops, but the output is wrong:  
*Sorted: None*  
`sort()` sorts the list **in place** and returns `None`. You printed its return value, which is nothing. Sort first, then print the list.

FIXED

```
weights = [12, 5, 20, 8]
weights.append(15)
weights.sort()
print("Sorted:", weights)
```

```
Sorted: [5, 8, 12, 15, 20]
```

The second error is the one worth remembering: **a method that changes a thing in place gives back None**. `sort()`, `reverse()` and `append()` are all like this. If you want a sorted copy and the original kept, that is `sorted(weights)`, which returns a new list.

## ANSWER

Q2 (C)

```
9
3
True
```

- 1 `stock["pen"] = stock["pen"] - 3` reads 12, takes 3, and writes 9 back to the **existing** key. The dictionary still has two keys.
- 2 `stock["bag"] = 7` uses a key that is not there, so it is **created**. The same square brackets both update and add: which one happens depends entirely on whether the key already exists.
- 3 `print(stock["pen"])` gives **9**.
- 4 `len(stock)` counts the **keys**: pen, book, bag, so **3**. Not the values, and not the characters.
- 5 `"book" in stock` asks whether that is a **key**, and it is: **True**. On a dictionary, `in` looks at the keys, never the values. `12 in stock` would be False even though 12 is in there.

## ANSWER

Q2 (D)

PARCELS.PY

```
weights = []

for i in range(4):
 w = int(input(f"Weight of parcel {i + 1}: "))
 weights.append(w)

weights.sort()
print("Sorted:", weights)
print("Heaviest:", max(weights))
print("Lightest:", min(weights))
```

Entering 12, 5, 20, 8:

```
Sorted: [5, 8, 12, 20]
Heaviest: 20
Lightest: 5
```

"Appropriate built-in functions" means `max()` and `min()`. You could take `weights[-1]` and `weights[0]` after sorting and be right, but only because you sorted first; `max()` needs no such promise and says what it means.

PARCELS\_ANY.PY

```
line = input("Enter all the weights separated by spaces: ")

weights = []
for part in line.split():
 weights.append(int(part))

weights.sort(reverse=True)
average = sum(weights) / len(weights)

surcharge = 0
for w in weights:
 if w > 20:
 surcharge += 1

print("Sorted:", weights)
print("Heaviest:", max(weights))
print("Lightest:", min(weights))
print(f"Average: {average:.2f}")
print("Surcharged:", surcharge)
```

Entering 12 5 20 8 25 :

```
Sorted: [25, 20, 12, 8, 5]
Heaviest: 25
Lightest: 5
Average: 14.00
Surcharged: 1
```

- ▶ **Descending** is `sort(reverse=True)` . It still returns `None` , so it is still a statement on its own line.
- ▶ **Only one parcel is surcharged.** The rule is *above* 20, so the 20 kg parcel does not qualify: 25 does. If you answered 2 you read `>` as `>=` , which is the single most expensive habit in this paper.
- ▶ The average is 70 divided by 5, which is 14.0 exactly. `{average:.2f}` shows it as 14.00: two decimal places for money and measurements, always.

## Question Three

### ANSWER

Q3 (A)

- ▶ **Parameter against argument.** The **parameter** is the name in the definition, `def fee(hours)`. The **argument** is the actual value you pass, `fee(3)`. One is the label, the other is the thing.
- ▶ **return** hands a value back to whoever called the function, and ends the function there and then. Leave it out and the function still returns something: it returns **None**. That is not an error, which is exactly why it is dangerous, and it is what part (b) is about.
- ▶ **A default parameter** supplies a value for an argument the caller does not give: `def pay(hours, rate=500)`. It **must come last** because Python matches arguments to parameters by position. If a default came first, then `pay(8)` could not say whether the 8 is the defaulted one or the required one. Python refuses the definition outright rather than guess.

### ANSWER

Q3 (B)

**Two errors, and the second is invisible until the first is fixed.**

- 1 `def fee(hours)` has **no colon**. Nothing runs at all:  
*SyntaxError: expected ':'*
- 2 Add it and the program runs cleanly and prints the wrong thing:  
*None*  
The function works out `total` and then **throws it away**. There is no `return`, so the function gives back `None`, and `print` faithfully prints it. No error, no warning, no marks.

FIXED

```
def fee(hours):
 return 200 + (hours - 1) * 150

print(fee(3))
```

500

200 for the first hour, then 2 more hours at 150, so  $200 + 300 = 500$ . **A function that computes and does not return is the most common quiet failure in this course.** If your function prints `None`, this is why, every time.

## ANSWER

Q3 (C)

```
4000
5600
```

- ▶ `pay(8)` gives one argument, so `hours` is 8 and `rate` falls back to its default of 500:  $8 * 500 = \mathbf{4000}$ .
- ▶ `pay(8, 700)` gives two, so the second **replaces** the default:  $8 * 700 = \mathbf{5600}$ .

The default is not a constant. It is the value used *only* when the caller stays silent. That is what makes `rate` optional without making it fixed, and it is the whole reason default parameters exist.

## ANSWER

Q3 (D)

CARPARK.PY

```
def fee(hours):
 return 200 + (hours - 1) * 150

for i in range(3):
 plate = input("Plate number: ")
 hours = int(input("Hours parked: "))
 print(f"Plate: {plate} | Hours: {hours} | Fee: N{fee(hours)}")
```

Entering ABC123/3, XYZ789/1 and KJA55/5:

```
Plate: ABC123 | Hours: 3 | Fee: N500
Plate: XYZ789 | Hours: 1 | Fee: N200
Plate: KJA55 | Hours: 5 | Fee: N800
```

Check the one-hour case by hand, because it is the one the formula can get wrong:  $200 + (1 - 1) * 150 = 200 + 0 = 200$ . Correct. Write the formula as `200 + hours * 150` and that vehicle is charged 350 for its first hour, and every single answer is 150 too high.

The **sentinel loop**: while True, ask, and break on the stop word. The function itself does not change at all.

CARPARK\_ANY.PY

```
def fee(hours):
 return 200 + (hours - 1) * 150

count = 0
takings = 0

while True:
 plate = input("Plate number (or 'done'): ")
 if plate == "done":
 break
 hours = int(input("Hours parked: "))
 amount = fee(hours)
 takings += amount
 count += 1
 print(f"Plate: {plate} | Hours: {hours} | Fee: N{amount}")

print("Vehicles charged:", count)
print(f"Total takings: N{takings}")
```

Entering ABC123/3, XYZ789/1, then done :

```
Plate: ABC123 | Hours: 3 | Fee: N500
Plate: XYZ789 | Hours: 1 | Fee: N200
Vehicles charged: 2
Total takings: N700
```

**Test the stop word before you ask for the hours.** Put the `int(input(...))` above the check and typing `done` makes the program demand the hours for a vehicle that does not exist, and then crash on `int("")`. The check goes immediately after the input it checks.

## Question Four

### ANSWER

Q4 (A)

A **string** is text: a sequence of characters in quotes. It is **immutable**, meaning it can never be changed after it is made.

**What follows from that** is the examinable half: a string method can never change the string it is called on, so it hands you a **new** one. `name.upper()` leaves `name` exactly as it was. If you do not catch the result, with `name = name.upper()`, the work is thrown away and nothing happens. This is the opposite of a list's `sort()`, which changes the list and hands back nothing.

| METHOD                  | DOES                                    | EXAMPLE                       | GIVES                   |
|-------------------------|-----------------------------------------|-------------------------------|-------------------------|
| <code>upper()</code>    | Returns a capitalised copy              | <code>"ada".upper()</code>    | <code>'ADA'</code>      |
| <code>strip()</code>    | Returns a copy with the ends trimmed    | <code>" hi ".strip()</code>   | <code>'hi'</code>       |
| <code>split(sep)</code> | Cuts it into a list                     | <code>"a,b".split(",")</code> | <code>['a', 'b']</code> |
| <code>isalpha()</code>  | <b>Asks</b> a question, changes nothing | <code>"Ada".isalpha()</code>  | <code>True</code>       |

The fourth is the one the question is fishing for. `isalpha()`, `startswith()` and `find()` interrogate a string rather than transform it; the rest hand back a changed copy.

### ANSWER

Q4 (B)

**Two errors, the second wearing the first as a disguise.**

- 1 `split("@")` gives a **list**, `['Ada', 'CU.edu.ng']`, not a string. So `domain` is a list, and a list has no `lower`:

*AttributeError: 'list' object has no attribute 'lower'*

You want the piece after the `@`, which is index **1**: `email.split("@")[1]`.

- 2 Take the index and the crash goes away, and the output is still wrong:

*Domain: <built-in method lower of str object at 0x...>*

`domain.lower` has **no brackets**, so it never calls the method. It names it. Python prints the method itself, which is a real object and prints happily. (The `0x...` is a memory address, and it is a different number every time the program runs.)

FIXED

```
email = "Ada@CU.edu.ng"
domain = email.split("@")[1]
print("Domain:", domain.lower())
```

```
Domain: cu.edu.ng
```

**The missing brackets again.** This is the third time this manual has shown you the same fault: `file.close`, `conn.cursor`, and now `domain.lower`. **A method without brackets is not called, and Python never complains.** When output looks like `<built-in method ...>` or `<bound method ...>`, you have found it.

```
CSC241-PYTHON
['csc241', 'python']
13
```

- 1 `code.strip()` gives `'csc241-python'`, and `.upper()` capitalises that copy. The chain reads left to right: strip first, then capitalise the result.
- 2 `split("-")` cuts the stripped string at the hyphen into **two** pieces. They stay lowercase, because `upper()` on the line above returned a new string and did not touch `code`.
- 3 `len(code.strip())` is **13**: c-s-c-2-4-1 is 6, the hyphen is 1, p-y-t-h-o-n is 6. `len(code)` without the strip would be **17**, counting the four spaces.

`code` itself is unchanged throughout, spaces and all. Three methods were called on it and not one of them altered it: that is immutability doing exactly what part (a) says it does.

```
DOMAINS.PY
```

```
emails = ["ada@cu.edu.ng", "bola@gmail.com", "chidi@yahoo.com",
 "dele@cu.edu.ng", "efe@outlook.com"]

for email in emails:
 domain = email.split("@")[1]
 print(f>Email: {email} | Domain: {domain}")
```

```
Email: ada@cu.edu.ng | Domain: cu.edu.ng
Email: bola@gmail.com | Domain: gmail.com
Email: chidi@yahoo.com | Domain: yahoo.com
Email: dele@cu.edu.ng | Domain: cu.edu.ng
Email: efe@outlook.com | Domain: outlook.com
```

Index **1**, not 0. Splitting `"ada@cu.edu.ng"` at the `@` gives `['ada', 'cu.edu.ng']`: the name is 0 and the domain is 1. The dots in `cu.edu.ng` are harmless, because we split on `@` and the dots are all inside a piece we never cut again.

CODES.PY

```
line = input("Enter the codes separated by commas: ")

for code in line.split(","):
 code = code.strip()
 if code == "":
 continue
 code = code.upper()

 letters = 0
 for ch in code:
 if ch.isalpha():
 letters += 1

 others = len(code) - letters
 csc = code.startswith("CSC")
 print(f"Code: {code} | Letters: {letters} | Not letters: {others} | CSC: {csc}")
```

Entering csc241, mat121x, , abc:

```
Code: CSC241 | Letters: 3 | Not letters: 3 | CSC: True
Code: MAT121X | Letters: 4 | Not letters: 3 | CSC: False
Code: ABC | Letters: 3 | Not letters: 0 | CSC: False
```

- ▶ **Four pieces went in and three came out.** The empty one between the two commas is skipped by the `continue`, exactly as the requirement asks. This is the same guard as Mock One's paragraph, and it exists because real input has gaps in it.
- ▶ **Capitalise before you test.** `startswith("CSC")` is case sensitive, so `"csc241"` would answer False if you tested it before `upper()`. Order matters, and the requirement lists it in the right order for a reason.
- ▶ **Not letters is `len(code) - letters`**, not a second loop. You have already counted the letters, and everything else is what remains.

## Question Five

### ANSWER

Q5 (A)

`with open(...)` closes the file for you, whatever happens, even if the code inside raises an error. A plain `open()` leaves that to you, and if an exception fires before your `close()`, the file is never closed and the last thing you wrote may never reach the disk. `with` is not a convenience; it is the guarantee.

| MODE | DOES                                        | IF THE FILE DOES NOT EXIST             |
|------|---------------------------------------------|----------------------------------------|
| "r"  | Read only. Cannot write.                    | <b>Error.</b> It refuses to create it. |
| "w"  | Write, and <b>empty the file first.</b>     | Creates it.                            |
| "a"  | Append: add to the end, keep what is there. | Creates it.                            |

**Reading against appending:** "r" takes data out and changes nothing; "a" puts data in at the end and destroys nothing. The dangerous one is "w", which silently throws away the whole file the instant you open it.

### ANSWER

Q5 (B)

**Two errors.**

- 1 "r" is **read only**, and the program is trying to write. The mode should be "a" to add a name and keep the others.
- 2 `f.close` has **no brackets**, so the file is never closed. Same fault as Q4(b), in a different coat.

**The first error gives two different messages, and which one you get depends on something the code does not show you:**

| IF NAMES.TXT   | PYTHON SAYS                                                         |
|----------------|---------------------------------------------------------------------|
| exists         | UnsupportedOperation: not writable                                  |
| does not exist | FileNotFoundError: [Errno 2] No such file or directory: 'names.txt' |

With the file missing, `open()` itself fails and `write` is never reached, so you never hear about writability at all. Both answers are correct, and saying **why there are two** is worth more than either.

FIXED

```
f = open("names.txt", "a")
f.write("Ada\n")
f.close()
```

Better still, let `with` do the closing and the second error cannot happen at all:

```
with open("names.txt", "a") as f:
 f.write("Ada\n")
```

## ANSWER

Q5 (C)

```
second
```

Followed by a blank line, because the file's contents end in `\n` and `print` adds another.

**The first line is gone.** Opening in `"w"` empties the file immediately, before a single byte is written. The second `with` block therefore starts from nothing, and `"first"` is destroyed without a word.

Had the second block used `"a"`, the file would hold both lines. This one question is the whole difference between the two modes, and it is why a program that saves records in a loop must open the file **once**, outside the loop: reopen it in `"w"` each time round and you finish with a file containing only the last record.

## ANSWER

Q5 (D)

ATTENDANCE.PY

```
count = int(input("How many students are present? "))

with open("attendance.txt", "w") as f:
 for i in range(count):
 name = input(f"Name of student {i + 1}: ")
 f.write(name + "\n")

print("Attendance saved successfully.")
```

Entering 3, then Ada, Bola, Chidi:

```
Attendance saved successfully.
```

and `attendance.txt` holds:

```
Ada
Bola
Chidi
```

`write()` adds **no** newline of its own, unlike `print`. Leave off the  `+ "\n"` and the whole register lands on one line as `AdaBolaChidi`. The file is opened once, outside the loop, for the reason part (c) just demonstrated.

SCORES.PY

```
f = open("scores.txt", "w")

while True:
 name = input("Name (or 'done'): ")
 if name == "done":
 break
 score = input("Score: ")
 f.write(name + ", " + score + "\n")

f.close()

f = open("scores.txt", "r")
total = 0
count = 0
passes = 0

for line in f:
 name, raw = line.strip().split(", ")
 score = int(raw)
 total += score
 count += 1
 if score >= 50:
 passes += 1
 print(f"{name}: {score}")

f.close()
print(f"Average: {total / count:.2f}")
print("Passed:", passes)
```

Entering Ada/72, Bola/45, Chidi/58, then done :

```
Ada: 72
Bola: 45
Chidi: 58
Average: 58.33
Passed: 2
```

- ▶ **Bola fails.** A pass is 50 **and above**, and 45 is not. Two of the three passed. The average is  $175 / 3 = 58.33$ , which is not a pass mark for anybody: it is just the class average.
- ▶ `line.strip()` before `split(", ")`, or the newline sticks to the score and `int("72\n")` ... in fact survives, because `int()` ignores surrounding whitespace. Strip anyway: relying on that is a habit that breaks the moment you compare the text instead of converting it.
- ▶ **Split on ", " including the space**, because that is exactly how you wrote it. Split on `","` and the score arrives as `" 72"`. The file format and the parser are one decision, made twice.

## Question Six

### ANSWER

Q6 (A)

i. A **table** is one kind of thing being stored, like products. A **row** is one of them: one product. A **column** is one fact held about every one of them: the price. A table is a grid, rows down, columns across.

ii. `conn.commit()` makes your changes **permanent**. Until you call it, inserts live in an open transaction and belong only to your connection. Forget it and the program runs, prints nothing wrong, exits, and **the database is unchanged**. No error is ever raised. It is the single most common way to lose work in this topic.

iii. `fetchone()` returns **one** row and moves on, or `None` if there are none left. `fetchall()` returns **every** remaining row, as a list of tuples, and an empty list if there are none. Use `fetchone()` when you expect a single answer, `fetchall()` when you want to loop over the lot.

### ANSWER

Q6 (B)

#### Two errors.

- 1 `VALUE` should be **VALUES**. This is SQL, not Python, so Python cannot help you: the string is perfectly good Python and the database is the one that objects.  
*OperationalError: near "VALUE": syntax error*
- 2 `conn.close` has **no brackets**, so the connection is never closed. The third appearance of this fault in two mocks. Note that the program never even reaches it here: the error above stops everything first.

#### FIXED

```
import sqlite3

conn = sqlite3.connect("shop.db")
cursor = conn.cursor()
fields = "(id INTEGER PRIMARY KEY, name TEXT, price INTEGER)"
cursor.execute("CREATE TABLE IF NOT EXISTS products " + fields)
cursor.execute("INSERT INTO products (name, price) VALUES ('Pen', 150)")
conn.commit()
conn.close()
```

`IF NOT EXISTS` is added for a reason worth a mark on its own: run the original twice and the second run dies with `OperationalError: table products already exists`. A program that only works once is not finished.

i. It opens `shop.db`, asks for the `name` column of every row in `products`, prints all the rows it gets back, and closes the connection.

ii. With the single `Pen` row from (b) in the table:

```
[('Pen',)]
```

A **list of tuples**, one tuple per row, even though only one column was asked for. That trailing comma in `('Pen',)` is Python's way of writing a tuple of one item, and it is not a typo. To print plain names you would loop and take `row[0]`.

iii. Three assumptions:

1. `shop.db` can be opened. `connect()` **creates** the file if it is absent, so this never fails, and that is the trap.
2. A table called `products` exists in it, with a `name` column. If not: `OperationalError: no such table: products` which is exactly what you get if `connect()` just made you a fresh, empty database because you misspelled the filename.
3. The rows were **committed** by whoever inserted them. Uncommitted inserts from another connection are invisible here.

INVENTORY.PY

```
import sqlite3

conn = sqlite3.connect("inventory.db")
cursor = conn.cursor()
fields = "(id INTEGER PRIMARY KEY, name TEXT, price INTEGER)"
cursor.execute("CREATE TABLE IF NOT EXISTS products " + fields)

for i in range(3):
 name = input("Product name: ")
 price = int(input("Price: "))
 cursor.execute("INSERT INTO products (name, price) VALUES (?, ?)", (name, price))

conn.commit()
conn.close()
print("Three products saved.")
```

Three products saved.

- ▶ **The ? placeholders.** Pass the values as a tuple second and let `sqlite3` put them in. Build the SQL by gluing strings together and a product named `O'Brien` ends your statement early and crashes it.
- ▶ **id is not inserted.** `INTEGER PRIMARY KEY` fills itself in: 1, 2, 3.
- ▶ **`conn.commit()` is not optional.** Leave it out and this program prints "Three products saved." and saves nothing at all.

The sentinel loop from Q3(e), with the database from (d) behind it.

INVENTORY\_ANY.PY

```
import sqlite3

conn = sqlite3.connect("inventory.db")
cursor = conn.cursor()
fields = "(id INTEGER PRIMARY KEY, name TEXT, price INTEGER)"
cursor.execute("CREATE TABLE IF NOT EXISTS products " + fields)

while True:
 name = input("Product name (or 'done'): ")
 if name == "done":
 break
 price = int(input("Price: "))
 cursor.execute("INSERT INTO products (name, price) VALUES (?, ?)", (name, price))

conn.commit()

cursor.execute("SELECT * FROM products")
for row in cursor.fetchall():
 print(row)

conn.close()
```

Entering Pen/150, Book/1200, then done :

```
(1, 'Pen', 150)
(2, 'Book', 1200)
```

**Commit before you select.** The order is: loop, commit, then read back. And `close()` comes last of all, once: put it inside the loop and the second product is inserted through a closed connection.

`SELECT *` gives every column, so each row arrives as the full tuple including the id the database made for you. That is why these tuples have three items and Q6(c)'s had one.

Two papers, sixty parts, and underneath them **four** patterns. Learn these and you have not learned two mocks, you have learned the exam.

| PART | ALWAYS ONE OF                                                                                                                                         |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| (b)  | A first error that hides a second. Fix the crash and read the output again: it is usually None, or a method printed instead of called.                |
| (c)  | A rule stated from one side. Mock One broke the loop so else was skipped; Mock Two did not, so it ran.                                                |
| (d)  | A banding decided by boundaries. 49 against 50, 199 against 200, "under 13" against "13 and above".                                                   |
| (e)  | <b>Sentinel loop</b> (while True, break on 'done') or <b>split and convert</b> (one line, split(), int() each piece). Nothing else has ever been set. |

**The one habit worth more than any of them:** read `<` as "strictly less than" and `>` as "strictly more than", out loud, every single time. A parcel above 20 kg is not a parcel of 20 kg. A pass of 50 and above includes 50. Every (d) and every (e) in both papers turns on that, and it is the cheapest mark on the paper to lose.

## MOCK EXAMINATION THREE

# Sit It In The New Shape

*Built on the 2025/2026 pattern, not the older one: the marks vary from part to part, the questions have different numbers of parts, and one question mixes topics. Fresh problems throughout. Sit it cold, on paper, with the book shut.*

Q1 or, strings, dict, interest, temp

Q2 loops, control structures, PIN

Q3 functions, list trace, collections

Q4 mutability, a function, SQLite

Q5 files, modes, an inventory 

---

 pick 4

Q6 nesting, void, math.pi

# Question Paper

## ● MUST MEMORISE

## THE NEW RUBRIC

Course CSC241, Python Programming Language I

Credit unit 3

Time 3 hours

Instruction **Answer any Four (4) questions**

Marks Six offered, four attempted, 70 in total. The marks vary from part to part: read each one and spend your time to match.

This is the 2025/2026 shape. A part worth 10.5 marks deserves five times the effort of one worth 2. Watch for questions that stop at part (c), and for a single question that visits more than one topic.

## Question One

### QUESTION ONE

17.5 MARKS

**(a)** Briefly describe how the `or` operator works in Python. Use examples. [2.5 marks]

**(b)** What is a string in Python? List and briefly explain four common string methods, with an example for each. [5 marks]

**(c)** Identify the errors in the following code segment. [3 marks]

```
book = { 'title': 'Things Fall Apart', 'author': 'Achebe',
 'year': 1958, 'Genre': 'Fiction' }
print("Title:", book['Title'])
print("Author:", book['Author'])
print("Genre:", book['genre'])
book['Year'] = 1959
book['pages'] = 209
del book['genre']
print("Updated Book:")
print(book)
```

**(d)** Write a Python program that calculates **simple interest** using  $I = P * R * T / 100$ , then displays the interest and the total amount (principal plus interest), each to two decimal places. [3.5 marks]

**(e)** Write a Python program that converts a Celsius temperature to Fahrenheit using  $F = C * 9 / 5 + 32$ , then prints a band: Freezing for 32 and below, Cold below 60, Warm below 85, Hot otherwise. [3.5 marks]

## Question Two

### QUESTION TWO

17.5 MARKS

- (a) What is a **condition-controlled loop** in Python? [3.5 marks]
- (b) Explain the three main types of control structure in Python, giving one example of each. [4.5 marks]
- (c) The following code is meant to add up the even numbers from 1 to 10. Identify the errors and write the corrected version. [3 marks]

```
total = 0
for i in range(1, 10)
 if i % 2 = 0:
 total + i
print("Sum of evens:" total)
```

- (d) The correct PIN for a system is 2468. Write a program that uses appropriate control structures to: [6.5 marks]
1. keep asking for the PIN until it is correct, printing "Wrong PIN" each time and "PIN Accepted" once;
  2. read an age, validate that it is between 0 and 120, and print an error if not;
  3. if valid, print the category: Child under 13, Teenager under 20, Adult under 65, Senior otherwise;
  4. read a positive integer and count down from it to 1, then print "Liftoff!";
  5. read numbers until -1 is entered, then print their average.

## Question Three

### QUESTION THREE

17.5 MARKS

- (a) List five benefits of building a program out of functions. [2.5 marks]
- (b) What is a function in Python, and why are functions important? Differentiate between a **parameter** and an **argument**, with an example. [6 marks]
- (c) What will `a`, `b` and `len(b)` contain after these statements? [3 marks]

```
a = [0] * 3
b = [1, 2, 3]
b.append([4, 5])
```

- (d) A coding club keeps the following records. Write a program that stores each in the most suitable collection, then answers the questions below. [5 marks]
- ▶ Enrolled: Ada, Ben, Chidi, Dami, Ada, Efe, Ben, Femi
  - ▶ Club details: "Coding Club", "Lab 3", "Friday", 25
  - ▶ Present today: Ada, Ben, Grace, Femi, Ada, Chidi
  - ▶ Points: Ada 88, Ben 54, Chidi 72, Dami 45, Efe 63, Femi 91
- Display: the club details; the unique enrolled list; those enrolled but absent; those present but not enrolled; and those with 60 points or more.

## Question Four

### QUESTION FOUR

17.5 MARKS

**(a)** "Lists are mutable." Explain what this means, with reasons and an example. [5 marks]

**(b)** A program contains this function. [1 + 1 marks]

```
def half(n):
 return n / 2
```

(i) Write a statement that passes 10 to the function and stores its result in `result`. (ii) What value will `result` hold?

**(c)** A library computerizes its catalogue with an SQLite database named `library.db` and a table `book` with fields `BookID` (INTEGER), `Title` (TEXT), `Author` (TEXT) and `Copies` (INTEGER). Write a program that: [10.5 marks]

1. creates the database and table if they do not exist;
2. inserts these books: 1 Half of a Yellow Sun / Adichie / 4; 2 Things Fall Apart / Achebe / 7; 3 The Joys of Motherhood / Emecheta / 3; 4 Arrow of God / Achebe / 5; 5 Efuru / Nwapa / 2;
3. displays all books;
4. displays only the books by Achebe;
5. displays the number of titles, the most copies of any one, and the average copies;
6. updates Efuru to 6 copies;
7. deletes the book with BookID 3.

## Question Five

### QUESTION FIVE

17.5 MARKS

**(a)** What is file handling in Python? [2 marks]

**(b)** List and explain four file access modes used in `open()`. [6 marks]

**(c)** State the difference between reading from a file and writing to a file. [2 marks]

**(d)** A store keeps its stock in a text file. Write a program that: [7.5 marks]

1. prompts for 3 items (name, code, quantity) and saves them to `inventory.txt`;
2. prompts for 2 more items and appends them without erasing the first three;
3. reads and displays every item in the file;
4. reads and displays only the first two items.

## Question Six

### QUESTION SIX

17.5 MARKS

(a) Given the code below, state each result. [3.5 marks]

```
data = [[100, 200], [300, [400, 500, 600]],
 [[700, 800], 900], ["a", "b", "c"]]
```

(i) `data[1][1][2]`   (ii) `len(data[1])`   (iii) `data[2][0][1]`

(b) A school prints exam result summaries. Write a program that reads a student's name, their number of subjects, and each subject score, then passes the name, the total score and a pass mark of 250 to a **void** function `print_result_summary(student, total_score, pass_mark)` that displays the name, the total, and PASS or FAIL. [7 marks]

(c) A water company sizes cylindrical tanks. Using **value-returning** functions and `math.pi`, write a program that reads a radius and a height, then computes and displays the volume `pi * r ** 2 * h` and prints "Large tank" if the volume exceeds 1000, otherwise "Small tank". [7 marks]

## Answers

Every output below was produced by running the code, not by reading it. Sit the paper first, then mark yourself honestly against these.

### Question One

#### ANSWER

#### Q1 (A) THE OR OPERATOR

**or** is a boolean operator that joins two conditions and is true when **at least one** of them is true. It is false only when both are false, the mirror of **and**.

```
>>> age = 15
>>> age < 13 or age > 65
False
>>> age < 18 or age > 65
True
```

Like **and**, it short-circuits: if the left is already true, Python does not bother looking at the right, because the answer is settled.

#### ANSWER

#### Q1 (B) STRINGS AND METHODS

A **string** is an ordered, immutable sequence of characters in quotes. Four methods, each returning a new string:

| METHOD   | DOES             | EXAMPLE          | GIVES      |
|----------|------------------|------------------|------------|
| upper()  | Capitals         | "go".upper()     | 'GO'       |
| strip()  | Trims ends       | " hi ".strip()   | 'hi'       |
| split()  | Splits to a list | "a b".split()    | ['a', 'b'] |
| count(x) | Counts x         | "aba".count("a") | 2          |

The keys are 'title', 'author', 'year' and 'Genre'. The faults are all case mismatches:

- ▶ `book['Title']` and `book['Author']`: the keys are lower case. `KeyError`, and 'Title' stops the program first.
- ▶ `book['genre']` twice (line 5 and the `del`): the key is 'Genre' with a capital G.
- ▶ `book['Year'] = 1959` makes a new key 'Year' instead of updating 'year'. Silent, and wrong.

Corrected so the case matches, it prints:

```
Title: Things Fall Apart
Author: Achebe
Genre: Fiction
Updated Book:
{'title': 'Things Fall Apart', 'author': 'Achebe', 'year': 1959, 'pages': 209}
```

'year' is updated to 1959, 'pages' is added, and 'Genre' is deleted.

```
P = float(input("Principal: "))
R = float(input("Rate (%): "))
T = float(input("Time (years): "))

interest = P * R * T / 100

print(f"Interest: {interest:.2f}")
print(f"Total amount: {P + interest:.2f}")
```

For 50000 at 8% over 3 years:

```
Interest: 12000.00
Total amount: 62000.00
```

```
C = float(input("Celsius: "))
F = C * 9 / 5 + 32

if F <= 32:
 band = "Freezing"
elif F < 60:
 band = "Cold"
elif F < 85:
 band = "Warm"
else:
 band = "Hot"

print(f"Fahrenheit: {F:.1f}")
print("Band:", band)
```

For 25 degrees Celsius:

```
Fahrenheit: 77.0
Band: Warm
```

77 is below 85 but not below 60, so the ladder lands on Warm. The order high-to-low is what makes the bare bounds work.

## Question Two

A **condition-controlled loop** runs an unknown number of times, repeating as long as a condition stays true and stopping when it turns false. In Python it is the `while` loop. You use it when you do not know in advance how many passes are needed, such as reading until a sentinel.

```
total = 0
while total < 100:
 total += int(input("Add: "))
```

It contrasts with a count-controlled `for` loop, which runs a known, fixed number of times.

- ▶ **Sequence:** statements run in order, top to bottom. `x = 1; y = 2; print(x + y)`
- ▶ **Selection:** a choice with `if/elif/else`. `if age >= 18: print("Adult")`
- ▶ **Repetition:** a block many times, with `for` or `while`. `while n > 0: n -= 1`

Four errors, three of them syntax:

- 1 **No colon** after the `for` line.
- 2 `if i % 2 = 0` uses `=` (assignment) where it needs `==` (comparison).
- 3 `total + i` computes and throws the result away; it should be `total += i`.
- 4 `print("Sum of evens:" total)` is missing the comma between its two arguments. And `range(1, 10)` stops at 9, so 10 is missed: use `range(1, 11)`.

FIXED

```
total = 0
for i in range(1, 11):
 if i % 2 == 0:
 total += i
print("Sum of evens:", total)
```

```
Sum of evens: 30
```

The evens 2, 4, 6, 8, 10 add to 30. Answer 20 and you kept the off-by-one in the range.

SYSTEM.PY

```
while True:
 pin = input("PIN: ")
 if pin == "2468":
 print("PIN Accepted")
 break
 print("Wrong PIN")

age = int(input("Age: "))
if age < 0 or age > 120:
 print("Error: age must be 0 to 120")
else:
 if age < 13:
 print("Child")
 elif age < 20:
 print("Teenager")
 elif age < 65:
 print("Adult")
 else:
 print("Senior")

start = int(input("Count down from: "))
for i in range(start, 0, -1):
 print(i)
print("Liftoff!")

total = 0
count = 0
while True:
 n = int(input("Number (-1 to stop): "))
 if n == -1:
 break
 total += n
 count += 1
print(f"Average: {total / count:.2f}")
```

PIN 1111, then 9999, then 2468 gives "Wrong PIN" twice then "PIN Accepted". Age 30 is an Adult; age 200 gives the error. A countdown from 5 prints 5 4 3 2 1 then "Liftoff!". Numbers 20, 40, 60, then -1 give:

```
Average: 40.00
```

`range(start, 0, -1)` counts down and stops before 0, so it ends at 1. The step of -1 is what reverses it.

## Question Three

Reuse; readability; maintenance; decomposition into small testable pieces; and abstraction, using a function by what it does rather than how. A sixth if needed: collaboration, since a team can split the functions between them.

## ANSWER

## Q3 (B) FUNCTIONS, PARAMETER VS ARGUMENT

A **function** is a named block of code that runs when called. Functions matter for reuse, readability, maintenance and decomposition.

A **parameter** is the name in the definition; an **argument** is the actual value you pass when you call. In `def greet(name)`, `name` is the parameter; in `greet("Ada")`, `"Ada"` is the argument. Parameter is the empty slot, argument is what you drop into it.

## ANSWER

## Q3 (C) THE LIST TRACE

- ▶ `[0] * 3` repeats the list, so `a` is `[0, 0, 0]`.
- ▶ `b.append([4, 5])` adds the list `[4, 5]` as a **single** item, so `b` is `[1, 2, 3, [4, 5]]`.
- ▶ `len(b)` is 4: three numbers and one inner list, which counts as one.

```
a = [0, 0, 0]
b = [1, 2, 3, [4, 5]]
len(b) = 4
```

The trap is expecting `append` to spread `[4, 5]` into two items. It does not: it appends the list whole. That is why `len(b)` is 4, not 5. To spread, you would use `b += [4, 5]` or `b.extend([4, 5])`.

## ANSWER

## Q3 (D) THE COLLECTIONS PROGRAM

A set for the enrolled (unique, unordered), a tuple for the fixed club details, a set for attendance, a dictionary for points. The two "absent" and "not enrolled" questions are set difference both ways.

CLUB.PY

```
enrolled = {"Ada", "Ben", "Chidi", "Dami", "Efe", "Femi"}
details = ("Coding Club", "Lab 3", "Friday", 25)
present = {"Ada", "Ben", "Grace", "Femi", "Chidi"}
points = {"Ada": 88, "Ben": 54, "Chidi": 72,
 "Dami": 45, "Efe": 63, "Femi": 91}

print("Club info:", details)
print("Enrolled:", sorted(enrolled))
print("Enrolled, absent:", sorted(enrolled - present))
print("Present, not enrolled:", sorted(present - enrolled))
print("Points 60+:", sorted(nm for nm, p in points.items() if p >= 60))
```

```
Club info: ('Coding Club', 'Lab 3', 'Friday', 25)
Enrolled: ['Ada', 'Ben', 'Chidi', 'Dami', 'Efe', 'Femi']
Enrolled, absent: ['Dami', 'Efe']
Present, not enrolled: ['Grace']
Points 60+: ['Ada', 'Chidi', 'Efe', 'Femi']
```

Dami and Efe are enrolled but were absent; Grace turned up without enrolling. `sorted` around each set makes the printed order stable and markable.

## Question Four

### ANSWER

### Q4 (A) LISTS ARE MUTABLE

**Mutable** means a list can be changed in place after it is made: you can reassign an item, add one, or remove one, and it is still the same list.

```
>>> nums = [10, 20, 30]
>>> nums[0] = 99
>>> nums
[99, 20, 30]
```

**The reason it matters:** a string is immutable and refuses `s[0] = "x"` with a `TypeError`, while a list allows it. So a list is the structure you reach for when the data must change as the program runs.

### ANSWER

### Q4 (B) THE HALF FUNCTION

(i) `result = half(10)`. (ii) `half(10)` returns `10 / 2`, and `/` always gives a float, so:

```
result = 5.0
```

Note `5.0`, not `5`: true division `/` returns a float even when the result is whole. That decimal point is worth a mark.

LIBRARY.PY

```

import sqlite3

conn = sqlite3.connect("library.db")
cursor = conn.cursor()
cursor.execute("""CREATE TABLE IF NOT EXISTS book
 (BookID INTEGER PRIMARY KEY, Title TEXT,
 Author TEXT, Copies INTEGER)""")

books = [(1, "Half of a Yellow Sun", "Adichie", 4),
 (2, "Things Fall Apart", "Achebe", 7),
 (3, "The Joys of Motherhood", "Emecheta", 3),
 (4, "Arrow of God", "Achebe", 5),
 (5, "Efuru", "Nwapa", 2)]

cursor.executemany("INSERT INTO book VALUES (?, ?, ?, ?)", books)
conn.commit()

print("All books:")
cursor.execute("SELECT * FROM book")
for row in cursor.fetchall():
 print(row)

print("By Achebe:")
cursor.execute("SELECT * FROM book WHERE Author = ?", ("Achebe",))
for row in cursor.fetchall():
 print(row)

cursor.execute("SELECT COUNT(*), MAX(Copies), AVG(Copies) FROM book")
titles, most, average = cursor.fetchone()
print("Titles:", titles, " Most:", most, " Average:", average)

cursor.execute("UPDATE book SET Copies = ? WHERE Title = ?", (6, "Efuru"))
conn.commit()
cursor.execute("DELETE FROM book WHERE BookID = ?", (3,))
conn.commit()
conn.close()

```

```

All books:
(1, 'Half of a Yellow Sun', 'Adichie', 4)
(2, 'Things Fall Apart', 'Achebe', 7)
(3, 'The Joys of Motherhood', 'Emecheta', 3)
(4, 'Arrow of God', 'Achebe', 5)
(5, 'Efuru', 'Nwapa', 2)
By Achebe:
(2, 'Things Fall Apart', 'Achebe', 7)
(4, 'Arrow of God', 'Achebe', 5)
Titles: 5 Most: 7 Average: 4.2

```

Copies is INTEGER, so the rows show 4 and 7, not 4.0. The average, 4.2, is a real number because an average can be fractional even when the column is whole. After the update Efuru holds 6, and after the delete BookID 3 is gone.

## Question Five

### ANSWER

Q5 (A), (B), (C)

**(a)** File handling is reading from and writing to files on disk, so data lasts after the program ends, using `open()`, the read and write methods, and `close()`.

**(b)** Four modes:

| MODE | MEANS          | EXISTS         | MISSING           |
|------|----------------|----------------|-------------------|
| "r"  | Read           | Reads          | FileNotFoundError |
| "w"  | Write          | Erased first   | Created           |
| "a"  | Append         | Adds at end    | Created           |
| "r+" | Read and write | Both, no erase | FileNotFoundError |

**(c)** Reading takes data out and changes nothing. Writing ( "w" ) puts data in and **erases** whatever was there first. One looks, the other replaces.

INVENTORY.PY

```
filename = "inventory.txt"

with open(filename, "w") as f:
 for i in range(3):
 name = input("Item: ")
 code = input("Code: ")
 qty = input("Quantity: ")
 f.write(f"Item: {name}, Code: {code}, Quantity: {qty}\n")

with open(filename, "a") as f:
 for i in range(2):
 name = input("Item: ")
 code = input("Code: ")
 qty = input("Quantity: ")
 f.write(f"Item: {name}, Code: {code}, Quantity: {qty}\n")

print("All items:")
with open(filename, "r") as f:
 for line in f:
 print(line.strip())

print("First two:")
with open(filename, "r") as f:
 for i, line in enumerate(f):
 if i >= 2:
 break
 print(line.strip())
```

Entering Pencil/P01/120, Eraser/E01/75, Ruler/R01/60, then two more, the reads give:

```
All items:
Item: Pencil, Code: P01, Quantity: 120
Item: Eraser, Code: E01, Quantity: 75
Item: Ruler, Code: R01, Quantity: 60
First two:
Item: Pencil, Code: P01, Quantity: 120
Item: Eraser, Code: E01, Quantity: 75
```

The append opens with "a", so the first three survive. Had it opened with "w", the file would have been wiped before the extra items were added.

## Question Six

### ANSWER

### Q6 (A) NESTED INDEXING

- ▶ `data[1][1][2]`: `data[1]` is `[300, [400, 500, 600]]`, its `[1]` is `[400, 500, 600]`, and `[2]` of that is `600`.
- ▶ `len(data[1])`: `data[1]` holds two items, so `2`.
- ▶ `data[2][0][1]`: `data[2]` is `[[700, 800], 900]`, its `[0]` is `[700, 800]`, and `[1]` of that is `800`.

```
600
2
800
```

Follow the brackets strictly left to right, writing down each layer as you peel it.

### ANSWER

### Q6 (B) THE VOID SUMMARY

RESULTS.PY

```
def print_result_summary(student, total_score, pass_mark):
 print(f"=== Result for {student} ===")
 print("Total Score:", total_score)
 if total_score >= pass_mark:
 print("Result: PASS")
 else:
 print("Result: FAIL")

name = input("Student name: ")
subjects = int(input("Number of subjects: "))
total_score = 0
for i in range(subjects):
 total_score += int(input(f"Score for subject {i + 1}: "))

print_result_summary(name, total_score, 250)
```

For Bola with five scores 70, 65, 80, 55, 70 (total 340):

```
=== Result for Bola ===
Total Score: 340
Result: PASS
```

340 clears the 250 pass mark. The function only prints and returns nothing, which is what "void" means: called for its effect, not for a value.

TANK.PY

```
import math

def volume(r, h):
 return math.pi * r ** 2 * h

radius = float(input("Radius: "))
height = float(input("Height: "))

v = volume(radius, height)
print(f"Volume: {v:.2f}")
if v > 1000:
 print("Large tank")
else:
 print("Small tank")
```

For radius 7 and height 10:

```
Volume: 1539.38
Large tank
```

$\text{math.pi} * 7 ** 2 * 10$  is  $\text{math.pi} * 49 * 10 = 1539.38$ . The  $** 2$  binds before the multiplications, so the radius is squared first, no brackets needed. 1539.38 exceeds 1000, so it is a large tank.